THE INFLUENCES OF MONEY · BOOK NOTES

Thinking, Fast and Slow

Daniel Kahneman (2011)

A chapter-by-chapter deep-dive study guide for investors, persuaders and content creators

Prepared for Niko Stavros · October 2026

Contents

How to use this guide ... 3

Opening Overview ... 4

Introduction ... 7

Part 1 · Two Systems ... 10

Chapter 1 · The Characters of the Story ... 11

Chapter 2 · Attention and Effort ... 14

Chapter 3 · The Lazy Controller ... 16

Chapter 4 · The Associative Machine ... 19

Chapter 5 · Cognitive Ease ... 22

Chapter 6 · Norms, Surprises, and Causes ... 25

Chapter 7 · A Machine for Jumping to Conclusions ... 28

Chapter 8 · How Judgments Happen ... 31

Chapter 9 · Answering an Easier Question ... 33

Part 2 · Heuristics and Biases ... 36

Chapter 10 · The Law of Small Numbers ... 37

Chapter 11 · Anchors ... 40

Chapter 12 · The Science of Availability ... 43

Chapter 13 · Availability, Emotion, and Risk ... 45

Chapter 14 · Tom W's Specialty ... 48

Chapter 15 · Linda: Less Is More ... 51

Chapter 16 · Causes Trump Statistics ... 54

Chapter 17 · Regression to the Mean ... 57

Chapter 18 · Taming Intuitive Predictions ... 60

Part 3 · Overconfidence ... 63

Chapter 19 · The Illusion of Understanding ... 64

Chapter 20 · The Illusion of Validity ... 67

Chapter 21 · Intuitions vs. Formulas ... 71

Chapter 22 · Expert Intuition: When Can We Trust It? ... 74

Chapter 23 · The Outside View ... 77

Chapter 24 · The Engine of Capitalism ... 80

Part 4 · Choices ... 83

Chapter 25 · Bernoulli's Errors ... 84

Chapter 26 · Prospect Theory ... 87

Chapter 27 · The Endowment Effect ... 91

Chapter 28 · Bad Events ... 93

Chapter 29 · The Fourfold Pattern ... 96

Chapter 30 · Rare Events ... 99

Chapter 31 · Risk Policies ... 102

Chapter 32 · Keeping Score ... 105

Chapter 33 · Reversals ... 108

Chapter 34 · Frames and Reality ... 111

Part 5 · Two Selves ... 114

Chapter 35 · Two Selves ... 115

Chapter 36 · Life as a Story ... 118

Chapter 37 · Experienced Well-Being ... 120

Chapter 38 · Thinking About Life ... 123

Conclusions ... 126

Appendix A · Judgment Under Uncertainty: Heuristics and Biases ... 129

Appendix B · Choices, Values, and Frames ... 132

Master List of Biases & Effects, with Investing Applications ... 135

One-Page Cheat Sheet ... 140

How to use this guide

This guide covers the whole book in order: the Introduction, all 38 chapters in Parts 1–5, the Conclusions, and the two original papers reprinted as Appendix A (*Judgment Under Uncertainty: Heuristics and Biases*, 1974) and Appendix B (*Choices, Values, and Frames*, 1984).

Every chapter follows the same structure:

1. **The argument.** What Kahneman is saying, in plain paragraphs.
2. **Key quotes.** Verbatim, with page numbers. Each one was checked word for word against the text of the source PDF.
3. **Concepts and experiments.** Each idea explained for a smart reader with no psychology background, with the original studies described.
4. **Connections.** How the chapter builds on earlier chapters and sets up later ones.
5. **Markets, money and people.** Real events and investor behaviour. Anything marked *Hypothetical* is an illustration I made up, not a historical fact.
6. **Takeaways** and **How to apply it**, under four headings: 💰 Investing & money · 🤝 People & persuasion · 🎬 Content · 🧠 Thinking.

About the page numbers. "p. 85" means page 85 of the source PDF (the 533-page ebook edition you supplied). That ebook has no printed page numbers, so page numbers from a hardcover or paperback will not match. To find a quote in another edition, search for a distinctive phrase. In a handful of quotes, an ellipsis (…) marks a place where the ebook's text contains a scanning artefact (stray characters inside a sentence). Those characters were left out; every word that is shown matches the source exactly.

About the science. The book came out in 2011. Since then, psychology's "replication crisis" has called some of the studies it cites into doubt, especially in the priming chapter (Chapter 4) and the ego-depletion section (Chapter 3). Kahneman acknowledged this himself. Wherever it matters, a **Replication note** tells you what has held up and what has not. The core of the book (heuristics, overconfidence, prospect theory, loss aversion, framing, regression, the planning fallacy) has held up well.

Why this book matters for a macro investor. Druckenmiller often says the hardest part of investing is managing yourself: sizing up when you are right, cutting fast when you are wrong, and not marrying a story. Dalio built Bridgewater around writing principles down, using algorithms, and "believability-weighted" disagreement, because he learned (most painfully in 1982) that his own confident intuition could be catastrophically wrong. *Thinking, Fast and Slow* is the scientific map of why those practices work. It explains which of your instincts you can trust, which you cannot, and how to build procedures that protect you from yourself.

Opening Overview

The thesis in one paragraph

Our minds run two kinds of thinking. One is fast, automatic and effortless; Kahneman calls it **System 1**. The other is slow, effortful and deliberate: **System 2**. We identify with System 2, the conscious reasoner, but System 1 does most of the work and is "the secret author of many of the choices and judgments" we make. System 1 is brilliant at what it evolved for: reading faces, avoiding threats, recognizing patterns in familiar, regular environments. It also makes **systematic, predictable errors** (biases) when a question is statistical, novel or abstract. It swaps hard questions for easy ones (*heuristics*), builds coherent stories from scraps of evidence (*what you see is all there is*), and feels confident whether or not that confidence is justified. Standard economics assumes **Econs**: rational, consistent, selfish agents. Real people are **Humans**. We judge gains and losses relative to a reference point, losses hurt about twice as much as gains please, we overweight small probabilities, and our choices flip with the wording of the problem. Finally, there are **two selves**. The *experiencing self* lives through each moment. The *remembering self* keeps the score and makes the decisions, and it judges experiences by their peak and their ending while ignoring how long they lasted. The book offers no magic cure. Its realistic remedy is to learn the vocabulary, recognize the "cognitive minefields," slow down when the stakes are high, and, most powerfully, build **organizations and procedures** that catch the errors individuals cannot see in themselves.

System 1 vs System 2

System 1: Fast

- Automatic, quick, effortless, always on
- Associative: one idea activates many others
- Builds coherent stories; suppresses doubt and ambiguity
- Answers easy questions in place of hard ones (substitution)
- Driven by emotion, vividness and familiarity
- Sensitive to changes and to losses; ignores base rates
- Expert intuition, but only in regular environments with feedback
- *Examples:* 2+2, reading a face, driving an empty road, sensing anger in a voice

System 2: Slow

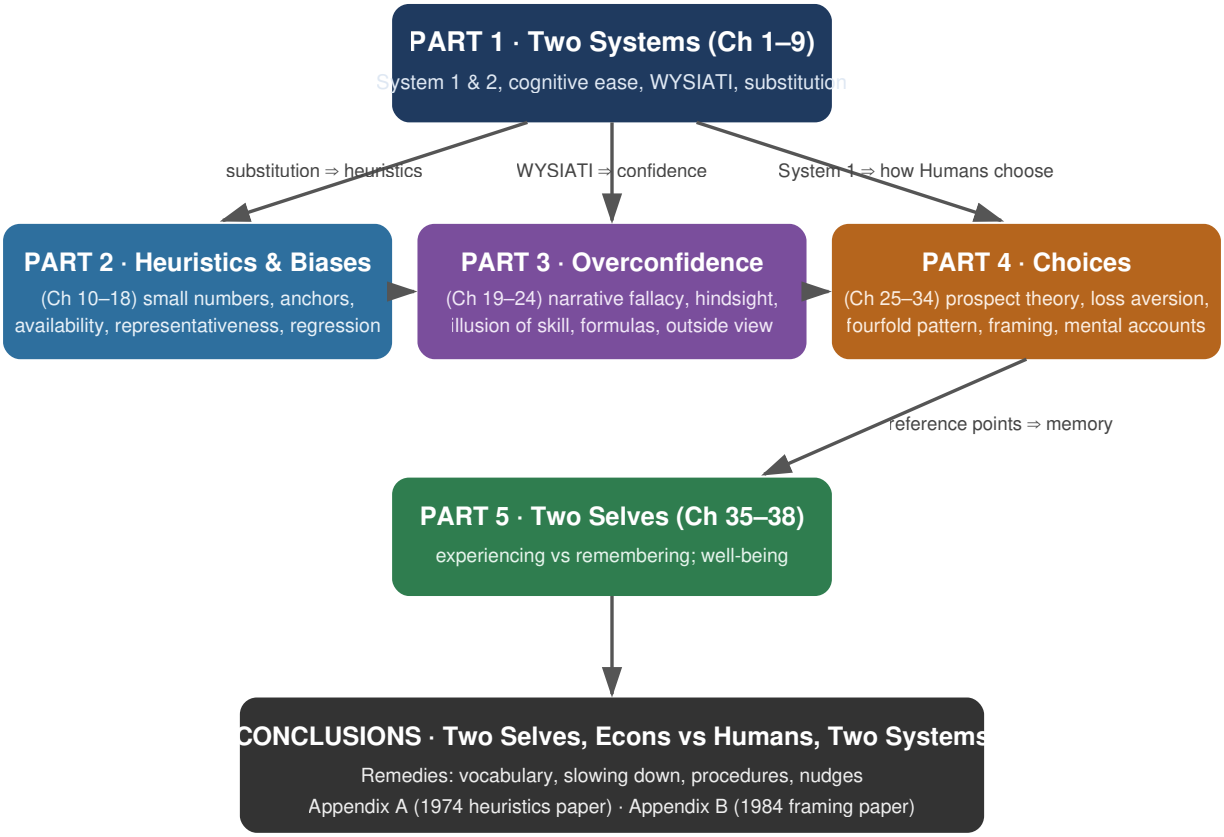
- Deliberate, effortful, sequential, limited capacity
- Follows rules, compares options, computes
- Monitors and can override System 1, but is lazy
- Gets depleted by fatigue, distraction, hunger and good mood
- Usually endorses System 1's suggestions ("an endorser rather than an enforcer")
- Needed for statistics, logic, self-control and novel problems
- *Examples:* 17 × 24, filling in a tax form, checking an argument, parking in a tight spot

Kahneman stresses that the two systems are **useful fictions**, characters in a story rather than brain regions. Saying "System 1 did X" is shorthand for "X happened automatically and fast."

The two species and the two selves

- **Econs vs Humans** (Part 4): Econs are the rational agents of economic theory, who are consistent, unaffected by framing, and value final states of wealth. Humans value changes, hate losses, and are moved by wording.
- **Experiencing self vs remembering self** (Part 5): the experiencing self answers "Does it hurt now?" The remembering self answers "How was it, on the whole?" The remembering self follows the **peak-end rule** and **duration neglect**, and it is the one that chooses what we do next.

Map of the book: how the parts build on each other



How the parts build on each other. Part 1 introduces the machinery. Part 2 shows the shortcuts it takes with numbers and probabilities. Part 3 shows why we don't notice those shortcuts (overconfidence). Part 4 applies the same machinery to risky choices. Part 5 applies it to happiness and memory.

The logic chain, in one line per part:

- **Part 1, the machinery.** System 1 runs on association and coherence. System 2 is lazy. When a question is hard, System 1 quietly answers an easier one.
- **Part 2, the shortcuts.** Those substitutions are the *heuristics*: representativeness, availability, anchoring and affect. They produce predictable biases, above all ignoring base rates, sample sizes and regression.

- **Part 3, the blindness.** Because System 1 builds coherent stories from whatever it sees (WYSIATI), we feel confident in our explanations of the past and predictions of the future even where the world is unpredictable. Formulas, the outside view and premortems are the antidotes.
- **Part 4, the choices.** The same System 1 turns decisions about money into decisions about gains and losses relative to a reference point. That gives us loss aversion, the fourfold pattern of risk attitudes, framing effects, mental accounting and regret.
- **Part 5, the selves.** We choose on the basis of memories, and memories distort experience (peak-end, duration neglect). So even "what makes us happy" is subject to the same illusions, especially the focusing illusion.
- **Conclusions.** Humans are not irrational, but they are not Econs either. Individuals can learn to spot minefields; organizations can build procedures (checklists, premortems, formulas, nudges) that slow thinking down where it matters.

The four big ideas to carry with you

1. **WYSIATI: What You See Is All There Is.** System 1 builds the best story it can from the information in front of it and never asks what is missing. Confidence reflects the coherence of the story, not the quality of the evidence.
2. **Substitution.** Faced with a hard question ("Will this stock outperform?"), we answer an easier one ("Do I like this company?") without noticing.
3. **Loss aversion and reference dependence.** We evaluate outcomes as gains or losses from a reference point, and losses weigh roughly 1.5–2.5 times as much as equivalent gains.
4. **Regression and the outside view.** Extreme results are partly luck and tend to be followed by less extreme ones. Start every forecast from the base rate of similar cases, then adjust modestly.

Introduction

PDF pp. 7–18

The argument

Kahneman opens with office gossip. He wants to improve the conversations people have at the "watercooler" about other people's judgments and choices, because it is easier to spot a mistake in someone else than in yourself. To talk about errors precisely, you need a vocabulary, the way a doctor needs names for diseases. The book's purpose is to supply that vocabulary: names such as *anchoring*, *availability*, *halo effect* and *loss aversion* that let you recognize a recurring pattern of error when you see it.

He then tells the origin story of his partnership with Amos Tversky, which began at the Hebrew University in 1969. Their first finding was embarrassing: even trained statisticians and research psychologists had poor statistical intuitions. They trusted small samples and drew big conclusions from little data. Kahneman and Tversky spent years inventing short puzzles that revealed how intuition works, and in 1974 published the *Science* paper on **heuristics and biases** (reprinted as Appendix A). Its message was that people use a few mental shortcuts (representativeness, availability, anchoring) which usually work but cause **systematic** errors. The errors are predictable, not random. That is what made the work influential far beyond psychology, in economics, medicine, law and policy. A second line of work, on decisions under risk, produced **prospect theory** (1979) and the framing paper (Appendix B).

Kahneman stresses that the book is not an argument that people are stupid. Most of our judgments are sound, and expert intuition is often impressive. His example is the firefighter commander who got his crew out of a burning house moments before the floor collapsed without consciously knowing why. Following Herbert Simon, he treats such skilled intuition as **recognition**: the expert's memory has learned to recognize cues. The book's goal is to tell valid intuition apart from the confident but baseless kind, and to show how the "secret author" of our choices, System 1, sometimes leads us astray.

Key quotes

> *"Why be concerned with gossip? Because it is much easier, as well as far more enjoyable, to identify and label the mistakes of others than to recognize our own."* p. 7

> *"So this is my aim for watercooler conversations: improve the ability to identify and understand errors of judgment and choice, in others and eventually in ourselves, by providing a richer and more precise language to discuss them."* p. 8

> *"We are often confident even when we are wrong, and an objective observer is more likely to detect our errors than we are."* p. 8

> *“Systematic errors are known as biases, and they recur predictably in particular circumstances.”* p. 7

> *“The situation has provided a cue; this cue has given the expert access to information stored in memory, and the information provides the answer. Intuition is nothing more and nothing less than recognition.”* p. 15

(The first quote above actually appears on p. 30, near the end of Chapter 1, where Kahneman restates the premise. It belongs here because it is the thesis of the whole project.)

Concepts and experiments

- **Heuristic.** A mental shortcut: a quick rule of thumb that gives an adequate answer most of the time.
- **Bias.** A *systematic* error, one that leans in the same direction, in predictable circumstances. Random noise cancels out on average; bias does not.
- **The statisticians' survey.** Kahneman and Tversky asked sophisticated researchers questions about replicating experiments. The experts were far too confident that small samples would repeat their results. Experts are fooled too. This became Chapter 10.
- **Steve the librarian.** "Steve is very shy and withdrawn, invariably helpful but with little interest in people or in the world of reality..." Is he more likely a librarian or a farmer? People say librarian because he fits the stereotype. But there are vastly more male farmers than male librarians, so on the statistics he is probably a farmer. This is the **representativeness** heuristic overriding **base rates** (Chapters 14–16).
- **The letter K.** Is K more likely to appear as the first or the third letter of English words? It is easier to *think of* words that start with K, so people say first. In fact K is more common in third position. This is the **availability** heuristic (Chapter 12).
- **Skilled intuition vs heuristics.** The firefighter's "sixth sense" and the chess master's instant read are cases of recognition built through long practice with feedback. Heuristic judgments look like intuition but do not rest on expertise (Chapter 22).

Connections

- **Sets up:** the two characters (Chapter 1), the heuristics of Part 2, the expert-intuition debate (Chapter 22), prospect theory (Part 4) and the two selves (Part 5).
- **Recurring idea:** the book's goal is diagnostic vocabulary. Each chapter ends with "Speaking of..." sentences you can use in real conversations.

Markets, money and people

- **Investment committees.** The "watercooler" is the investment committee. A colleague can see your anchoring on a purchase price far more easily than you can, which is why

Bridgewater built a culture of open disagreement and "believability-weighted" decisions, and why Druckenmiller has kept trusted partners whose job is to challenge his views.

- **Experts fooled by small samples.** Before the dot-com crash, many professional analysts extrapolated a few years of internet growth into permanent trends. From its March 2000 peak, the Nasdaq Composite lost roughly three-quarters of its value by October 2002. Sophistication does not protect against statistical intuition errors.
- **Hypothetical:** an analyst says a stock "looks like the next Amazon." That is representativeness. The relevant base rate is how many early-stage tech companies ever become dominant platforms, and the answer is very few.

Takeaways

- Errors of intuition are **systematic**, so they can be predicted and prepared for.
- You will see these errors in others before you see them in yourself, so use other people as error detectors.
- Intuition is valid when it is **recognition** learned in a regular environment, and suspect otherwise.
- Learning the names of biases is the first defence, because a named error is easier to catch.

How to apply it

- 💰 **Investing:** keep a decision journal recording your thesis, the confidence you felt, and the evidence. Review it with someone who will tell you when you're anchoring or story-telling.
- 🤝 **People:** criticize decisions, not people. A shared vocabulary ("Are we anchoring on last quarter?") makes criticism impersonal and acceptable.
- 🎬 **Content:** the "watercooler" framing is a natural series format. Each episode can name one bias, show a famous market example, and give viewers a phrase to use.
- 🧠 **Thinking:** ask "Is this recognition from real practice, or a feeling of confidence?"

Part 1 · Two Systems

Part 1 introduces the two characters, System 1 and System 2, and the mechanics of the automatic mind: effort, laziness, association, cognitive ease, causal thinking, jumping to conclusions (WYSIATI), and substitution, which is the engine behind every heuristic in Part 2.

Chapter 1 · The Characters of the Story

PDF pp. 21–32

The argument

Look at a photo of an angry woman and you know instantly that she is angry and about to shout. You didn't decide to know this; it happened to you. Now look at 17 × 24. You know it's a multiplication problem and that the answer isn't 12,609 or 123, but getting the exact answer (408) takes deliberate, effortful work. These two experiences are Kahneman's two characters. **System 1** operates automatically and quickly, without effort or a sense of control. **System 2** handles effortful mental activities such as complex calculation, and it is linked to our feeling of agency and choice.

We identify with System 2, the conscious self that believes it is in charge. The book's real hero, though, is System 1. It continuously generates impressions, intuitions, intentions and feelings. When System 2 endorses them, they become beliefs and actions. Usually this division of labour works well. System 1 is good at familiar situations and short-term predictions. System 2 steps in when System 1 runs into trouble (a surprise, a problem it cannot solve) and is responsible for self-control.

The conflict between the systems shows in tasks like naming the font case of words, or saying "left" or "right" for words printed on the wrong side, where an automatic response must be overridden. Some System 1 errors are **illusions** that cannot be switched off. In the Müller-Lyer illusion, two lines with fins pointing in different directions look unequal even after you have measured them and *know* they are equal. The remedy is not to stop seeing the illusion but to learn to distrust your impression in specific situations. Cognitive illusions work the same way. Since constant vigilance is impossible, the best we can do is learn to recognize the situations where mistakes are likely and slow down when the stakes are high.

Key quotes

> *"System 1 operates automatically and quickly, with little or no effort and no sense of voluntary control."* p. 22

> *"System 2 allocates attention to the effortful mental activities that demand it, including complex computations."* p. 22

> *"Although System 2 believes itself to be where the action is, the automatic System 1 is the hero of the book."* p. 23

> *"The gorilla study illustrates two important facts about our minds: we can be blind to the obvious, and we are also blind to our blindness."* p. 26

> *"The best we can do is a compromise: learn to recognize situations in which mistakes are likely and try harder to avoid significant mistakes when the stakes are high."* p. 30

> *"This is your System 1 talking. Slow down and let your System 2 take control."* p. 32

Concepts and experiments

- **System 1 examples:** detecting that one object is farther away than another, completing "bread and...", making a disgust face at a horrible picture, reading words on billboards, driving a car on an empty road, understanding simple sentences.
- **System 2 examples:** bracing for the starter gun in a race, looking for a woman with white hair in a crowd, counting the occurrences of the letter *a* on a page, giving your phone number, parking in a narrow space, checking the validity of a complex argument.
- **Attention is a budget.** "Pay attention" is apt. Attention is limited, and if you overspend, you fail. You cannot compute 17 × 24 while making a left turn into dense traffic.
- **The invisible gorilla (Chabris & Simons).** Viewers watch a video of two teams passing basketballs and count the white team's passes. Partway through, a person in a gorilla suit crosses the court, thumps their chest and leaves, nine seconds on screen. About half the viewers don't notice. Those who missed it are sure it wasn't there. Intense focus makes us blind to the obvious, and we don't know we're blind.
- **Müller-Lyer illusion.** The lines are equal but look unequal. Knowing the truth does not change the perception. You must *learn to mistrust* your impressions when you see "fins."
- **Psychopath charm (illusions of thought).** Kahneman recalls a teacher telling clinicians that a patient who tells a moving story of mistreatment by previous therapists, and who seems to feel instant rapport with you, may well be a psychopath. Feeling drawn to the patient is the warning sign. The lesson is to recognize the *situation* in which your intuition is untrustworthy.
- **Useful fictions.** System 1 and System 2 are not brain regions. They are characters that make the mind easier to think about, because "System 1 does X" is easier to remember than "X occurs automatically."

Connections

- **Earlier:** develops the Introduction's distinction between skilled intuition and heuristics.
- **Later:** System 2's laziness (Chapters 2–3) is why System 1's suggestions so often go unchecked. The Müller-Lyer lesson ("learn to mistrust your impression") returns throughout, most clearly in the corrections for intuitive prediction (Chapter 18), the outside view (Chapter 23) and the Conclusions.

Markets, money and people

- **The gorilla in 2006–07.** Many bank risk managers concentrated on volatility-based models (VaR) that showed low risk because recent volatility was low, and many missed the growing

exposure to U.S. housing through mortgage-backed securities. Focus on one metric can blind you to a huge risk.

- **Fast reactions on a trading floor.** System 1 is useful when a headline hits and you need to know immediately whether it is good or bad for your position. It is dangerous when it also decides the *size* of your reaction.
- **Hypothetical:** you see a stock fall 15% pre-market, feel immediate fear and want to sell. That is System 1. The System 2 question is "What do I know now that I didn't know yesterday, and does it change my thesis?"

Takeaways

- Most of what you think and do starts in System 1. System 2 is the slower checker.
- Focused attention creates blindness. You will miss things you aren't looking for.
- Some errors are like optical illusions: they persist even when you know about them. Learn the *situations* that call for distrust.

How to apply it

- 💰 **Investing:** write pre-defined "slow down" triggers, for example: any trade larger than X% of the portfolio, any trade made within an hour of big news, or any trade that reverses a position taken this week.
- 🤝 **People:** in a negotiation, the person who forces you to answer *fast* is keeping you in System 1. "Let me come back to you tomorrow" brings in System 2.
- 🎬 **Content:** the gorilla video is one of the best hooks in psychology, so build an episode around "the gorilla in the market." Interactive demos (the bat-and-ball problem, Müller-Lyer) make viewers *feel* System 1 working.
- 🧠 **Thinking:** when you feel instant certainty in an unfamiliar, high-stakes situation, treat that certainty itself as a warning.

Chapter 2 · Attention and Effort

PDF pp. 33–40

The argument

If the book were a movie, System 2 would be a supporting character who thinks she's the star. Its defining feature is that its work is **effortful**, and its main trait is laziness. Kahneman demonstrates effort with the **Add-1** task: hear a string of four digits and, in rhythm, say each one increased by 1 (5294 becomes 6305). Add-3 is harder still. Within seconds you reach the limit of your mental capacity.

The best measure of effort turned out to be the **pupil**. Working with Jackson Beatty, Kahneman found that the pupil widens in proportion to mental effort: it dilates as the digits are loaded, peaks at the hardest moment and contracts as the load lifts, an "inverted V." Under strain, people become effectively blind to other things. Observers noticed that people in the middle of hard mental work didn't register things right in front of them.

Mental effort is costly, so the brain follows a **law of least effort**: given several ways to reach a goal, we drift toward the least demanding one. Skill reduces effort. Experts and highly intelligent people use less effort for the same problem, and training lets more of the work shift to System 1. Effort also shows in what System 2 alone can do: hold several ideas in mind at once, follow rules, compare objects on several attributes, and make deliberate choices between options.

Key quotes

> *"System 2 would be a supporting character who believes herself to be the hero."* p. 33

> *"The law asserts that if there are several ways of achieving the same goal, people will eventually gravitate to the least demanding course of action."* p. 37

> *"As you become skilled in a task, its demand for energy diminishes."* p. 37

> *"The law of least effort is operating here. He will think as little as possible."* p. 39

Concepts and experiments

- **Add-1 / Add-3 task.** A tightly paced memory-and-transform task that drives System 2 to its limit within seconds.
- **Pupillometry (Hess; Kahneman & Beatty).** The pupils dilate with mental effort, a physical "meter" of System 2 use. During a casual conversation, the pupil barely moves.
- **Effort and blindness.** At peak effort, people miss stimuli they would otherwise notice (the gorilla effect again).

- **Law of least effort.** Cognitive effort is a cost, and people minimize it. Laziness is built into the design.
- **Task switching and time pressure.** The most effortful kinds of thinking require holding several things in memory at once or switching quickly between tasks, especially under time pressure.

Connections

- **Earlier:** explains *why* System 2 often fails to override System 1 (Chapter 1).
- **Later:** effort is limited and shared with self-control (Chapter 3). Cognitive strain versus ease (Chapter 5). Because effort is costly, substitution (Chapter 9) is the default.

Markets, money and people

- **Prospectus complexity.** Complex structured products (synthetic CDOs before 2008, for example) asked buyers to make System 2 calculations few would actually do. The law of least effort pushed investors to rely on the AAA rating, an easy cue that stood in for real analysis. Many of those AAA tranches were later downgraded and lost heavily.
- **Fee disclosures.** Fund fees are buried in percentages that take effort to translate into dollars. Few investors calculate that a 1% annual fee can take a substantial share of a portfolio's ending value over decades.
- **Hypothetical:** a trader who checks positions while on a demanding call has no System 2 capacity left for either task.

Takeaways

- Mental effort is limited and measurable; your pupils show it.
- Under heavy load you become blind to other information.
- People default to the least effortful route, so design matters.

How to apply it

- 💰 **Investing:** don't make portfolio decisions while multitasking. Batch important decisions into protected, uninterrupted time.
- 🤝 **People:** if you want people to evaluate something carefully, reduce their load (one idea at a time). If someone is trying to rush you through complex terms, recognize the tactic.
- 🎬 **Content:** viewers obey the law of least effort too. One idea per video, a clear structure and visual aids lower the effort needed to follow you.
- 🧠 **Thinking:** write intermediate steps down instead of holding them in working memory, as Kahneman advises (taking time, using paper).

Chapter 3 · The Lazy Controller

PDF pp. 41–51

The argument

Kahneman likes a leisurely walk at about 17 minutes per mile. At that pace he can think. If he speeds up, he loses the ability to think coherently, because self-control and deliberate thought draw on the same limited budget. Psychologist Mihaly Csikszentmihalyi's **flow** is the exception: a state of effortless concentration where no self-control is needed.

Research on the "busy" System 2 shows that cognitive load weakens self-control. People asked to memorize a seven-digit number were more likely to pick chocolate cake over fruit salad. Roy Baumeister's research on **ego depletion** suggested that exerting self-control (resisting cookies, suppressing emotion) leaves people less able to persist on the next demanding task, as if willpower were a tank that runs dry, possibly linked to glucose. The striking real-world study cited is of **Israeli parole judges**: approval rates spiked to about 65% after each meal break and fell toward zero just before the next one. The interpretation offered is that tired and hungry judges retreat to the default of denial.

The core of the chapter is System 2's laziness. In the **bat-and-ball** problem ("A bat and ball cost $1.10. The bat costs one dollar more than the ball. How much does the ball cost?"), the intuitive answer, 10¢, is wrong (it's 5¢). More than half of students at Harvard, MIT and Princeton gave the intuitive answer, and over 80% at less selective universities. They could have checked it but didn't. Similarly, with the syllogism "All roses are flowers. Some flowers fade quickly. Therefore some roses fade quickly," most college students accept the invalid conclusion because it *feels* true. Shane Frederick's **Cognitive Reflection Test** measures this tendency to accept the first answer. Keith Stanovich distinguishes intelligence from *rationality*: the willingness to engage System 2. Walter Mischel's marshmallow studies link childhood self-control to later life outcomes.

Key quotes

> *"Self-control and deliberate thought apparently draw on the same limited budget of effort."* p. 42

> *"The bat-and-ball problem is our first encounter with an observation that will be a recurrent theme of this book: many people are overconfident, prone to place too much faith in their intuitions."* p. 46

> *"The best possible account of the data provides bad news: tired and hungry judges tend to fall back on the easier default position of denying requests for parole."* p. 45

> *“Those who avoid the sin of intellectual sloth could be called "engaged."”* p. 48

> *“His ego was depleted after a long day of meetings. So he just turned to standard operating procedures instead of thinking through the problem.”* p. 50

Concepts and experiments

- **Flow.** Deep, effortless concentration (Csikszentmihalyi) in which attention needs no self-control.
- **Cognitive load and temptation (Shiv & Fedorikhin).** People holding seven digits in memory were more likely to choose chocolate cake.
- **Ego depletion (Baumeister).** An initial act of self-control reduces performance on a subsequent one.
- **Parole judges (Danziger, Levav & Avnaim-Pesso, 2011).** Eight Israeli judges, more than 1,100 decisions. Approvals were highest after food breaks.
- **Bat-and-ball and the CRT (Frederick).** Measures "cognitive reflection," the habit of checking intuitive answers.
- **Syllogism with roses.** Belief bias: plausibility overrides logic.
- **Intelligence vs rationality (Stanovich).** Being smart does not mean you engage System 2.
- **Marshmallow test (Mischel).** Four-year-olds choosing between one treat now or two later. The ability to wait correlated with later outcomes.

Replication note. Ego depletion has largely failed to replicate. A pre-registered replication run in 23 labs (Hagger et al., 2016) found an effect close to zero, and the glucose explanation is widely doubted. The parole-judges study has been challenged: critics (Weinshall-Margel & Shapard, 2011) argued that the order of cases was not random, since prisoners without lawyers tended to be heard last before a break. Treat both as suggestive at best. Later marshmallow replications (Watts, Duncan & Quan, 2018) found much smaller effects once family background is controlled for. The bat-and-ball and CRT findings, by contrast, are robust.

Connections

- **Earlier:** the limited attention budget from Chapter 2.
- **Later:** System 2's laziness is why System 1's substitutions (Chapter 9) and heuristics (Part 2) usually go unchallenged. Overconfidence returns in Part 3. "Engaged" thinkers are the readers the book hopes to create.

Markets, money and people

- **GameStop, January 2021.** Social media made a short squeeze emotionally vivid and easy to understand ("hedge funds lose, we win"). GameStop rose from about $20 at the start of the month to an intraday high above $480 on January 28. Many late buyers never did the simple

System 2 check of what valuation the price implied, and the stock fell back sharply within weeks. This was a bat-and-ball problem at market scale: an intuitive answer that felt right and went unchecked.

- **Decision fatigue in trading.** Many professional traders limit trading after long sessions. Whatever the replication status of ego depletion, fatigue and long hours plainly degrade attention.
- **Hypothetical:** a "10¢ answer" in investing: "The stock fell 50%, so it's half price." A 50% fall needs a 100% gain to recover, and the fall may reflect a permanent change in value.

Takeaways

- System 2 is capable but lazy and often fails to check System 1.
- Intelligence does not guarantee rational thinking. The habit of checking matters.
- Whatever the precise mechanism, fatigue, load and distraction make defaults and impulses win.

How to apply it

- 💰 **Investing:** run a "CRT check" on every thesis. What is the obvious answer? What would make it wrong? Do the arithmetic, especially percentages, compounding and break-evens.
- 🤝 **People:** don't ask for important decisions late in the day or right before lunch. Make your proposal when people are fresh.
- 🎬 **Content:** bat-and-ball-style puzzles ("Most people get this wrong…") are irresistible openers and a good way to dramatize System 1 vs System 2.
- 🧠 **Thinking:** when an answer comes instantly to a quantitative question, recompute it once on paper.

Chapter 4 · The Associative Machine

PDF pp. 52–60

The argument

Read two words, *Bananas* and *Vomit*, and in a second or two your body reacts: a slight grimace, a moment of disgust, a temporary aversion to bananas. Your mind also *invented a story* linking the two words causally. This is **associative activation**: ideas trigger other ideas in a spreading cascade, automatically, and the mind responds with body, emotions and thoughts together. Cognition is *embodied*: we think with our body, not only our brain.

The chapter's central concept is **priming**. Exposure to one stimulus changes how we respond to the next without our awareness. After seeing EAT, you are more likely to complete SO_P as SOUP; after WASH, as SOAP. Kahneman goes further and reports that priming affects *actions*. In John Bargh's "**Florida effect**" study, students who built sentences from words associated with old age (Florida, forgetful, bald, grey, wrinkle) walked down the hall more slowly afterwards. The **ideomotor** effect works in reverse too: walking slowly makes old-age ideas more accessible, and holding a pencil in your teeth to force a smile makes cartoons seem funnier.

Kahneman reports **money priming** (Kathleen Vohs): people exposed to images of money became more independent and selfish, persisting longer on hard problems without asking for help but helping others less. He also cites the honesty box study at a British university: when a picture of eyes hung above the box, people paid nearly three times more for their tea or coffee than when flowers were shown. The broad claim is that we are influenced by our environment much more than we know, and that this feels alien because it corresponds to nothing in our experience.

Key quotes

> *"As cognitive scientists have emphasized in recent years, cognition is embodied; you think with your body, not only with your brain."* p. 53

> *"You cannot know this from conscious experience, of course, but you must accept the alien idea that your actions and your emotions can be primed by events of which you are not even aware."* p. 54

> *"Studies of priming effects have yielded discoveries that threaten our self-image as conscious and autonomous authors of our judgments and our choices."* p. 56

> *"Money-primed people become more independent than they would be without the associative trigger."* p. 57

> *"The world makes much less sense than you think. The coherence comes mostly from the way your mind works."* p. 60

Concepts and experiments

- **Associative coherence.** Activated ideas reinforce each other into a coherent pattern: cognitive, emotional and physical.
- **Priming of words.** EAT primes SOUP; WASH primes SOAP. This semantic priming is solidly established.
- **Florida effect (Bargh, Chen & Burrows, 1996).** Old-age words led to slower walking.
- **Ideomotor effect.** Ideas prime actions and actions prime ideas (nodding vs shaking your head while listening to messages; facial-feedback pencil studies).
- **Money priming (Vohs et al.).** Reminders of money led to more self-reliance and less helpfulness.
- **Voting location.** Support for school funding was higher when the polling station was in a school.
- **Honesty box (Bateson, Nettle & Roberts, 2006).** Images of eyes increased contributions.

Replication note. This is the most problematic chapter in the book. The Florida effect failed to replicate in a 2012 study (Doyen et al.), and money priming has failed in large replications. In 2012 Kahneman wrote an open letter to priming researchers warning of a "train wreck looming." In 2017, responding to a critical analysis of this chapter, he wrote that he had placed too much faith in underpowered studies. Basic semantic priming (EAT → SOUP) is solid. Dramatic *behavioural* priming claims should be treated as unproven. The eyes-on-the-honesty-box result has had mixed replications.

Connections

- **Earlier:** System 1 as automatic and effortless (Chapters 1–2).
- **Later:** association is the mechanism behind cognitive ease (Chapter 5), causal story-making (Chapter 6), the halo effect and WYSIATI (Chapter 7), anchoring as priming (Chapter 11) and availability (Chapter 12).

Markets, money and people

- **Narratives spread by association.** In 2020–21, terms like "EV," "SPAC" and "metaverse" triggered associations with the future, Tesla and innovation. Facebook renamed itself Meta in October 2021. Many SPACs that went public in 2020–21 then fell sharply. Labels activate stories before analysis starts.
- **Confusing tickers.** In March 2020, investors bid up Zoom Technologies (ticker ZOOM), an unrelated, tiny company, apparently confusing it with Zoom Video (ZM). The SEC suspended trading in ZOOM. In January 2021, after Elon Musk tweeted "Use Signal" (the messaging

app), an unrelated company, Signal Advance (SIGL), soared. Pure association, with no analysis.

- **Hypothetical:** a newsletter that pairs every pick with images of yachts and wealth may be designed to trigger associations with success that have nothing to do with the merits of the stock.

Takeaways

- Ideas spread automatically through networks of association, and the body joins in.
- Context shapes perception: a word, image or label can change what you notice next.
- Be sceptical of dramatic priming claims, but the core insight, that the environment of the moment influences us more than we realize, is sound.

How to apply it

- 💰 **Investing:** strip labels. Describe a company in neutral words ("a lossmaking retailer with $X revenue") before you read the story. Rename themes in your notes to avoid triggering hype associations.
- 🤝 **People:** the setting of a meeting and the words you use first shape the associations that follow. Open with the frame you want.
- 🎬 **Content:** your thumbnail, first line and music prime how viewers interpret everything after. Use that ethically: prime curiosity and clarity, not fear.
- 🧠 **Thinking:** when you notice a strong emotional reaction to a word ("bubble," "crash," "AI"), ask what associations it triggered and whether they bear on the facts.

Chapter 5 · Cognitive Ease

PDF pp. 61–71

The argument

System 1 constantly checks a dashboard of dials, and one of them measures **cognitive ease**, ranging from "easy" to "strained." Ease signals that all is well: no threats, no need to mobilize System 2. Strain signals a problem. Ease comes from repeated experience, clear display, primed ideas and good mood. It produces feelings of **familiarity, truth, goodness and effortlessness**. Strain comes from bad fonts, complex language and bad mood, and it makes people more vigilant and analytical.

The trouble is that System 1 cannot tell *why* something feels easy. Larry Jacoby's "becoming famous overnight" studies showed that names seen once before later felt familiar and were judged famous. The **illusion of truth**: statements that are repeated, or printed in clear, high-contrast fonts, or that rhyme ("Woes unite foes" vs "Woes unite enemies"), are judged more true. Kahneman draws practical lessons for writing persuasive messages: use clear print, simple language, verse where possible, and easily pronounced names.

Strain has an upside. When Shane Frederick's Cognitive Reflection Test was printed in a faint, hard-to-read font, error rates dropped sharply, because strain activated System 2. Robert Zajonc's **mere exposure effect** shows that repeated exposure to neutral stimuli (Turkish-sounding words in a student newspaper, even tones heard by chicks in the egg) increases liking, with no awareness required. Finally, good mood loosens System 2: people in a good mood are more intuitive and creative but also more gullible and error-prone.

Key quotes

> *"A reliable way to make people believe in falsehoods is frequent repetition, because familiarity is not easily distinguished from truth."* p. 64

> *"If you care about being thought credible and intelligent, do not use complex language where simpler language will do."* p. 65

> *"Put your ideas in verse if you can; they will be more likely to be taken as truth."* p. 65

> *"When you feel strained, you are more likely to be vigilant and suspicious, invest more effort in what you are doing, feel less comfortable, and make fewer errors, but you also are less intuitive and less creative than usual."* p. 62

> *"Mood evidently affects the operation of System 1: when we are uncomfortable and unhappy, we lose touch with our intuition."* p. 70

> *“We must be inclined to believe it because it has been repeated so often, but let's think it through again.”* p. 71

Concepts and experiments

- **Cognitive ease.** The feeling of fluency. It causes, and is caused by, familiarity, good mood, clarity and priming.
- **Becoming famous overnight (Jacoby).** Exposure to made-up names made them feel famous later.
- **Illusion of truth.** Repetition and fluency increase belief, even in partially familiar false statements.
- **Rhyme and fonts (McGlone & Tofighbakhsh).** Rhyming aphorisms were rated more insightful. Bold, clear statements were judged more true.
- **Pronounceable names.** Stocks with easily pronounced names or tickers outperformed in the first days after an IPO (Alter & Oppenheimer). This is a direct finance result reported in the chapter.
- **Strain improves accuracy.** The CRT printed in a hard-to-read font produced fewer errors (Alter, Oppenheimer and colleagues).
- **Mere exposure (Zajonc).** Repetition breeds liking.
- **Remote Association Test (Mednick).** People could feel that a triad like "cottage, Swiss, cake" has a solution ("cheese") before finding it. Good mood improved this intuition, and sad mood wiped it out.

Connections

- **Earlier:** priming and association (Chapter 4) are a major source of ease.
- **Later:** fluency underlies availability (Chapters 12–13: ease of retrieval), confidence by coherence (Chapters 7, 20) and the anchoring-as-priming account (Chapter 11). Cognitive strain as a debiasing tool returns in Chapter 22's discussion of validity.

Markets, money and people

- **Fluent names.** Alter & Oppenheimer's finding on pronounceable tickers is a real market example. The dot-com era showed fluency's power: a study by Cooper, Dimitrov & Rau (2001) found that firms that added ".com" to their names in 1998–99 saw large abnormal stock returns around the announcement, even when the business didn't change.
- **Repetition as truth.** "Stocks for the long run," "housing prices never fall nationally" (a widely repeated belief before 2006–08), "inflation is transitory" (the Fed's widely repeated framing in 2021, before U.S. CPI inflation peaked at 9.1% in June 2022). Repeated claims feel true.
- **Simplicity sells.** Warren Buffett's shareholder letters are famously plain-English. Clear writing earns credibility.

- **Hypothetical:** a pitch deck with clean fonts and a confident story feels more true than a messy spreadsheet with the same numbers. Read the messy one carefully.

Takeaways

- Feelings of truth and familiarity come partly from fluency, which may have nothing to do with truth.
- Repetition makes claims feel true. Be aware of this in yourself and in markets.
- Strain triggers System 2; ease lowers your guard. Good mood makes you more creative and more gullible.

How to apply it

- 💰 **Investing:** be extra sceptical of ideas you have heard repeatedly (consensus trades) and of beautifully presented pitches. Deliberately reformat key numbers into a plain table before deciding.
- 🤝 **People:** to persuade, be clear, simple, memorable and repeated. To evaluate, slow down when something feels too easy.
- 🎬 **Content:** this chapter is a creator's manual. Use simple words, high-contrast thumbnails, rhythmic and rhyming titles, consistent catchphrases and repetition of the core idea.
  Ethical line: use fluency to make true ideas easy, never to make false ones feel true.
- 🧠 **Thinking:** when you feel in a great mood and very sure, that's when to double-check.

Chapter 6 · Norms, Surprises, and Causes

PDF pp. 72–78

The argument

System 1's main job is to maintain and update a model of your personal world that represents what is *normal*. Surprise is a measure of how well that model works. There are two kinds of surprise: events you actively expected that didn't happen, and events you didn't expect at all. The model adjusts very quickly. Kahneman tells of meeting an acquaintance, a psychologist named Jon, at a tiny resort on the Great Barrier Reef, and then, two weeks later, finding Jon in the next seat at a London theatre. The second coincidence was objectively even less likely, but it felt *less* surprising. A single earlier meeting had made "running into Jon in odd places" feel almost normal.

Norms are built automatically and shared widely. The **Moses illusion** ("How many animals of each kind did Moses take into the ark?") shows that we don't notice an error if it fits the context. Moses and Noah share a biblical context, so few people object (it was Noah). We detect abnormality instantly when context is violated ("Earth is a planet, and so is Mars" vs a semantically odd sentence).

Above all, System 1 sees **causes and intentions**. Albert Michotte showed that people perceive physical causality directly: watching one square "launch" another, we *see* the cause, without inferring it. Heider and Simmel's animation of triangles and a circle moving around a box is perceived as a story of a bully and victims. Infants show this sensitivity. This is why we are so poor at statistical thinking: statistics deals with chance and aggregates, while System 1 wants a causal story for everything. Kahneman retells a story from Nassim Taleb's *The Black Swan* about the headlines on the day Saddam Hussein was captured in December 2003. Bond prices rose, and Bloomberg explained the rise by the capture ("may not curb terrorism"). Half an hour later prices fell, and the same capture was used to explain the fall ("boosts the allure of risky assets"). One event "explained" opposite outcomes.

Key quotes

> *"The main function of System 1 is to maintain and update a model of your personal world, which represents what is normal in it."* p. 72

> *"We are evidently ready from birth to have impressions of causality, which do not depend on reasoning about patterns of causation."* p. 77

> *"She can't accept that she was just unlucky; she needs a causal story. She will end up thinking that someone intentionally sabotaged her work."* p. 78

> *"When we survey the reaction to these products, let's make sure we don't focus exclusively on the average. We should consider the entire range of normal reactions."* p. 78

Concepts and experiments

- **Norm theory (Kahneman & Miller).** Every event is compared with the norms it evokes. Surprise is the gap.
- **Passive vs active expectations.** Active expectations are conscious ("I'm waiting for the bus"). Passive ones are not conscious, but their violation still surprises.
- **Moses illusion.** Contextually consistent errors slip by.
- **Michotte's launching displays.** Causality is perceived, not inferred.
- **Heider & Simmel (1944).** Moving shapes are perceived as agents with intentions.
- **Causes vs statistics.** System 1 thinks in agents and causes, which is why "chance" feels unsatisfying (Chapter 10) and why base rates are ignored unless causal (Chapter 16).

Connections

- **Earlier:** associative coherence (Chapter 4) builds the norms.
- **Later:** the hunger for causes drives the misreading of small samples (Chapter 10), "causes trump statistics" (Chapter 16), causal stories for regression (Chapter 17) and the narrative fallacy (Chapter 19). Norms are the basis of regret (Chapter 32), which depends on what was normal.

Markets, money and people

- **Daily market commentary.** The Saddam-capture bond example is in the book. Every day financial media attach a cause to every move ("stocks fell on profit-taking," "rose on rate optimism"). Much day-to-day movement is noise, but System 1 demands a reason.
- **Normalization of extremes.** After years of near-zero interest rates (2009–2015 and again in 2020–21), many investors came to treat ultra-low rates as normal. The rapid hikes of 2022 surprised many who had built portfolios on that norm.
- **Hypothetical:** a fund posts two bad months, and investors invent a story (the manager "lost his touch") when the result is within normal variation for the strategy.

Takeaways

- Your sense of "normal" updates fast, sometimes after a single repetition.
- System 1 perceives causes and intentions automatically, even in random events.
- Explanations offered after the fact, especially for market moves, are often just stories.

How to apply it

- 💰 **Investing:** before explaining a price move, ask whether it falls inside normal volatility. Know the standard deviation of your positions so you can tell signal from noise.
- 🤝 **People:** what someone sees as "normal" is their reference. Reset expectations early (by explaining the range of possible outcomes) so normal variation doesn't feel like betrayal.
- 🎬 **Content:** audiences crave causes. You can satisfy that honestly by explaining causes where they exist and teaching viewers that "sometimes there's no reason, just noise," which is itself a compelling contrarian hook.
- 🧠 **Thinking:** when you catch yourself explaining a random outcome, try saying "that could have been chance" out loud.

Chapter 7 · A Machine for Jumping to Conclusions

PDF pp. 79–88

The argument

Jumping to conclusions is efficient when the conclusions are likely to be right, mistakes are cheap and time is short. It is risky in unfamiliar, high-stakes situations. System 1 resolves ambiguity without telling you there was any. You read "A B C" and "12 13 14" with an identical middle symbol, and a sentence about Ann approaching the bank as being about money, not a river, unless context says otherwise. You are not aware that a choice was made.

Following Daniel Gilbert, Kahneman argues that understanding a statement starts with *believing* it. Unbelieving is a System 2 operation, and when System 2 is busy, people are more likely to believe false statements. Then comes **confirmation bias**: we test hypotheses by looking for confirming evidence ("Is Sam friendly?" brings friendly examples to mind).

The **halo effect** colours everything: if you like a person's politics, you're more likely to like their voice and looks. First impressions weigh heavily. Solomon Asch showed that "intelligent, industrious, impulsive, critical, stubborn, envious" (Alan) seems much nicer than the same list in reverse order (Ben). As a teacher, Kahneman found that his grading of an essay was swayed by the student's first essay, so he started grading all students' answers to one question before moving to the next, and hid the scores. That is the principle of **decorrelating errors**: get independent judgments before discussion, as in Galton's ox-weighing crowd or a meeting where everyone writes their view first.

The chapter's key idea is **WYSIATI: What You See Is All There Is**. System 1 builds the most coherent story it can from available information and does not account for missing information. In a study by Brenner, Koehler & Tversky, people who heard only one side of a legal case were *more* confident of their judgment than those who heard both sides, even though they knew they had heard only one side. WYSIATI explains overconfidence (the story's coherence produces confidence), framing effects (different presentation of the same facts produces different conclusions) and base-rate neglect (statistics aren't part of the story).

Key quotes

> *"Jumping to conclusions is efficient if the conclusions are likely to be correct and the costs of an occasional mistake acceptable, and if the jump saves much time and effort."* p. 79

> *"System 1 is gullible and biased to believe, System 2 is in charge of doubting and unbelieving, but System 2 is sometimes busy, and often lazy."* p. 81

> *"The tendency to like (or dislike) everything about a person—including things you have not observed—is known as the halo effect."* p. 81

> *"The procedure I adopted to tame the halo effect conforms to a general principle: decorrelate error!"* p. 83

> *"The measure of success for System 1 is the coherence of the story it manages to create. The amount and quality of the data on which the story is based are largely irrelevant."* p. 85

> *"System 1 is radically insensitive to both the quality and the quantity of the information that gives rise to impressions and intuitions."* p. 85

> *"They made that big decision on the basis of a good report from one consultant. WYSIATI—what you see is all there is. They did not seem to realize how little information they had."* p. 88

Concepts and experiments

- **Neglect of ambiguity.** Context silently picks one interpretation.
- **Believing first (Gilbert).** Comprehension involves initial belief. Under load, false statements are more often judged true.
- **Confirmation bias / positive test strategy.** We search for evidence consistent with what we're testing.
- **Halo effect.** Liking (or disliking) spreads across all attributes. Asch's order effect shows first impressions dominate.
- **Decorrelating errors.** Independent judgments, averaged, are accurate (the wisdom of crowds), provided the errors are not correlated. Discussion before independent views correlates errors.
- **WYSIATI.** The story is built from what is available. Absent evidence is not considered. One-sided evidence produces *more* confidence (Brenner, Koehler & Tversky).
- **"Will Mindik be a good leader? She is intelligent and strong..."** You answer yes before hearing "...corrupt and cruel."

Connections

- **Earlier:** associative coherence (Chapter 4), cognitive ease (Chapter 5) and causal thinking (Chapter 6).
- **Later:** WYSIATI is one of the most cited ideas in the book. It returns in base-rate neglect (Chapters 14–16), the narrative fallacy (Chapter 19), the illusion of validity (Chapter 20), the planning fallacy and inside view (Chapter 23), overconfidence (Chapter 24), narrow framing

(Chapter 31) and framing (Chapter 34). Decorrelating errors underlies formulas (Chapter 21) and the premortem (Chapter 24).

Markets, money and people

- **Theranos.** The company attracted prominent board members and investors on the strength of a charismatic founder and a compelling story while its technology did not work. Founder Elizabeth Holmes was convicted of fraud in January 2022. This is the halo effect plus WYSIATI: the absence of peer-reviewed data was not part of the story.
- **Madoff.** Bernard Madoff's reputation (he was a former Nasdaq chairman) created a halo. Investors accepted remarkably smooth returns without the missing information, such as independent custody or a credible auditor. The fraud collapsed in December 2008.
- **Investment committees.** Discussing before writing down independent views lets the most senior or vocal person anchor everyone. Dalio's emphasis on recording people's independent views before debate, and weighting them by track record, is a form of decorrelating error.
- **Hypothetical:** you read one bullish deep-dive on a stock and feel very confident. The confidence comes from the coherence of the one story you read, not from completeness.

Takeaways

- System 1 believes first and asks questions later, if ever.
- First impressions and halos spread across unrelated judgments.
- **WYSIATI:** confidence comes from coherence, not from the amount or quality of evidence. Always ask "What am I *not* seeing?"
- Independent judgments, collected before discussion, are much more informative.

How to apply it

- 💰 **Investing:** for every thesis, write the strongest bear case (or bull case, if you're short) and list the information you'd want but don't have. Ask "What would I need to see to change my mind?"
- 🤝 **People:** in meetings, have people write their estimates before anyone speaks. In hiring, score each attribute separately to block the halo.
- 🎬 **Content:** a one-sided video produces overconfident viewers, which is good for engagement and bad for them. A brand built on intellectual honesty ("here's the strongest counter-argument") is rarer and more trusted.
- 🧠 **Thinking:** make WYSIATI a reflex. "What would I believe if I saw the other half of the evidence?"

Chapter 8 · How Judgments Happen

PDF pp. 89–96

The argument

System 2 asks questions and directs attention. System 1 continuously makes **basic assessments** of the situation with no effort and no intention: Is this good or bad? Threat or opportunity? Normal or surprising? Should I approach or avoid? Evolution shaped these assessments for survival, including judging a stranger's dominance and trustworthiness from their face in a fraction of a second.

Alex Todorov showed that people's split-second ratings of *competence* from candidates' faces predicted the winners of about 70% of U.S. Senate, House and governor races. The effect was about three times larger among politically uninformed voters who watch a lot of TV. System 1 also represents categories by **prototypes** and averages, not sums. It can instantly estimate the average length of a set of lines but not their total length. This leads to errors in the economics of value (Chapter 15's dinnerware sets; the "willingness to pay to save birds" study in which people would pay about the same whether 2,000, 20,000 or 200,000 birds were saved).

Intensity matching: System 1 can translate intensity across dimensions ("If Sam were as tall as he is intelligent, how tall would he be?"), and it matches punishments to crimes the same way. The **mental shotgun**: when you answer one question, System 1 computes more than you asked for. Asked whether word pairs *rhyme*, people were slowed when the *spellings* differed (VOTE–GOAT vs VOTE–NOTE). Asked whether a sentence was literally true, people were slowed by sentences that were metaphorically true.

Key quotes

> *"These basic assessments play an important role in intuitive judgment, because they are easily substituted for more difficult questions—this is the essential idea of the heuristics and biases approach."* p. 89

> *"There are circuits in the brain that evaluate dominance from the shape of the face. He looks the part for a leadership role."* p. 96

> *"This was a clear instance of a mental shotgun. He was asked whether he thought the company was financially sound, but he couldn't forget that he likes their product."* p. 96

Concepts and experiments

- **Basic assessments.** Automatic evaluations of threat, normality, liking and similarity.
- **Faces and elections (Todorov).** Competence ratings from brief exposure to faces predicted about 70% of race outcomes.

- **Prototypes and averages vs sums.** System 1 represents sets by averages and is insensitive to quantity. This is *scope insensitivity* (Desvousges et al.'s birds study).
- **Intensity matching.** Cross-dimension matching, which later produces extreme predictions (Chapter 18).
- **Mental shotgun.** Excess, unintended computation that interferes with the question at hand.

Connections

- **Earlier:** WYSIATI and the halo (Chapter 7).
- **Later:** basic assessments are the "easy answers" substituted in Chapter 9. Prototypes drive representativeness (Chapters 14–15). Intensity matching causes nonregressive predictions (Chapter 18). Insensitivity to quantity reappears in duration neglect (Chapter 35).

Markets, money and people

- **CEO looks.** Research has linked CEO facial "competence" ratings to compensation and selection. Boards also judge leaders by appearance and presence. Look for track records, not looks.
- **Mental shotgun with products.** Investors who love a company's products (Apple, Tesla, Peloton) may conflate "great product" with "great investment at this price." Peloton's share price rose more than 400% in 2020 as pandemic demand surged, then fell more than 90% from its January 2021 peak by 2022. Loving the bike was not a valuation.
- **Scope insensitivity in risk.** People react similarly to a "large loss" whether it is 5% or 15% of a portfolio unless they convert it to concrete numbers.

Takeaways

- System 1 automatically judges threat, liking, similarity and dominance.
- It thinks in averages and prototypes and neglects quantity and sums.
- When you answer one question, you also unconsciously answer others, and those answers leak in.

How to apply it

- 💰 **Investing:** separate the questions "Do I like the product/CEO?" and "Is the stock mispriced?" Rate them on separate lines in your notes.
- 🤝 **People:** first impressions are formed from faces and presence in a fraction of a second. Make them work for you, and when evaluating others, demand evidence beyond presence.
- 🎬 **Content:** faces in thumbnails convey emotion and trust instantly. Expressive, direct faces are a basic assessment you can design for.
- 🧠 **Thinking:** when asked a narrow question, list what *else* your mind is assessing (liking, fear, status) and set it aside.

Chapter 9 · Answering an Easier Question

PDF pp. 97–105

The argument

You are rarely stumped. You have opinions about almost everything: people you've barely met, companies you've never analysed. How? Kahneman's answer is **substitution**. If a satisfactory answer to a hard question isn't found quickly, System 1 finds a related easier question and answers it instead, usually without your noticing. The **target question** is the one you intend to answer. The **heuristic question** is the easier one you actually answer.

Examples: "How much would you contribute to save an endangered species?" becomes "How much emotion do I feel when I think of dying dolphins?" "How happy are you with your life?" becomes "What's my mood right now?" "How popular will the president be six months from now?" becomes "How popular is the president right now?" The answer is then **intensity-matched** onto the scale of the original question. Kahneman shows the "3-D heuristic" with a picture of three identical figures on a corridor drawn in perspective: the farthest one looks larger because you answer a 3-D question (how big would it be in reality?) instead of the 2-D one asked.

In a survey of students, the questions "How happy are you these days?" and "How many dates did you have last month?" were unrelated when asked in that order. When the dating question came first, the correlation was high: students substituted their love life for overall happiness. Paul Slovic's **affect heuristic** is substitution at its most powerful: "What do I think about it?" becomes "How do I feel about it?" Liking a technology makes you believe its benefits are high and its risks low. Political preferences determine which arguments seem compelling. System 2 often acts as an apologist for System 1's emotions, an endorser rather than an enforcer. The chapter ends with a summary table of System 1's characteristics, effectively the book's index of mechanisms.

Key quotes

> *"The target question is the assessment you intend to produce. The heuristic question is the simpler question that you answer instead."* p. 97

> *"The technical definition of heuristic is a simple procedure that helps find adequate, though often imperfect, answers to difficult questions."* p. 97

> *"The present state of mind looms very large when people evaluate their happiness."* p. 102

> *"The dominance of conclusions over arguments is most pronounced where emotions are involved."* p. 102

> *"In the context of attitudes, however, System 2 is more of an apologist for the emotions of System 1 than a critic of those emotions—an endorser rather than an enforcer."* p. 103

> *"The question we face is whether this candidate can succeed. The question we seem to answer is whether she interviews well. Let's not substitute."* p. 103

> *"We are using last year's performance as a heuristic to predict the value of the firm several years from now. Is this heuristic good enough? What other information do we need?"* p. 104

Concepts and experiments

- **Substitution.** The core of the heuristics-and-biases approach. A hard question (probability, value, future) is replaced by an easier one (similarity, ease of recall, feeling).
- **Intensity matching onto the target scale.** The intensity of the easy answer is translated into dollars, probabilities or ratings.
- **3-D heuristic.** The perspective illusion: a 3-D size judgment is substituted for 2-D size.
- **Mood heuristic for happiness (Strack, Martin & Schwarz).** The order of the dating and happiness questions changed the correlation.
- **Affect heuristic (Slovic).** Feelings drive judgments of risk and benefit, and they move together (good = low risk + high benefit).
- **System 1's characteristics table.** Generates impressions; operates automatically; creates coherent patterns; links a sense of cognitive ease to illusions of truth; distinguishes surprise from normal; infers causes; neglects ambiguity; is biased to believe and confirm; exaggerates emotional consistency (halo); focuses on existing evidence (WYSIATI); computes basic assessments; represents sets by averages; matches intensities; computes more than intended (mental shotgun); substitutes easier questions; is more sensitive to changes than states; overweights low probabilities; shows diminishing sensitivity; and responds more strongly to losses than gains. The last few points preview Part 4.

Connections

- **Earlier:** pulls Part 1 together: lazy System 2 (Chapter 3), basic assessments and intensity matching (Chapter 8).
- **Later:** every heuristic in Part 2 is a substitution. Representativeness substitutes similarity for probability, availability substitutes ease of recall for frequency, and anchoring works through priming and adjustment. The affect heuristic returns in Chapter 13. Mood and happiness substitution returns in Chapters 37–38.

Markets, money and people

- **"Is this a good investment?" becomes "Is this a good company?"** The classic substitution. In the Nifty Fifty era of the early 1970s, investors bought great companies at very high valuations, and many of those stocks performed poorly after the 1973–74 bear market.
- **"Will this go up?" becomes "Has it been going up?"** Momentum chasing. ARK Innovation ETF (ARKK) rose about 150% in 2020 and attracted large inflows, then fell roughly 75% from its February 2021 peak by late 2022. Many investors had substituted recent performance for future prospects.
- **Affect in crypto and ESG.** People who love a technology or cause tend to see its risks as low and its rewards as high, exactly Slovic's pattern. That held for crypto enthusiasts in 2021, before Bitcoin fell about 75% from its November 2021 high and FTX collapsed in November 2022.
- **Hypothetical:** a viewer asks, "Should I buy gold?" A quick answer substitutes "Am I worried about the world?" A real answer asks about allocation, correlation, real rates and time horizon.

Takeaways

- When you have a quick answer to a hard question, you probably answered a different, easier question.
- Feelings become judgments of risk and benefit (the affect heuristic).
- System 2 often rationalizes System 1's emotional conclusions instead of testing them.

How to apply it

- 💰 **Investing:** write the exact target question at the top of your analysis ("Is the expected 5-year return at today's price above my hurdle?"). After deciding, check whether you answered *that* question.
- 🤝 **People:** people buy feelings first and reasons second. In persuasion, the emotional association comes first. In negotiation, notice when the other side's "reasons" are rationalizations of a feeling.
- 🎬 **Content:** "the question you think you're answering vs the one you're actually answering" is a strong recurring segment. Use the "Speaking of…" lines as on-screen captions.
- 🧠 **Thinking:** ask "What question am I really answering?" This may be the most useful habit in the whole book.

Part 2 · Heuristics and Biases

Part 2 asks why we find it so hard to think statistically. Each chapter shows a substitution at work: small samples treated as representative, irrelevant numbers anchoring estimates, ease of recall standing in for frequency, stereotypes standing in for base rates, and causal stories standing in for regression to the mean. It ends with a practical method for correcting intuitive predictions.

Chapter 10 · The Law of Small Numbers

PDF pp. 108–117

The argument

U.S. counties with the *lowest* kidney cancer rates are mostly rural, sparsely populated and in Republican states. Your mind immediately offers a story: clean living, fresh air, no pollution. But the counties with the *highest* rates are also mostly rural, sparsely populated and Republican, and your mind can just as easily explain that (poverty, high-fat diet, poor access to care). Neither story is right. Small counties have small populations, and **small samples produce extreme results more often** than large ones, in both directions. The pattern is an artefact of sampling.

The fact is simple: small samples are more variable than large ones (the law of large numbers). Yet even trained researchers, as Kahneman and Tversky found in their first collaboration, trust results from small samples far too much and choose sample sizes that give them a high chance of failing to detect real effects. Kahneman admits he did the same. People pay more attention to the *content* of a message than to information about its reliability, such as sample size.

The deeper problem is our hunger for causes (Chapter 6). We see patterns in randomness. Of the birth sequences BBBGGG, GGGGGG and BGBBGB, the third "looks" more random, but all are equally likely. During the rocket bombing of London in World War II, people believed the hits clustered around German spies' locations. Statistical analysis showed the pattern was consistent with random hits. Basketball fans believe in the **hot hand**, but Gilovich, Vallone and Tversky's analysis found no evidence of it. The Gates Foundation invested about $1.7 billion in small schools, partly because the *best* schools in surveys were disproportionately small. The *worst* schools were also disproportionately small, for the same statistical reason.

Key quotes

> *"The exaggerated faith in small samples is only one example of a more general illusion—we pay more attention to the content of messages than to information about their reliability"* p. 116

> *"Causal explanations of chance events are inevitably wrong."* p. 116

> *"Many facts of the world are due to chance, including accidents of sampling."* p. 116

> *"We are far too willing to reject the belief that much of what we see in life is random."* p. 115

> *"I won't believe that the new trader is a genius before consulting a statistician who could estimate the likelihood of his streak being a chance event."* p. 117

Concepts and experiments

- **Law of large numbers vs "law of small numbers."** The real law: large samples are more precise. The "law of small numbers" is the false belief that small samples resemble the population just as closely.
- **Kidney cancer counties (Wainer & Zwerling).** Extremes at both ends come from small populations.
- **Researchers' sample sizes.** Psychologists routinely chose samples that risked a 50% failure rate for real hypotheses.
- **Bias of confidence over doubt.** System 1 suppresses doubt and builds a coherent story from a few observations.
- **Perceiving patterns in randomness.** Birth sequences, London bombing (from William Feller), and the hot hand in basketball.
- **Small schools (Gates Foundation).** Small schools are over-represented at both the top and the bottom.

Replication note. In 2018, Joshua Miller and Adam Sanjurjo showed that the original hot-hand analysis contained a subtle statistical bias. Correcting for it suggests a modest hot hand may exist in some settings. The main point, that people see far more streakiness than exists and draw causal conclusions from short runs, still stands.

Connections

- **Earlier:** WYSIATI and suppression of doubt (Chapter 7), and the search for causes (Chapter 6).
- **Later:** small samples feed representativeness (Chapter 14), illusions of skill (Chapter 20) and the misreading of short track records (Chapters 17–18).

Markets, money and people

- **Fund manager streaks.** A few years of beating the market is a small sample. S&P's SPIVA scorecards have consistently found that most active U.S. large-cap funds underperform the S&P 500 over long periods (roughly 9 in 10 over 15 years in recent reports), and few top-quartile funds stay top-quartile. Short track records reveal little.
- **Backtests.** A strategy tested on a few years of data, or tuned across many variables, can look brilliant by chance (data mining). The more variables tried, the more likely a spurious "pattern."
- **Small-cap and IPO "stories."** The best and worst one-year performers in any market are usually small, volatile companies, the kidney-cancer pattern in stock form.

- **Hypothetical:** a crypto trader posts six winning calls in a row on social media. If thousands of people post calls, some will have streaks by chance alone.

Takeaways

- Small samples produce extreme results. Expect them and discount them.
- Ask about sample size before you believe a finding, a track record or a trend.
- Random sequences contain streaks and clusters. Don't invent causes for them.

How to apply it

- 💰 **Investing:** before trusting any track record, ask how many observations it contains, how many people were trying, and what the result would look like if it were luck. Prefer long, out-of-sample records. Be suspicious of backtests with many parameters.
- 🤝 **People:** don't judge a new hire, partner or strategy on a few outcomes. Agree on the evaluation period in advance.
- 🎬 **Content:** "Why the best AND worst schools are small" is a perfect counterintuitive hook. Showing viewers both tails of a distribution teaches the idea quickly.
- 🧠 **Thinking:** when a pattern appears, ask "Would I see this by chance?" before asking "Why?"

Chapter 11 · Anchors

PDF pp. 118–127

The argument

Kahneman and Tversky rigged a wheel of fortune to stop only at 10 or 65, spun it in front of students, and then asked: Is the percentage of African nations among UN members higher or lower than the number you just wrote? What is your best guess? Students who saw 10 guessed 25% on average; those who saw 65 guessed 45%. An obviously meaningless number shifted their estimates. This is **anchoring**: estimates stay close to a number you considered beforehand.

There are two mechanisms. **Anchoring as adjustment** (System 2): you start from the anchor and adjust until you are unsure whether to go further, and you stop too early. Asked when George Washington became president, people start from 1776 and adjust upward, stopping short of the true date. Tired or busy people adjust less. **Anchoring as priming** (System 1): the anchor makes compatible information more accessible. Asked whether Gandhi was older than 144 when he died, you don't adjust down from 144, but the question brings to mind images of a very old man.

The **anchoring index** measures the effect. Visitors to the San Francisco Exploratorium asked about redwoods with a 1,200-foot or a 180-foot anchor gave average estimates of 844 and 282 feet, an index of 55%. Experts are not immune. Real estate agents shown different listing prices for the same house were anchored (an index of 41%) and denied it. Experienced German judges who rolled loaded dice (3 or 9) before sentencing a hypothetical shoplifter gave 5 months versus 8 months on average. Anchors are used against you everywhere: "limit of 12 per person" on Campbell's soup doubled average purchases (7 cans vs about half that), opening offers in negotiation, damages caps in court. Kahneman's advice: if the other side makes an outrageous offer, don't counter. Make a scene, walk away and refuse to negotiate from that number. A useful debiasing strategy is to **think the opposite**: deliberately generate arguments against the anchor.

Key quotes

> *"Any number that you are asked to consider as a possible solution to an estimation problem will induce an anchoring effect."* p. 118

> *"By now you should be convinced that anchoring effects—sometimes due to priming, sometimes to insufficient adjustment—are everywhere."* p. 124

> *"We saw in the discussion of the law of small numbers that a message, unless it is immediately rejected as a lie, will have the same effect on the associative system regardless of its reliability."* p. 126

> *"The main moral of priming research is that our thoughts and our behavior are influenced, much more than we know or want, by the environment of the moment."* p. 126

> *"Plans are best-case scenarios. Let's avoid anchoring on plans when we forecast actual outcomes. Thinking about ways the plan could go wrong is one way to do it."* p. 127

Concepts and experiments

- **Wheel of fortune / UN question.** A meaningless anchor shifted estimates (25% vs 45%).
- **Adjustment (Epley & Gilovich).** Insufficient adjustment, increased by cognitive load or alcohol. Shaking your head while adjusting makes people adjust *more*.
- **Priming (Mussweiler & Strack).** Anchors make compatible thoughts accessible. Thinking of a high annual mean temperature in Germany primes summer words.
- **Anchoring index.** The redwood question gave 55%, real estate agents 41%, students 48%.
- **Judges and dice (Englich, Mussweiler & Strack).** An anchor of 3 vs 9 produced sentences of 5 vs 8 months.
- **Arbitrary rationing.** "Limit 12 per person" anchored purchases.
- **Negotiation and law.** First offers anchor. Damages caps can raise small awards.
- **Think the opposite.** Searching for arguments against the anchor reduces its effect.

Connections

- **Earlier:** priming (Chapter 4) and lazy adjustment by System 2 (Chapter 3).
- **Later:** anchoring on plans drives the planning fallacy (Chapter 23). Reference points in prospect theory (Chapters 26–28) are a form of anchor. Appendix A's "adjustment and anchoring" section is the original 1974 statement.

Markets, money and people

- **Purchase price.** Investors anchor on what they paid and on 52-week highs. Research by George & Hwang (2004) found that stocks near their 52-week highs tend to keep outperforming, consistent with investors under-reacting because they anchor on the high. "It's down 60% from its high, so it's cheap" is anchoring. The high was not a measure of value.
- **IPO pricing and price targets.** Analyst price targets and IPO prices anchor expectations. Analysts tend to revise targets gradually from prior ones, under-adjusting.
- **Macro forecasting.** Consensus forecasts of inflation or GDP often move slowly from last year's number. In 2021 many forecasters anchored on a decade of roughly 2% inflation and adjusted too little as inflation surged.
- **Real estate.** The book's realtor study is direct evidence: the listing price anchors professional valuations.

- **Hypothetical:** a startup founder opens a funding talk with "comparable companies trade at 30× revenue." Even if you reject that number, it has already pulled your valuation upward.

Takeaways

- Any number you consider, however irrelevant, pulls your estimate toward it.
- Experts are anchored too, and they deny it.
- Counter anchors by generating reasons the anchor is wrong, or by setting your own number *before* you see theirs.

How to apply it

- 💰 **Investing:** write down your estimate of fair value *before* you look at the current price, analyst targets or the last funding round. Ignore your purchase price when deciding whether to hold. Ask, "Would I buy it today at this price?"
- 🤝 **People:** in negotiation, move first with a well-founded, ambitious anchor when you have good information. If the other side opens absurdly, don't counter. Reset the frame.
- 🎬 **Content:** numbers in titles anchor ("$1M," "90% of traders"). Use real numbers responsibly. The pricing of a course or membership is anchored by what viewers see first.
- 🧠 **Thinking:** "Is any number on the table influencing me?" The answer is almost always yes.

Chapter 12 · The Science of Availability

PDF pp. 128–135

The argument

In 1971–72 in Eugene, Oregon, Kahneman and Tversky studied how people estimate the frequency of categories: divorces after 60, dangerous plants, words starting with K. They proposed that we judge frequency by **the ease with which instances come to mind**. This is the **availability heuristic**, another substitution. Usually ease of recall is a good guide, since common things come to mind easily. But anything that makes instances easier to retrieve biases the judgment: salience (a celebrity divorce), dramatic events (a plane crash), personal experience (your own mistakes), vividness and recency.

The most surprising research comes from Norbert Schwarz, who asked people to list either six or twelve examples of times they had been assertive. Those who listed *twelve* rated themselves *less* assertive. Listing twelve is hard, the difficulty is felt as evidence ("if it's so hard to think of examples, I must not be very assertive"). **The fluency of retrieval matters more than the number of examples retrieved.** The effect reverses when people are given an explanation for the difficulty (such as background music said to impair recall). People who are personally involved in the judgment, or in a good mood, or who score low on depression scales, rely more on ease. Spouses asked to estimate their contribution to household chores give estimates that add up to more than 100%, because your own contributions are more available to you.

Key quotes

> *"The availability heuristic, like other heuristics of judgment, substitutes one question for another"* p. 129

> *"Because of the coincidence of two planes crashing last month, she now prefers to take the train. That's silly. The risk hasn't really changed; it is an availability bias."* p. 134

> *"The CEO has had several successes in a row, so failure doesn't come easily to her mind. The availability bias is making her overconfident."* p. 135

Concepts and experiments

- **Availability heuristic.** Frequency judged by the ease of recall.
- **Sources of bias:** salience, dramatic events, personal experience and vividness.
- **Self-serving estimates of contribution.** Spouses' and team members' estimates sum to more than 100%.
- **Ease vs amount (Schwarz et al.).** Six vs twelve examples of assertiveness. Difficulty of retrieval lowers judgments.

- **Reversals when difficulty is explained.** If the difficulty is attributed to something else, the effect disappears.
- **When people rely on ease:** when System 2 is engaged elsewhere, in good mood, when people feel powerful, or when they are inexperienced with the topic.

Connections

- **Earlier:** cognitive ease (Chapter 5) and substitution (Chapter 9).
- **Later:** availability plus emotion leads to risk misperception and availability cascades (Chapter 13). Vivid outcomes are overweighted in rare events (Chapter 30). The original 1974 version is in Appendix A.

Markets, money and people

- **Recency in risk.** After the 2008 crisis, surveys showed many investors underweighting stocks for years. The crisis was highly available, which may have contributed to their missing much of the long bull market from the March 2009 low.
- **Lottery-like stories.** Stories of people who got rich on Bitcoin, Tesla or Nvidia are highly available. The many who lost money are not.
- **CEO overconfidence.** The book's own example: a CEO with a run of successes can't easily imagine failure.
- **Hypothetical:** after reading three headlines about a recession, you put the chance of one at 70%. The headlines made it available. They didn't change the base rate.

Takeaways

- We judge how common something is by how easily examples come to mind.
- Vivid, recent, personal and dramatic events are overestimated.
- The difficulty of finding examples is itself treated as evidence.

How to apply it

- 💰 **Investing:** keep a base-rate file with recession frequency, typical drawdowns, average bear-market length and how often IPOs outperform. Look at it when headlines are loud.
- 🤝 **People:** in teams, everyone overestimates their own contribution. Remembering this prevents resentment. To persuade, make your point easy to recall (examples, stories).
- 🎬 **Content:** asking viewers to "list three reasons" works. Asking for ten backfires, because difficulty undermines belief. Easily recalled examples make your argument feel common and true.
- 🧠 **Thinking:** ask "Is this common, or just memorable?"

Chapter 13 · Availability, Emotion, and Risk

PDF pp. 136–143

The argument

Howard Kunreuther noticed that after earthquakes and floods people buy insurance and take precautions, then let them lapse as memories fade. Paul Slovic, Sarah Lichtenstein and Baruch Fischhoff found that people's estimates of causes of death are distorted by media coverage. Strokes cause almost twice as many deaths as all accidents combined, yet 80% of respondents judged accidental death more likely. Tornadoes were seen as deadlier than asthma, which kills 20 times more people. **Availability is shaped by emotion and media**, and the **affect heuristic** (Chapter 9) makes judgments of risk and benefit move together: if you like a technology, you think its risks are low.

Kahneman lays out a debate between two scholars he respects. **Paul Slovic** argues that experts' narrow, statistical definitions of risk (lives lost) miss what the public cares about, such as good and bad deaths and voluntary versus involuntary risk, and that "risk" is not an objective quantity. **Cass Sunstein** worries that public fear, amplified by the media, distorts policy and wastes resources. Sunstein and Timur Kuran's **availability cascade** describes a self-sustaining chain: a media story triggers public concern, which produces more stories, and politicians and "availability entrepreneurs" respond. Examples are the 1989 Alar scare over a chemical sprayed on apples (the risk was tiny, but the panic hit apple growers) and Love Canal. Sunstein also describes **probability neglect**: when emotions are vivid, people react to the *possibility* of harm and ignore its probability. Kahneman sides partly with both. Policy should be guided by expert analysis but must take public fears into account, because fear itself is a real cost.

Key quotes

> *“The world in our heads is not a precise replica of reality”* p. 137

> *“The implication is clear: as the psychologist Jonathan Haidt said in another context, "The emotional tail wags the rational dog.””* p. 138

> *“"Risk" does not exist "out there," independent of our minds and culture, waiting to be measured.”* p. 139

> *“An availability cascade is a self-sustaining chain of events”* p. 140

> *“Terrorism speaks directly to System 1.”* p. 142

> *"This is an availability cascade: a nonevent that is inflated by the media and the public until it fills our TV screens and becomes all anyone is talking about."* p. 143

Concepts and experiments

- **Disaster cycles (Kunreuther).** Protective behaviour spikes after disasters and decays.
- **Causes of death (Slovic, Lichtenstein & Fischhoff).** Media coverage distorts estimates.
- **Affect heuristic (Slovic, Finucane and colleagues).** Learning that a technology has high benefits lowers the perceived risk, even with no information about risk.
- **Experts vs the public (Slovic vs Sunstein).** Different definitions of risk.
- **Availability cascade (Kuran & Sunstein).** Media-driven feedback loops.
- **Probability neglect (Sunstein).** Vivid harms make probability irrelevant.

Connections

- **Earlier:** availability (Chapter 12) and the affect heuristic (Chapter 9).
- **Later:** rare events and vivid outcomes (Chapter 30), terror risk (Chapter 30's Israeli bus bombings) and loss aversion in policy (Chapter 28). Libertarian paternalism in the Conclusions builds on Sunstein.

Markets, money and people

- **Silicon Valley Bank, March 2023.** As worries about SVB's bond losses spread over social media and group chats among tech founders and VCs, depositors tried to withdraw about $42 billion in a single day (March 9). Regulators closed the bank on March 10. This was an availability cascade at internet speed.
- **COVID crash, February–March 2020.** The S&P 500 fell about 34% in roughly a month as vivid images spread, and then recovered to new highs by August 2020. Fear moved faster than fundamentals in both directions.
- **Meme stocks.** Social media cascades in 2021 (GameStop, AMC) were availability cascades in reverse, built on excitement rather than fear.
- **Affect in sector bets.** Investors who love a sector (clean energy in 2020, AI in 2023–24) tend to see its risks as low. The clean-energy ETF ICLN roughly doubled in 2020, then fell substantially in 2021–22.
- **Hypothetical:** a viral video claims a bank is about to fail. Even if it's false, if enough people act on it, it can come true. Cascades can be self-fulfilling in finance.

Takeaways

- Media coverage and emotion distort our sense of risk.
- When we like something, we see its risks as low and its benefits as high, and vice versa.

- Availability cascades can turn small risks into panics, and in finance they can become self-fulfilling.

How to apply it

- 💰 **Investing:** when "everyone is talking about" a risk or an opportunity, check the actual data. Panics create opportunity for those with liquidity and a base rate. Bank-run dynamics are real, so know your counterparty exposure.
- 🤝 **People:** public fear is a real cost. Address emotions, not just statistics. Don't dismiss people's fears; reframe them with concrete, relatable comparisons.
- 🎬 **Content:** creators are *availability entrepreneurs*. Your power to start cascades is real, so use it responsibly, and consider "Is this risk overhyped?" episodes that explain base rates calmly.
- 🧠 **Thinking:** when you feel strong fear or excitement, ask "What is the probability?" and not just "Could it happen?"

Chapter 14 · Tom W's Specialty

PDF pp. 144–152

The argument

Rank nine graduate fields by how many students they have, and you'll use **base rates**: more people study humanities and education than library science. Now read a personality sketch of Tom W: "high intelligence, although lacking in true creativity… need for order and clarity… corny puns… little sympathy for other people." Rank the fields by how similar Tom is to the typical student, and computer science and engineering come first. Asked to rank the fields by the *probability* that Tom is enrolled in them, people give the *same* ranking as by similarity. They substitute **representativeness** (similarity to a stereotype) for probability and ignore base rates, even though the sketch was described as based on tests "of uncertain validity." Kahneman's own graduate students, trained in statistics, did the same.

Representativeness is not useless. Stereotypes often contain some truth, and people who look like their category are often more likely to belong to it. But it leads to two sins. The first is **predicting rare events from weak evidence**: someone reading the *New York Times* on the subway is more likely to have a PhD than to be a non-graduate, people say, but there are far more non-graduates riding the subway. The second is **insensitivity to the quality of evidence**: WYSIATI means System 1 treats unreliable evidence as if it were true. Instructions to "think like a statistician" increase use of base rates. Frowning (cognitive strain) also helps. The discipline of Bayesian reasoning comes down to two rules: **anchor your judgment on a plausible base rate, and question the diagnosticity of your evidence**.

Key quotes

> *"Anchor your judgment of the probability of an outcome on a plausible base rate."* p. 152

> *"Question the diagnosticity of your evidence."* p. 152

> *"This start-up looks as if it could not fail, but the base rate of success in the industry is extremely low. How do we know this case is different?"* p. 152

> *"The lawn is well trimmed, the receptionist looks competent, and the furniture is attractive, but this doesn't mean it is a well-managed company. I hope the board does not go by representativeness."* p. 152

> *"They keep making the same mistake: predicting rare events from weak evidence. When the evidence is weak, one should stick with the base rates."* p. 152

Concepts and experiments

- **Base rate.** The prior frequency of a category (how many marbles of each colour in the urn).
- **Representativeness.** Judging probability by similarity to a prototype.
- **Tom W (Kahneman & Tversky, 1973).** Probability rankings matched similarity rankings, and base rates were ignored.
- **Bayes's rule, made simple.** Start from the base rate and move it only as far as the evidence is diagnostic. With weak evidence, stay close to the base rate.
- **Think like a statistician (Schwarz et al.).** Instructions and frowning improve base-rate use.
- **Two sins of representativeness:** excessive willingness to predict unlikely events, and insensitivity to the quality of evidence.

Connections

- **Earlier:** WYSIATI (Chapter 7), prototypes (Chapter 8) and substitution (Chapter 9).
- **Later:** the conjunction fallacy (Chapter 15), causal vs statistical base rates (Chapter 16), regression and the correction of predictions (Chapters 17–18), and the outside view (Chapter 23). Appendix A has the original treatment, including Steve the librarian.

Markets, money and people

- **"This looks like the next Amazon/Tesla/Nvidia."** Representativeness. Few companies with a similar story become giants. Of the companies that went public in the dot-com era, many lost most or all of their value. Pets.com, which IPO'd in February 2000, liquidated that November.
- **SPACs, 2020–21.** Hundreds of SPACs went public with stories that "looked like" disruptive winners. Many of the companies they took public later traded far below the standard $10 SPAC price.
- **Fund managers who "look like" winners.** Confident, articulate managers with a good story attract money regardless of process.
- **Hypothetical:** a biotech has an inspiring founder, a Nobel laureate on its board and a slick website. The base rate of drugs that start clinical trials and reach approval is low (often cited at roughly one in ten).

Takeaways

- Similarity to a stereotype is not probability.
- Always start with the base rate, then adjust only as much as the evidence justifies.
- Weak evidence should move you very little.

How to apply it

- 💰 **Investing:** for every new idea, find the base rate for the reference class (IPOs, turnarounds, biotech trials, early-stage startups). Write it down before you read the pitch.
- 🤝 **People:** don't hire or trust on "looks like a leader." Use structured evidence (Chapter 21).
- 🎬 **Content:** "The base rate nobody mentions" makes a great recurring segment. Show the stereotype, then reveal the statistics.
- 🧠 **Thinking:** two questions. What's the base rate? How diagnostic is this evidence?

Chapter 15 · Linda: Less Is More

PDF pp. 153–162

The argument

"Linda is thirty-one years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in antinuclear demonstrations." Which is more probable: (a) Linda is a bank teller, or (b) Linda is a bank teller and is active in the feminist movement? About 85–90% of undergraduates at several major universities chose (b). But (b) is a *subset* of (a), so it cannot be more probable. This is the **conjunction fallacy**: a more detailed scenario seems more probable because it is more *representative* or *plausible*, even though adding detail can only reduce probability.

The fallacy was robust. It survived even when the two options were shown side by side, and it appeared among graduate students in decision science. Kahneman notes the controversy the problem generated, and that the error is reduced when the question is stated in frequencies ("Of 100 people like Linda, how many are…?"). A forecasting version: experts judged "a massive flood somewhere in North America next year, in which more than 1,000 people drown" *less* probable than "an earthquake in California sometime next year, causing a flood in which more than 1,000 people drown." Adding a cause makes a story more persuasive and less likely.

Less is more shows up in valuation too. Christopher Hsee asked people to price two dinnerware sets. Set A had 40 pieces, including some broken ones; Set B had 24 intact pieces, all of which were also in Set A. Seen together (joint evaluation), A was valued slightly higher ($32 vs $30). Seen separately (single evaluation), B was priced much higher ($33 vs $23), because System 1 judges by the *average* quality, not the sum. John List found the same with baseball cards. Even in a dice-sequence version with no words, people chose the more "representative" but less probable sequence.

Key quotes

> *"The word fallacy is used, in general, when people fail to apply a logical rule that is obviously relevant."* p. 155

> *"The most coherent stories are not necessarily the most probable, but they are plausible, and the notions of coherence, plausibility, and probability are easily confused by the unwary."* p. 156

> *"The uncritical substitution of plausibility for probability has pernicious effects on judgments when scenarios are used as tools of forecasting."* p. 156

> *"This is a trap for forecasters and their clients: adding detail to scenarios makes them more persuasive, but less likely to come true."* p. 156

> *"They constructed a very complicated scenario and insisted on calling it highly probable. It is not—it is only a plausible story."* p. 162

Concepts and experiments

- **Conjunction fallacy.** P(A and B) ≤ P(A), yet people rank the conjunction higher.
- **Linda problem.** 85–90% of undergraduates committed the fallacy. Frequency formats reduce it.
- **Plausibility vs probability.** Coherent details increase plausibility and reduce probability.
- **Scenario forecasting (flood in North America vs flood caused by a California earthquake).**
- **Less is more (Hsee's dinnerware; List's baseball cards).** Single evaluation uses averages (prototype), so adding lower-value items can reduce perceived value.
- **Joint vs single evaluation.** Joint evaluation makes the logical relation clear (this returns in Chapter 33).

Connections

- **Earlier:** representativeness (Chapter 14) and averages vs sums (Chapter 8).
- **Later:** joint vs single evaluation and preference reversals (Chapter 33). The narrative fallacy (Chapter 19), where detailed stories feel true. Planning-fallacy scenarios (Chapter 23).

Markets, money and people

- **Macro scenarios.** "The Fed cuts, the dollar falls, emerging markets rally, and copper breaks out" is a compelling chain. Each link reduces the probability. Multi-step macro theses need every step to come true.
- **The 2008 crisis in reverse.** Ratings models assumed mortgage defaults across regions were largely independent, so a nationwide collapse looked extremely unlikely. In reality the housing markets were linked by a common cause. Conjunction and correlation errors cut both ways.
- **Product bundling.** Less-is-more pricing appears in practice: adding low-value extras to a product can reduce what people will pay for it in isolation.
- **Hypothetical:** a newsletter pitches a ten-step "inevitable" thesis. Even with 80% confidence in each step, the chance that all ten come true is only about 11% ($0.8^{10} \approx 0.107$).

Takeaways

- More detail makes a story more convincing and less probable.

- Coherence and plausibility are not probability.
- Single evaluation judges by averages; adding cheap extras can reduce value.

How to apply it

- 💰 **Investing:** break every thesis into its steps, assign each a probability, and multiply. Prefer theses with few necessary conditions. Remember that steps are often correlated (one common cause can break several at once).
- 🤝 **People:** to persuade, detailed stories work. To evaluate, strip the story to its necessary conditions.
- 🎬 **Content:** detailed scenarios grab attention ("Here's exactly how the next crash will unfold"). Pair them with honest probabilities. "Less is more" applies to bundles too: one premium offer beats a bloated package.
- 🧠 **Thinking:** when a story feels very convincing, ask "How many things have to be true for this to happen?"

Chapter 16 · Causes Trump Statistics

PDF pp. 163–171

The argument

A cab was involved in a hit-and-run accident at night. In the city, 85% of cabs are Green and 15% are Blue. A witness says the cab was Blue, and witnesses are correct 80% of the time under those conditions. What is the probability the cab was Blue? Most people answer 80%. The correct Bayesian answer is 41%. People ignore the **statistical base rate**. But change the wording: "The two companies operate the same number of cabs, but Green cabs are involved in 85% of accidents." Now people weight the base rate heavily, because it implies a **causal** story (Green drivers are reckless). **Causal base rates** are used. **Statistical base rates** are neglected.

Causal base rates turn into stereotypes, which System 1 uses readily, sometimes unfairly. This creates a tension: we have social norms against stereotyping, but ignoring valid base rates leads to worse judgments. The most troubling finding concerns **teaching psychology**. Nisbett and Borgida told students about the famous "helping experiment," in which only 4 of 15 participants responded immediately to a stranger having an apparent seizure when they believed others could also hear him. The students learned the statistic but did not apply it: shown videos of two "nice" participants, they predicted those individuals would quickly help. Knowing the base rate didn't change their view of individuals. But when they were shown individual cases and then told that these individuals did *not* help, they correctly generalized. People will not deduce the particular from the general, but they will readily infer the general from the particular. Surprising *individual cases* change minds; statistics usually don't.

Key quotes

> *"Subjects' unwillingness to deduce the particular from the general was matched only by their willingness to infer the general from the particular."* p. 170

> *"People who are taught surprising statistical facts about human behavior may be impressed to the point of telling their friends about what they have heard, but this does not mean that their understanding of the world has really changed."* p. 170

> *"You are more likely to learn something by finding surprises in your own behavior than by hearing surprising facts about people in general."* p. 171

> *"We can't assume that they will really learn anything from mere statistics. Let's show them one or two representative individual cases to influence their System 1."* p. 171

Concepts and experiments

- **Cab problem.** A base rate of 15% Blue and a witness who is 80% reliable give a posterior of 41% Blue. Intuition says 80%.
- **Statistical vs causal base rates.** Causal framing ("Green drivers cause 85% of accidents") makes base rates count.
- **Stereotypes.** Causal base rates become stereotypes about groups.
- **Helping experiment (Darley & Latané) and Nisbett & Borgida's teaching study.** Students' predictions about individuals ignored the base rate they had just learned.
- **Learning from cases.** Surprising individual cases teach. Statistics alone rarely do.

Connections

- **Earlier:** base-rate neglect (Chapter 14) and causal thinking (Chapter 6).
- **Later:** regression (Chapter 17), where causal stories are invented for statistical effects. The outside view (Chapter 23): "people who have information about an individual case rarely feel the need to know the statistics."

Markets, money and people

- **Statistics vs stories in market risk.** Investors are told that the S&P 500 has historically had a drawdown of 10% or more in most years, but many still panic when one happens to them. The statistic didn't reach System 1.
- **Causal base rates in credit.** Lenders and investors who hear a causal story ("this sector is reckless") weight it heavily, sometimes unfairly to individual borrowers.
- **Teaching investors.** Financial educators find that a single vivid story of someone who sold at the March 2009 bottom and missed the recovery often changes behaviour more than charts of long-term returns.
- **Hypothetical:** you tell a client that 80% of day traders lose money. They nod, then argue that *they* are different. Showing them their own trading record changes their mind.

Takeaways

- We use base rates when they tell a causal story and ignore them when they're "mere statistics."
- Learning a statistic rarely changes our view of individual cases.
- Surprising individual cases, especially our own, teach better than statistics.

How to apply it

- 💰 **Investing:** turn base rates into stories you can't ignore. Keep a "graveyard file" of real failed companies and trades similar to your idea. Review your *own* trading statistics, since your own surprises teach more than other people's.

- 🤝 **People:** to change minds, use a well-chosen individual case, then the statistic. To be fair, apply base rates to decisions, not to judgments of individual character.
- 🎬 **Content:** this chapter explains why stories outperform statistics online. Lead with a person, then reveal the data. "What would you have done?" makes the viewer the case.
- 🧠 **Thinking:** when you learn a surprising statistic, ask "What would this mean for *me* in that situation?"

Chapter 17 · Regression to the Mean

PDF pp. 172–181

The argument

Kahneman was teaching Israeli Air Force flight instructors that rewarding improvement works better than punishing mistakes. A senior instructor objected: when he praised a cadet for a clean manoeuvre, the next attempt was usually worse, and when he screamed at a cadet for a bad one, the next was usually better. Kahneman realized the instructor was right about the observations and wrong about the cause. Performance fluctuates randomly. An exceptionally good attempt is likely to be followed by a less good one, and a bad one by a better one, whatever the instructor does. This is **regression to the mean**. Because we praise after good performance and punish after bad, life teaches us that punishment works and praise backfires.

The simple formulas: **success** = **talent** + **luck**, and **great success** = **a little more talent** + **a lot of luck**. A golfer who shoots an exceptional first round probably had above-average luck and will likely do less well on day two. Regression is everywhere: the "*Sports Illustrated* jinx," the ski jumper who does worse after a great first jump (the TV commentator invented "nervousness" to explain it), depressed children who "improve" after any treatment, and highly intelligent women who tend to marry less intelligent men. That last pattern is not a story about mate choice. It follows mathematically from the imperfect correlation of spouses' intelligence. Francis Galton discovered regression in the 19th century with sweet-pea seeds and the heights of parents and children. **Whenever the correlation between two scores is imperfect, there will be regression to the mean.** Regression has no cause. That is why it is so hard for System 1, which demands causes, to accept.

Key quotes

> *"I had stumbled onto a significant fact of the human condition: the feedback to which life exposes us is perverse."* p. 173

> *"success = talent + luck"* p. 173

> *"great success = a little more talent + a lot of luck"* p. 173

> *"The general rule is straightforward but has surprising consequences: whenever the correlation between two scores is imperfect, there will be regression to the mean."* p. 178

> *"She says experience has taught her that criticism is more effective than praise. What she doesn't understand is that it's all due to regression to the mean."* p. 181

> *“Our screening procedure is good but not perfect, so we should anticipate regression. We shouldn't be surprised that the very best candidates often fail to meet our expectations.”* p. 181

Concepts and experiments

- **Flight instructors.** Regression creates the illusion that punishment works.
- **Talent and luck.** Extreme outcomes contain extreme luck, and luck doesn't persist.
- **Golf day 1 vs day 2.** Leaders on day 1 tend to do worse on day 2, and laggards better.
- **Galton.** Regression in heredity, where children of very tall parents tend to be shorter than their parents.
- **Correlation and regression are the same concept** from two angles.
- **Control groups.** Regression is why treatment studies need control groups: the "improvement" of extreme groups may be regression.

Connections

- **Earlier:** the hunger for causes (Chapter 6) and small samples (Chapter 10).
- **Later:** Chapter 18 shows how to build regression into predictions. The "outcome bias" and hindsight (Chapter 19), management fads after streaks of success (*Built to Last*, Chapter 19), and fund manager rankings (Chapter 20).

Markets, money and people

- **Fund performance chasing.** Investors pour money into last year's top funds, which often regress. Morningstar studies have shown investors' actual returns lagging their funds' reported returns because of badly timed buying and selling.
- **Mean reversion in valuations.** Robert Shiller's work shows that high CAPE ratios have historically been followed by lower long-run returns on average. That is regression in aggregate. It is a weak signal for timing but a meaningful one for long-run expectations.
- **"Hot" sectors.** Energy was the worst S&P 500 sector in 2020 and the best in 2021 and 2022. Extremes tend not to persist.
- **CEO "magazine cover" effect.** Research by Malmendier & Tate (2009) found that firms whose CEOs won prominent business awards subsequently underperformed, consistent with regression and complacency.
- **Hypothetical:** a trader has a record month and is promoted with more capital. Expect the next month to be less spectacular even if the trader is skilled.

Takeaways

- Extreme performance is partly luck. Expect the next result to be closer to average.
- Regression has no cause, so don't invent one.

- Praise after success and criticism after failure will seem to "backfire" and "work" because of regression alone.

How to apply it

- 💰 **Investing:** don't chase last year's winners or dump last year's losers just because of their results. Ask how much of the result was luck. Use long-run averages (margins, growth, valuations) as the default for forecasts.
- 🤝 **People:** don't conclude that harsh criticism works. Evaluate people over many observations. When hiring the "best" from a pool, expect some disappointment.
- 🎬 **Content:** a viral hit is partly luck, so expect the next video to do less well and don't overreact by changing everything. Judge your content strategy on averages over many uploads.
- 🧠 **Thinking:** whenever you see an extreme result followed by a change, ask "Would regression alone explain this?"

Chapter 18 · Taming Intuitive Predictions

PDF pp. 182–192

The argument

"Julie is currently a senior in a state university. She read fluently when she was four years old. What is her grade point average?" Most people answer about 3.7 or 3.8. System 1 finds a causal link (early reading suggests talent), evaluates the evidence (precocious reading is impressive) and **matches intensity** (impressive evidence means an impressive GPA). The prediction is as extreme as the evidence. But early reading is only weakly correlated with college GPA. When asked to *evaluate* evidence and to *predict* outcomes, people give the same answers: **prediction matches evaluation**. That is substitution again.

Kahneman offers a four-step correction for intuitive predictions:

1. Start with an estimate of average GPA (the **baseline**).
2. Determine the GPA that matches your impression of the evidence (the **intuitive** prediction).
3. Estimate the **correlation** between the evidence and the outcome.
4. Move from the baseline toward the intuitive prediction by that proportion. If the correlation is .30, move 30% of the way.

The same logic works for probabilities and for choosing between candidates. The corrected prediction is unbiased but less extreme, so you will rarely predict extreme outcomes, and that has costs: a venture capitalist looking for the next Google needs to spot rare winners. Kahneman accepts that sometimes the cost of missing a rare success outweighs the cost of error, but insists you should know what you're doing. Correcting requires effort (System 2), and intuitive extreme predictions are the default.

Key quotes

> *"As was the case with Julie, the prediction of the future is not distinguished from an evaluation of current evidence—prediction matches evaluation."* p. 185

> *"The two procedures are intended to address the same bias: intuitive predictions tend to be overconfident and overly extreme."* p. 188

> *"Extreme predictions and a willingness to predict rare events from weak evidence are both manifestations of System 1."* p. 191

> *"Our intuitive prediction is very favorable, but it is probably too high. Let's take into account the strength of our evidence and regress the prediction toward the mean."* p. 191

> *"The investment may be a good idea, even if the best guess is that it will fail. Let's not say we really believe it is the next Google."* p. 192

Concepts and experiments

- **Nonregressive intuitions.** Predictions match the intensity of the evidence.
- **Prediction = evaluation.** Students asked to evaluate or to predict gave identical responses (descriptions of freshmen).
- **The correction procedure:** baseline, intuitive estimate, correlation, then regress the estimate toward the baseline.
- **Probability version:** start from the base rate and adjust toward your intuitive probability according to the diagnosticity of the evidence.
- **Trade-off.** Unbiased predictions sacrifice the thrill (and occasional value) of correctly calling extremes.

Connections

- **Earlier:** intensity matching (Chapter 8), substitution (Chapter 9), base rates (Chapters 14–16) and regression (Chapter 17).
- **Later:** the illusion of validity (Chapter 20), formulas vs intuition (Chapter 21) and the outside view (Chapter 23), which applies the same "start from the baseline" principle to projects.

Markets, money and people

- **Earnings forecasts.** Analysts tend to extrapolate strong recent growth. Research on long-run earnings growth (for example by Chan, Karceski & Lakonishok, 2003) has found that few firms sustain very high growth for long. Regressing growth forecasts toward the economy-wide average is a sensible default.
- **Venture capital.** VC returns are dominated by a few extreme winners, the exception Kahneman discusses. A rational VC may knowingly back long shots, but should be honest that the base case for any single startup is failure.
- **Macro calls.** Forecasters who called the 2008 crisis became famous, but many later extreme calls did not come true. Calling one extreme correctly says little about the next.
- **Worked example (hypothetical):** a stock's revenue grew 50% last year. The baseline (average growth for similar firms) is 10%. You think the correlation between last year's growth and the next five years' growth is about 0.3. Your regressed forecast is 10% + 0.3 × (50% − 10%) = 22%, not 50%.

Takeaways

- Intuitive predictions are as extreme as the evidence feels, which is too extreme.

- Correct them by moving from the baseline toward your intuition only as far as the evidence's validity justifies.
- Sometimes betting on long shots is rational, but know that's what you're doing.

How to apply it

- 💰 **Investing:** apply the four-step correction to growth, margins and returns. For conviction calls, write both the "regressed" forecast and the "story" forecast, and size the position on the regressed one.
- 🤝 **People:** for hiring and promotions, expect stars to regress somewhat. Balance first impressions with base rates of success in the role.
- 🎬 **Content:** explain the four-step method on a whiteboard with a real company. It's a practical, rare tool that makes viewers feel smarter.
- 🧠 **Thinking:** before giving any forecast, ask "What's the average outcome, and how much does my evidence really tell me?"

Part 3 · Overconfidence

Part 3 asks why we fail to notice the errors described in Part 2. Our minds build coherent stories of the past (the narrative fallacy, hindsight), so we feel we understand the world and can predict it. This chapter group covers the illusions of validity and skill (stock-picking and punditry), the superiority of simple formulas, the conditions for trustworthy expert intuition, the outside view against the planning fallacy, and the optimism that drives capitalism.

Chapter 19 · The Illusion of Understanding

PDF pp. 195–203

The argument

Nassim Taleb's **narrative fallacy** names our habit of building simple, coherent stories about the past. Those stories emphasize talent, stupidity and intentions, neglect luck, and focus on a few striking events that happened rather than the many that didn't. Kahneman takes Google as an example. Its founders made a series of good decisions, and the story reads like inevitable success. But luck played a large role, and any of many events could have derailed it. The story *feels* like understanding. It isn't.

We believe we understand the past, which implies the future should be knowable. That is an illusion. **Hindsight bias**: Baruch Fischhoff asked people before Nixon's 1972 trips to China and Russia to assign probabilities to possible outcomes. Afterwards, they "remembered" assigning higher probabilities to the events that happened. Once we know the outcome, we revise our memory of what we believed. This leads to **outcome bias**: we judge decisions by their results instead of by the quality of the decision when it was made. Doctors, CIA analysts before 9/11 and financial advisers are blamed for bad outcomes that were unforeseeable, and rewarded for reckless decisions that happened to work. The consequence is that decision makers become bureaucratic and risk-averse to protect themselves.

Business books like *Built to Last* and *In Search of Excellence* sell the idea that management practices reliably produce success. Kahneman points out that the gap in profitability between the "visionary" companies and their comparison firms shrank to almost nothing after the study period, and the excellent companies' profitability fell sharply. That is regression to the mean. A CEO's influence on firm performance is real but modest: a correlation of about .30 means the stronger CEO leads the more successful firm in only about 60% of pairs, against 50% by chance. Because of the halo effect, we attribute success to the CEO's brilliance and failure to arrogance, using the same traits (bold vs reckless, flexible vs erratic) depending on the outcome.

Key quotes

> *"Narrative fallacies arise inevitably from our continuous attempt to make sense of the world."* p. 195

> *"You cannot help dealing with the limited information you have as if it were all there is to know."* p. 196

> *"Our comforting conviction that the world makes sense rests on a secure foundation: our almost unlimited ability to ignore our ignorance."* p. 196

> *"Hindsight bias has pernicious effects on the evaluations of decision makers."* p. 198

> *"The illusion that one has understood the past feeds the further illusion that one can predict and control the future."* p. 200

> *"She has no evidence for saying that the firm is badly managed. All she knows is that its stock has gone down. This is an outcome bias, part hindsight and part halo effect."* p. 203

> *"Let's not fall for the outcome bias. This was a stupid decision even though it worked out well."* p. 203

Concepts and experiments

- **Narrative fallacy (Taleb).** Simple, concrete, causal stories of the past that underweight luck.
- **Hindsight bias / "I-knew-it-all-along" (Fischhoff & Beyth).** Memories of earlier predictions shift toward known outcomes.
- **Outcome bias.** Judging decision quality by results.
- **Halo in business (Rosenzweig's *The Halo Effect*).** The same CEO traits are praised or blamed depending on outcome.
- **The CEO correlation.** At r = .30, the stronger CEO leads the more successful firm in about 60% of pairs.
- **Regression of "excellent" firms.** *Built to Last* and *In Search of Excellence* firms regressed. Kahneman also cites research finding that, over twenty years, stocks of *Fortune*'s "most admired" companies earned lower returns than stocks of the least admired.

Connections

- **Earlier:** WYSIATI (Chapter 7), causal stories (Chapter 6), regression (Chapter 17) and the halo effect (Chapter 7).
- **Later:** the illusion of validity and the illusion of skill in stock-picking (Chapter 20), unpredictability (Chapters 20–22), and outcome-driven blame that creates loss-averse organizations (Chapters 31–32).

Markets, money and people

- **Post-crisis narratives.** After 2008, many people "knew" the crisis was obvious. Before it, very few positioned for it. Michael Burry and the investors profiled in *The Big Short* were famous precisely because they were rare. Hindsight makes the crash look predictable.
- **Outcome bias in markets.** A trader who bet a fund on a single position and won is called a genius. One who sized a good, well-reasoned bet sensibly and lost is called a fool.

Druckenmiller has said that some of his best years came from decisions he would not repeat, and some of his best decisions lost money. Judge the process.

- **Dalio's 1982 call.** In 1982 Ray Dalio publicly predicted an economic collapse. Instead the U.S. began a historic bull market, and he lost so much that he had to let his staff go and borrow money from his father, a story he tells in *Principles*. His response was to institutionalize humility: "What if I'm wrong?" became a core question. It is a cure for the illusion of understanding.
- **Corporate "visionaries."** GE under Jack Welch was celebrated as a model of management. After his 2001 retirement, the company's later troubles (its share price fell more than 70% between 2016 and 2018) led to reassessments of the same practices once praised. The story followed the stock price.

Takeaways

- Stories of success and failure underweight luck and overweight skill and intent.
- Once we know the outcome, we can't remember how uncertain things were.
- Judge decisions by the process and the information available at the time, not by results.

How to apply it

- 💰 **Investing:** keep a decision journal recording, *before* the outcome, your reasoning, probability estimates and alternatives. It is the only reliable defence against hindsight bias. Evaluate your process separately from your returns.
- 🤝 **People:** don't blame people for unforeseeable bad outcomes or praise them for lucky good ones. Ask "Was it a good decision given what we knew?"
- 🎬 **Content:** "Was it really obvious?" is a great series format. Show what experts said *before* famous events. It's honest, fascinating and anti-hindsight.
- 🧠 **Thinking:** when a story of success seems inevitable, list three points where luck could have gone the other way.

Chapter 20 · The Illusion of Validity

PDF pp. 204–215

The argument

As a young psychologist in the Israeli army, Kahneman evaluated candidates for officer training. They worked in groups to get over a wall using a log, without touching it. He and his colleagues watched who led, who followed and who gave up, and they felt confident about who would make a good officer. Feedback from officer school later showed their predictions were barely better than chance. Yet the next time they watched a new group, they felt just as confident. He called this the **illusion of validity**: subjective confidence comes from the coherence of the story, not from evidence of predictive accuracy.

The core of the chapter is the **illusion of stock-picking skill**. Terry Odean analysed nearly 163,000 trades from 10,000 brokerage accounts. When investors sold one stock and bought another, the stock they sold did better over the following year than the one they bought, by an average of 3.2 percentage points, before trading costs. With Brad Barber, Odean found that the most active traders had the poorest results. Men traded more than women and did worse. Individual investors tend to sell winners and buy stocks in the news. Professionals do little better: most mutual funds underperform the market, and year-to-year fund performance is barely correlated. Kahneman examined data for twenty-five wealth advisers over eight years. The average correlation between their rankings in different years was .01, essentially zero, the results of a dice-rolling contest. Yet the firm paid bonuses as if skill determined the outcomes. When he told the executives, they didn't change anything.

The illusion persists because people use real skills (analysing financial statements, judging management), but the key question, whether information is already in the price, is one they can't answer. A culture of experts who all believe in their skill reinforces it. Philip Tetlock's study of 284 experts making over 80,000 political and economic forecasts found they did worse than simple extrapolation and only slightly better than chance. Those with the most fame and confidence did worst. Tetlock's **hedgehogs** (who know "one big thing") did worse than **foxes** (who know many little things and accept complexity). The world is fundamentally unpredictable beyond the short term, and confident pundits are not more accurate.

Key quotes

> *"Considering how little we know, the confidence we have in our beliefs is preposterous—and it is also essential."* p. 204

> *"The illusion that we understand the past fosters overconfidence in our ability to predict the future."* p. 212

> *"However, it is clear that for the large majority of individual investors, taking a shower and doing nothing would have been a better policy than implementing the ideas that came to their minds."* p. 208

> *"the most active traders had the poorest results, while the investors who traded the least earned the highest returns."* p. 208

> *"Still, I was surprised to find that the average of the 28 correlations was .01. In other words, zero."* p. 210

> *"The illusion of skill is not only an individual aberration; it is deeply ingrained in the culture of the industry."* p. 210

> *"Unfortunately, skill in evaluating the business prospects of a firm is not sufficient for successful stock trading, where the key question is whether the information about the firm is already incorporated in the price of its stock."* p. 211

> *"The first lesson is that errors of prediction are inevitable because the world is unpredictable. The second is that high subjective confidence is not to be trusted as an indicator of accuracy (low confidence could be more informative)."* p. 214

> *"What makes him believe that he is smarter than the market? Is this an illusion of skill?"* p. 215

Concepts and experiments

- **Leaderless group challenge (Israeli army).** Confident predictions that were close to worthless, and confidence that was unchanged by the feedback.
- **Odean's brokerage study.** Stocks sold outperformed stocks bought. Trading reduced returns.
- **Barber & Odean.** "Trading Is Hazardous to Your Wealth." The most active traders did worst, and men did worse than women.
- **Wealth advisers.** An average year-to-year rank correlation of .01.
- **Mutual fund persistence.** Weak. Skill effects are swamped by luck.
- **Tetlock's *Expert Political Judgment*.** Experts did worse than simple extrapolation. Hedgehogs vs foxes.
- **Why the illusion persists:** real skills applied to the wrong question, cultural reinforcement, and pay that rewards confidence.

Connections

- **Earlier:** confidence by coherence (Chapter 7), regression and luck (Chapters 17–19), and the narrative fallacy (Chapter 19).
- **Later:** formulas beat experts (Chapter 21). When intuition can be trusted, it depends on environment regularity (Chapter 22). Overconfidence among CFOs (Chapter 24). Mental accounting and the disposition effect in trading (Chapter 32).

Markets, money and people

- **Buffett's bet.** In 2007 Warren Buffett bet Protégé Partners that an S&P 500 index fund would beat a portfolio of hedge funds over ten years (2008–2017). The index fund returned about 7.1% a year against roughly 2.2% for the hedge funds, and Buffett won. The illusion of validity carries a price, mostly paid in fees.
- **SPIVA.** S&P's scorecards consistently show that the large majority of active funds underperform their benchmarks over 10–15 years.
- **Druckenmiller and the 2000 tech top.** By his own public account, in early 2000 Druckenmiller (then at Soros's Quantum Fund) bought into tech stocks near the peak against his own judgment and lost heavily. He has said he made the mistake out of emotion, after watching younger managers profit. He resigned from Soros Fund Management in April 2000. Even the greatest macro investors are not immune to the illusion of understanding a mania.
- **Tetlock's follow-up.** Tetlock's later Good Judgment Project (2011–2015) found that "superforecasters" (fox-like, numerate, constantly updating) could beat other forecasters, including analysts with access to classified information, on short-range geopolitical questions. Forecasting is possible when it is short-term, probabilistic and scored.
- **Hypothetical:** a YouTuber presents a single big theory that explains all market moves. That's a hedgehog, and great TV, but the evidence suggests poorer forecasts.

Takeaways

- Confidence comes from story coherence, not proof. High confidence is not evidence of accuracy.
- Most active traders and managers don't beat the market after costs, and performance persistence is weak.
- Being skilled at analysis is not the same as having an edge over prices.
- Forecasts beyond the short term are unreliable. Foxes beat hedgehogs.

How to apply it

- 💰 **Investing:** default to low-cost index exposure for core holdings. Treat active bets as a deliberate, sized edge with a clear answer to "Why is this not in the price?" Trade less. Track your own hit rate and compare yourself to a passive benchmark honestly.

- 🤝 **People:** be wary of the most confident expert in the room. Ask for track records with scores, not stories.
- 🎬 **Content:** be a fox on camera. Present scenarios with probabilities, revisit past calls publicly and score yourself. It builds rare credibility, which is the Tetlock brand.
- 🧠 **Thinking:** "How would I know if my predictions in this area are any good?" If you can't answer, lower your confidence.

Chapter 21 · Intuitions vs. Formulas

PDF pp. 216–226

The argument

In 1954, Paul Meehl reviewed about twenty studies comparing clinical predictions (by trained counsellors) with statistical predictions (simple formulas combining a few scores) of outcomes such as students' grades, parole violations and success in pilot training. The formulas matched or beat the experts in nearly every case. The number of such studies has since grown to about two hundred, and the score hasn't changed: in about 60% of studies the algorithm is significantly more accurate, and in the rest the two tie. Since algorithms are cheaper, a tie is a win.

Why do experts lose? They try to be clever, thinking outside the box and weighing complex combinations of features, which adds noise in low-validity environments. More importantly, they are **inconsistent**: experienced radiologists contradict themselves about 20% of the time when shown the same X-ray again, and auditors and psychologists are similarly inconsistent. Context, mood and priming influence each judgment. Orley Ashenfelter predicted the prices of Bordeaux wines from three weather variables (summer temperature, harvest rain, winter rain) with a correlation above .90, beating renowned wine experts. Robyn Dawes showed that **equal-weighted** formulas (just add standardized scores) are often as good as optimally weighted multiple regression, and sometimes better. His memorable example: marital stability is well predicted by frequency of lovemaking minus frequency of quarrels. Virginia Apgar's 1953 newborn score, five variables rated 0–2, saved countless babies.

People resist algorithms, preferring human judgment, especially when the stakes are high and the decisions personal. Kahneman tells how, as a 21-year-old lieutenant in 1955, he redesigned the Israeli army's interview system. Interviewers were to rate six traits (responsibility, sociability, masculine pride and others) one at a time, using factual questions. They complained ("You are turning us into robots!"), so he added a final "close your eyes" intuitive rating. The new system was much better than the old one, and the intuitive rating, given *after* the disciplined scoring, also had predictive value. Practical recipe for hiring: choose about six independent traits, make factual questions for each, score each before moving to the next, and hire the candidate with the highest total, even if you like another better.

Key quotes

> *"The number of studies reporting comparisons of clinical and statistical predictions has increased to roughly two hundred, but the score in the contest between algorithms and humans has not changed."* p. 216

> *"The research suggests a surprising conclusion: to maximize predictive accuracy, final decisions should be left to formulas, especially in low-validity environments."* p. 219

> *"In a memorable example, Dawes showed that marital stability is well predicted by a formula: frequency of lovemaking minus frequency of quarrels"* p. 220

> *"I learned from this finding a lesson that I have never forgotten: intuition adds value even in the justly derided selection interview, but only after a disciplined collection of objective information and disciplined scoring of separate traits."* p. 225

> *"Whenever we can replace human judgment by a formula, we should at least consider it."* p. 226

> *"Let's decide in advance what weight to give to the data we have on the candidates' past performance. Otherwise we will give too much weight to our impression from the interviews."* p. 226

Concepts and experiments

- **Clinical vs statistical prediction (Meehl, 1954).** Formulas match or beat experts.
- **Inconsistency.** The same expert gives different answers to the same case. Formulas don't.
- **Ashenfelter's wine formula.** Weather predicts Bordeaux prices better than experts.
- **Improper linear models (Dawes).** Equal weights work about as well as optimal ones.
- **Apgar score.** A simple checklist that standardized newborn assessment.
- **Hostility to algorithms.** People prefer human judgment and are more upset by algorithmic errors. Kahneman predicts the resistance will weaken as algorithms become more common.
- **Israeli army interview (1955).** Structured, trait-by-trait scoring plus a final intuitive rating.
- **"Do it yourself" hiring procedure.** Six traits, factual questions, independent scoring, then follow the total.

Connections

- **Earlier:** the illusion of validity (Chapter 20), the halo effect and decorrelating errors (Chapter 7), and correcting predictions (Chapter 18).
- **Later:** when expert intuition *can* be trusted (Chapter 22) and the outside view (Chapter 23) as a "formula" for project forecasting. Kahneman's later book *Noise* (2021, with Olivier Sibony and Cass Sunstein) expands the inconsistency argument at length.

Markets, money and people

- **Quant investing.** Renaissance Technologies' Medallion fund, run on systematic models, has had one of the best reported track records in finance. Many quant funds, by contrast, have had poor stretches. Formulas aren't magic, but rules remove inconsistency.

- **Dalio's systematized principles.** Bridgewater encoded its decision rules into algorithms. Dalio has explained that writing down criteria and testing them historically was how he protected himself from his own emotional judgment after 1982.
- **Credit scoring.** FICO-style scores replaced loan officers' gut feel for consumer lending. Rule-based checklists are also widely used for portfolio risk limits.
- **Hypothetical:** a personal "investment scorecard" with six criteria (valuation vs history, balance sheet, earnings revisions, insider activity, macro regime fit, and position correlation), each scored 1–5. Buy only if the total exceeds a threshold. Your gut gets one final vote, *after* the scores.

Takeaways

- Simple formulas usually beat expert judgment in noisy, low-validity environments.
- Experts are inconsistent, and that inconsistency destroys accuracy.
- Equal-weighted checklists of relevant factors work surprisingly well.
- Intuition helps *after* disciplined, structured evaluation, not instead of it.

How to apply it

- 💰 **Investing:** build a checklist and scoring system for entries, exits and sizing, and backtest it. Use rules for risk management (stop-loss levels, maximum position size, drawdown limits) so emotion can't override them.
- 🤝 **People:** use structured interviews for hiring: define the traits first, score each independently, and decide by total. It's fairer and more accurate.
- 🎬 **Content:** use a scorecard for content too. Define 5–6 factors (hook, clarity, novelty, emotion, thumbnail, title), rate every idea before producing it, and track the scores against views.
- 🧠 **Thinking:** in any repeated decision, ask "Could a simple rule do this better than me?"

Chapter 22 · Expert Intuition: When Can We Trust It?

PDF pp. 227–237

The argument

Kahneman disliked hostile academic exchanges ("sarcasm for beginners and advanced sarcasm"), so he tried "adversarial collaborations." The most important was with Gary Klein, leader of the "naturalistic decision making" movement. Klein admired expert intuition (firefighters, nurses, chess players); Kahneman was sceptical of it. Over seven or eight years they worked out where they agreed, publishing in 2009 an article titled "Conditions for Intuitive Expertise: A Failure to Disagree."

Klein's **recognition-primed decision model** describes how fireground commanders act. One option comes to mind through pattern recognition, they mentally simulate it (System 2), and if it works they act. Herbert Simon's definition is that intuition is recognition: the situation provides a cue, the cue gives access to memory, and memory provides the answer. Expertise takes a long time to build: roughly 10,000 hours of practice for chess mastery, by one estimate.

They agreed on the conditions under which intuition can be trusted. The environment must be **sufficiently regular to be predictable**, and there must be an **opportunity to learn its regularities through prolonged practice** with quick, clear feedback. Chess players, physicians, nurses, athletes and firefighters work in regular environments. Stock pickers and political pundits do not. Their "intuitions" are not valid, however confident they feel. Hogarth's "wicked" environments teach the wrong lessons. Feedback quality matters: anaesthesiologists get quick feedback, radiologists much slower feedback. **Subjective confidence is not a reliable guide** to validity, because confidence comes from coherence and ease (System 1), not from a check on accuracy. Experts may also not know the limits of their expertise: a psychotherapist may have good intuitions about a patient's next response but poor ones about the patient's progress years later.

Key quotes

> *"The situation has provided a cue; this cue has given the expert access to information stored in memory, and the information provides the answer. Intuition is nothing more and nothing less than recognition."* p. 15

> *"Klein and I eventually agreed on an important principle: the confidence that people have in their intuitions is not a reliable guide to their validity."* p. 232

> *"Remember this rule: intuition cannot be trusted in the absence of stable regularities in the environment."* p. 233

> *"This is why subjective confidence is not a good diagnostic of accuracy: judgments that answer the wrong question can also be made with high confidence."* p. 236

> *"The question is not whether these experts are well trained. It is whether their world is predictable."* p. 215

> *"Did he really have an opportunity to learn? How quick and how clear was the feedback he received on his judgments?"* p. 237

(The Simon quote appears in the Introduction and the "Is their world predictable?" line closes Chapter 20. Both are central to this chapter's argument.)

Concepts and experiments

- **Adversarial collaboration.** Opponents jointly design studies or arguments to settle their disagreements.
- **Recognition-primed decision (RPD) model (Klein).** Pattern match, mental simulation, then act.
- **Intuition = recognition (Simon).**
- **Conditions for valid intuition:** (1) a regular environment and (2) prolonged practice with rapid, clear feedback.
- **Wicked environments (Hogarth).** Feedback misleads.
- **Validity vs confidence.** Confidence is a feeling, not a measure of accuracy.
- **The kouros story (from Gladwell's *Blink*).** Experts sensed a statue was fake, an example of valid intuition. Kahneman and Klein agree such intuitions deserve investigation, not blind trust.

Connections

- **Earlier:** the Introduction's firefighter story, the illusion of validity (Chapter 20) and formulas (Chapter 21).
- **Later:** the outside view (Chapter 23) as a remedy where intuition has no valid basis. The Conclusions' advice on when to slow down.

Markets, money and people

- **Where trading intuition is valid.** Short-horizon execution, market making and reading order flow can provide quick, frequent feedback in a fairly regular environment. That's why experienced floor traders and market makers may have real intuitions about liquidity and price action. Long-term macro forecasting gives slow, ambiguous feedback in a wicked, changing environment.
- **Druckenmiller's "gut."** Druckenmiller speaks of trusting instinct, but his process also relies on tight risk control, fast reversal when wrong, and watching how markets *react* to news,

which provides quick feedback. Intuition backed by rapid feedback and disciplined exits is closer to Klein's conditions.

- **Regime shifts.** Intuitions honed in one regime (2010–2021 low rates) may fail in another (2022's inflation and rate shock). Bond investors suffered: the Bloomberg U.S. Aggregate bond index lost about 13% in 2022, its worst year on record. The environment changed, so learned regularities broke.
- **Hypothetical:** a real estate investor with 20 years in one city may have valid intuitions about local rents, and none about crypto or emerging-market currencies.

Takeaways

- Trust intuition only in regular environments where the expert has had long practice with quick, clear feedback.
- Confidence alone tells you nothing about validity.
- Experts may be valid in part of their domain and not in others.

How to apply it

- 💰 **Investing:** map your decisions by environment. Execution and short-term risk may be intuitive; long-term forecasts need base rates and models. Get faster feedback on predictions by writing them down with dates and scoring them.
- 🤝 **People:** when someone says "trust me," ask how often they've made this kind of call and how quickly they learned whether they were right.
- 🎬 **Content:** you have excellent feedback loops (views, retention, click-through), so your intuition about content can become expert. Track the data, and vary your approach on purpose to learn faster.
- 🧠 **Thinking:** two questions before trusting your gut: Is this world regular? Have I had enough feedback to learn it?

Chapter 23 · The Outside View

PDF pp. 238–247

The argument

In the 1970s Kahneman led a team writing a high-school curriculum and textbook on judgment and decision making. After a year, he asked everyone to estimate how long it would take to finish. Estimates ranged from about 1.5 to 2.5 years. Then he asked Seymour Fox, a curriculum expert, how long similar teams had taken. Fox paused and said that about 40% never finished, and none of those that finished took less than seven years (seven to ten was typical). He added that their team was below average in resources and skill. The team ignored this, carried on, and finished eight years later, by which time the Ministry of Education had lost interest and the textbook was never used.

Kahneman draws three lessons. There is a difference between the **inside view** (focusing on your specific case and plans) and the **outside view** (the statistics of the reference class of similar cases). The inside view produced wildly optimistic estimates. Even an expert who *had* the outside view information didn't apply it spontaneously. And the team made an irrational decision to continue, a form of **sunk-cost** thinking.

The **planning fallacy** describes plans and forecasts that are unrealistically close to best-case scenarios and that could be improved by consulting the statistics of similar cases. Examples: the Scottish Parliament building was estimated at up to £40 million in 1997 and was finished in 2004 at roughly £431 million. A 2005 study of rail projects worldwide (1969–1998) found that planners overestimated passenger numbers by 106% on average and that the average cost overrun was 45%. American homeowners who remodelled their kitchens expected to spend $18,658 on average and spent $38,769. Bent Flyvbjerg's cure is **reference class forecasting**: identify the reference class, obtain its statistics, and adjust from that baseline using case-specific information. Kahneman adds a decision-level view: executives make decisions based on **delusional optimism**, and organizations should reward precise planning and penalize failure to anticipate difficulties.

Key quotes

> *"Seymour estimated the base rate of success in that reference class: 40% failure and seven to ten years for completion."* p. 240

> *"This is a common pattern: people who have information about an individual case rarely feel the need to know the statistics of the class to which the case belongs."* p. 241

> *"The treatment for the planning fallacy has now acquired a technical name, reference class forecasting, and Flyvbjerg has applied it to transportation projects in several countries."* p. 244

> *"In its grip, they make decisions based on delusional optimism rather than on a rational weighting of gains, losses, and probabilities."* p. 245

> *"He's taking an inside view. He should forget about his own case and look for what happened in other cases."* p. 246

> *"We are making an additional investment because we do not want to admit failure. This is an instance of the sunk-cost fallacy."* p. 246

Concepts and experiments

- **Inside view vs outside view.** Case-specific reasoning vs reference-class statistics.
- **The curriculum project.** Forecasts of about 2 years, a reference class of 7–10 years and 40% failure, and an actual 8 years with no use.
- **Planning fallacy (Kahneman & Tversky, 1979).** Best-case plans and neglect of base rates and "unknown unknowns."
- **Evidence:** the Scottish Parliament, rail projects (Flyvbjerg) and kitchen remodels.
- **Reference class forecasting (Flyvbjerg).** Find the class, its distribution, and adjust.
- **Irrational perseverance.** Continuing after the outside view says stop. Sunk costs.

Connections

- **Earlier:** base rates and their neglect (Chapters 14–16), anchoring on plans (Chapter 11) and correcting intuitive predictions (Chapter 18).
- **Later:** optimism and the premortem (Chapter 24), the sunk-cost fallacy (Chapter 32) and broad framing (Chapter 31). The outside view is one of Kahneman's most practical remedies, repeated in the Conclusions.

Markets, money and people

- **Mega-projects.** Berlin Brandenburg Airport was planned to open in 2011 and finally opened in October 2020, far over budget. Boston's "Big Dig" was initially estimated at under $3 billion and cost about $14.6 billion. Classic planning fallacies.
- **Corporate M&A.** Acquirers project synergies from the inside view. Studies of mergers have repeatedly found that many fail to create value for acquirers' shareholders. AOL–Time Warner (2000) is a famous example: in 2002 the company recorded a goodwill write-down of about $99 billion.
- **Startup and fund forecasts.** Most founders and new fund managers project above-average outcomes. The reference class (survival rates, median returns) is sobering.
- **Personal finance:** people's retirement and home renovation budgets routinely miss. Use the reference class.

- **Hypothetical:** you plan to grow a newsletter to 100,000 subscribers in a year. Outside view: what share of newsletters ever reach that size, and how long did it take?

Takeaways

- Plans are best-case scenarios. Expect delays and overruns.
- Start every forecast with the outside view, the statistics of similar cases, and then adjust.
- Having the base-rate information isn't enough. You must deliberately use it.

How to apply it

- 💰 **Investing:** for every investment thesis, ask what happened to similar companies or situations (turnarounds, spin-offs, IPOs, rate-hiking cycles). Build the base rate into your expected return and timeline.
- 🤝 **People:** when someone gives you a timeline, ask "How long did the last three similar projects take?" Pad your own estimates using your track record.
- 🎬 **Content:** creators chronically underestimate production time. Track your actual time per video and plan from that reference class.
- 🧠 **Thinking:** before any estimate, ask "What's the reference class, and what usually happens?"

Chapter 24 · The Engine of Capitalism

PDF pp. 248–258

The argument

The planning fallacy is one manifestation of a pervasive **optimistic bias**: we see the world as more benign, our attributes as better and our goals as more achievable than they are. Optimism is a blessing. Optimists are happier, more resilient, live longer and take risks that drive innovation, which is why it is the "engine of capitalism." But optimistic entrepreneurs and executives take risks partly because they misjudge the odds. The chance that a small business in the U.S. survives five years is about 35%, yet surveyed American entrepreneurs put the odds of success for "any business like yours" at 60%, almost double the true value. For their *own* venture, 81% put their odds at 7 out of 10 or higher, and 33% said their chance of failing was zero. Thomas Åstebro studied inventors who received objective evaluations from Canada's Inventor's Assistance Program. Among those told their project was hopeless, 47% continued and doubled their losses on average. **Hubris** among CEOs leads to value-destroying acquisitions. Malmendier and Tate found that optimistic CEOs (identified by how much company stock they held personally) took excessive risks, and that CEOs who won prestigious awards subsequently underperformed.

Two sources of optimism are WYSIATI and **competition neglect**. We focus on our own plans and abilities and neglect competitors and the base rate. Camerer and Lovallo quote a Disney Studios chairman who explained why so many big-budget films are released on the same weekends: "Hubris. Hubris. If you only think about your own business…" This leads to **excess entry**: more firms enter than the market can sustain. **Overconfidence**: a Duke University survey of CFOs (by John Graham, Campbell Harvey and colleagues) collected more than 11,600 forecasts of the S&P 500's return over the next year. The correlation between their estimates and the actual outcomes was slightly *below zero*. Their 80% confidence intervals were missed 67% of the time, more than three times the expected 20%. CFOs, physicians and pundits are rewarded for confidence. An expert who admits uncertainty gets replaced by a more confident competitor.

The partial remedy is Gary Klein's **premortem**. Before committing to a decision, the team imagines that it is a year later, the plan has been implemented, and the outcome was a disaster. Everyone takes 5–10 minutes to write a brief history of that disaster. This legitimizes doubt, overcomes groupthink and surfaces threats that people were reluctant to raise.

Key quotes

> *"In terms of its consequences for decisions, the optimistic bias may well be the most significant of the cognitive biases."* p. 248

> *"The chances that a small business will … survive for five years in the United States are about 35%."* p. 249

> *"However, 47% of them continued development efforts even after being told that their project was hopeless, and on average these persistent (or obstinate) individuals doubled their initial losses before giving up."* p. 250

> *"The truly bad news is that the CFOs did not appear to know that their forecasts were worthless."* p. 254

> *"An unbiased appreciation of uncertainty is a cornerstone of rationality—but it is not what people and organizations want."* p. 255

> *"Experts who acknowledge the full extent of their ignorance may expect to be replaced by more confident competitors"* p. 255

> *"Imagine that we are a year into the future. We implemented the plan as it now exists. The outcome was a disaster. Please take 5 to 10 minutes to write a brief history of that disaster."* p. 257

> *"We should conduct a premortem session. Someone may come up with a threat we have neglected."* p. 258

Concepts and experiments

- **Optimistic bias.** Most people see themselves as above average and their plans as likely to succeed.
- **Entrepreneurial delusions.** Five-year survival of about 35% vs self-estimates of 60% or more.
- **Åstebro's inventors.** 47% persisted after being told their project was hopeless.
- **CEO overconfidence (Malmendier & Tate).** Excessive risk and value-destroying mergers.
- **Above-average effect and competition neglect (Camerer & Lovallo).** Excess entry.
- **CFO forecasts (Duke survey).** Zero accuracy and extreme overconfidence.
- **The social reward for confidence.** Clients and organizations prefer confident experts.
- **Premortem (Klein).** Prospective hindsight that legitimizes dissent.

Connections

- **Earlier:** WYSIATI (Chapter 7), the illusion of validity (Chapter 20) and the planning fallacy (Chapter 23).

- **Later:** loss aversion vs optimism, two forces that shape risk-taking (Part 4). In Chapter 31 Kahneman notes that optimism and loss aversion partly cancel out. Sunk costs (Chapter 32). Organizational remedies in the Conclusions.

Markets, money and people

- **LTCM, 1998.** Long-Term Capital Management, whose partners included Nobel laureates Robert Merton and Myron Scholes, ran very high leverage based on confident models. It lost about $4.6 billion in a few months in 1998, and the Federal Reserve Bank of New York organized a roughly $3.6 billion recapitalization by banks. Brilliance doesn't prevent overconfidence. It may increase it.
- **Archegos, 2021.** Bill Hwang's family office built highly concentrated, leveraged positions through swaps. It collapsed in March 2021, and its banks lost more than $10 billion combined (Credit Suisse alone about $5.5 billion).
- **IPO and SPAC booms.** Excess entry: 2021 saw a record number of SPAC IPOs (over 600 in the U.S.), far more than could find good targets.
- **Druckenmiller on sizing.** Druckenmiller often cites the lesson he credits to Soros: it's not whether you're right or wrong, but how much you make when right and lose when wrong. The premortem is the risk-control version: assume you're wrong and decide in advance what you'll do.
- **Hypothetical:** before going big on a trade, write the headline: "How [your name] lost 20% on this trade." What happened in that story? Set your stop and size accordingly.

Takeaways

- Optimism fuels effort and innovation but leads to underestimating risk and competition.
- Most entrepreneurs and executives overrate their odds and neglect competitors.
- Forecasters are overconfident and rewarded for it. Confidence is not accuracy.
- The premortem is a cheap, powerful way to surface hidden risks.

How to apply it

- 💰 **Investing:** run a premortem before every major position: "It's 12 months from now and this lost 30%. Why?" Turn the answers into risk limits and exit triggers. Widen your forecast ranges, since your "80% confident" intervals are probably more like 30–40%.
- 🤝 **People:** use premortems in team decisions to give quieter people permission to dissent. Value advisers who quantify uncertainty.
- 🎬 **Content:** stay optimistic about effort and realistic about odds. Premortem a video series ("Why did this flop?") before launching. Competition neglect also applies to topics: check what's already saturated.
- 🧠 **Thinking:** keep your optimism for persistence and drop it from probability estimates.

Part 4 · Choices

Part 4 moves from judgment to decision. It sets the Econs of economic theory against the Humans of psychology. Kahneman explains prospect theory, the work for which he received the 2002 Nobel Prize in economics (Tversky died in 1996 and so could not share it). Outcomes are judged as gains and losses relative to a reference point. Losses loom larger than gains. Sensitivity diminishes as amounts grow. Probabilities are weighted non-linearly. These principles explain the endowment effect, the fourfold pattern of risk attitudes, the overweighting of rare events, narrow framing, mental accounting, regret, preference reversals and framing effects.

Chapter 25 · Bernoulli's Errors

PDF pp. 261–269

The argument

In the early 1970s, Amos gave Kahneman an essay by the Swiss economist Bruno Frey whose first sentence read: "The agent of economic theory is rational, selfish, and his tastes do not change." To a psychologist this was astonishing. Richard Thaler later named the two species **Econs** and **Humans**. Kahneman and Tversky set out to study how Humans make risky choices, starting with simple gambles.

The dominant theory was **expected utility theory**, which goes back to Daniel Bernoulli in 1738. Bernoulli saw that people are risk averse: they prefer a sure $X to a gamble with the same expected value. He explained this with diminishing marginal utility of wealth: each extra ducat adds less happiness than the one before, so the pain of losing exceeds the joy of winning an equal amount. His theory explains insurance, since a poorer person will happily pay a richer one to take on risk. It held sway for nearly 300 years.

Bernoulli's error was to assume that utility depends on the **state of wealth**, not on **changes** in wealth. Jack and Jill each have $5 million today, but yesterday Jack had $1 million and Jill had $9 million. Bernoulli's theory says they are equally happy. Obviously Jack is elated and Jill despondent. Utility depends on the **reference point**. Similarly, Anthony (current wealth $1 million) and Betty ($4 million) choose between a gamble (equal chances to end up with $1 million or $4 million) and a sure $2 million. For Anthony the sure thing doubles his wealth, so he takes it. For Betty it means losing half her wealth, so she gambles. Bernoulli's model can't explain the difference. Kahneman calls the fact that this flaw was ignored for centuries **theory-induced blindness**: once you have accepted a theory and used it as a tool, it is very hard to notice its flaws.

Key quotes

> *“The agent of economic theory is rational, selfish, and his tastes do not change.”* p. 261

> *“The happiness that Jack and Jill experience is determined by the recent change in their wealth, relative to the different states of wealth that define their reference points”* p. 266

> *“The errors of a theory are rarely found in what it asserts explicitly; they hide in what it ignores or tacitly assumes.”* p. 266

> *“once you have accepted a theory and used it as a tool in your thinking, it is extraordinarily difficult to notice its flaws.”* p. 268

> *"Both candidates are willing to accept the salary we're offering, but they won't be equally satisfied because their reference points are different. She currently has a much higher salary."* p. 268

Concepts and experiments

- **Econs vs Humans (Thaler's terms).** The rational agent of theory vs real people.
- **Expected utility theory (Bernoulli; von Neumann & Morgenstern).** Choose the option with the highest probability-weighted utility.
- **Diminishing marginal utility.** Utility of wealth is concave (logarithmic in Bernoulli's version).
- **Psychophysics (Fechner, Weber).** Perceived magnitude follows ratios. Bernoulli anticipated this.
- **Jack & Jill; Anthony & Betty.** Reference dependence: the same final wealth gives different feelings, and different choices.
- **Theory-induced blindness.** Accepted theories blind experts to obvious counterexamples.

Connections

- **Earlier:** System 1's sensitivity to changes rather than states (Chapter 9's list).
- **Later:** prospect theory fixes Bernoulli's error (Chapter 26). Reference points drive the endowment effect (Chapter 27), loss aversion (Chapter 28) and framing (Chapter 34). Reference points also shape well-being (Part 5).

Markets, money and people

- **Theory-induced blindness in finance.** The efficient market hypothesis and models that assumed normally distributed returns guided risk management for decades. Episodes like the October 1987 crash (the Dow fell about 22.6% in one day) and 2008 exposed tail risks that the models treated as nearly impossible.
- **Reference points in wealth.** Two investors with $5 million feel very differently if one just made it and the other just lost $4 million. This explains why investors after a drawdown take on more risk ("trying to get back to even").
- **Hypothetical:** a CEO's bonus feels small if last year's was larger, even if it's objectively huge.

Takeaways

- People evaluate *changes* (gains and losses) relative to a reference point, not final wealth.
- The same outcome can feel like a triumph or a disaster depending on where you started.
- Theories can blind smart people to obvious facts.

How to apply it

- 💰 **Investing:** notice your reference point (purchase price, peak value, last statement). Decide on the basis of total wealth and future prospects, not on recovering to a reference. Be most careful when you're "below your reference."
- 🤝 **People:** in salary and price talks, find the other side's reference point. The same offer feels like a gain or a loss depending on where they start.
- 🎬 **Content:** "Jack and Jill both have $5 million…" is a perfect short-form explainer of why money doesn't equal happiness.
- 🧠 **Thinking:** ask "Which theory am I using, and what does it ignore?"

Chapter 26 · Prospect Theory

PDF pp. 270–280

The argument

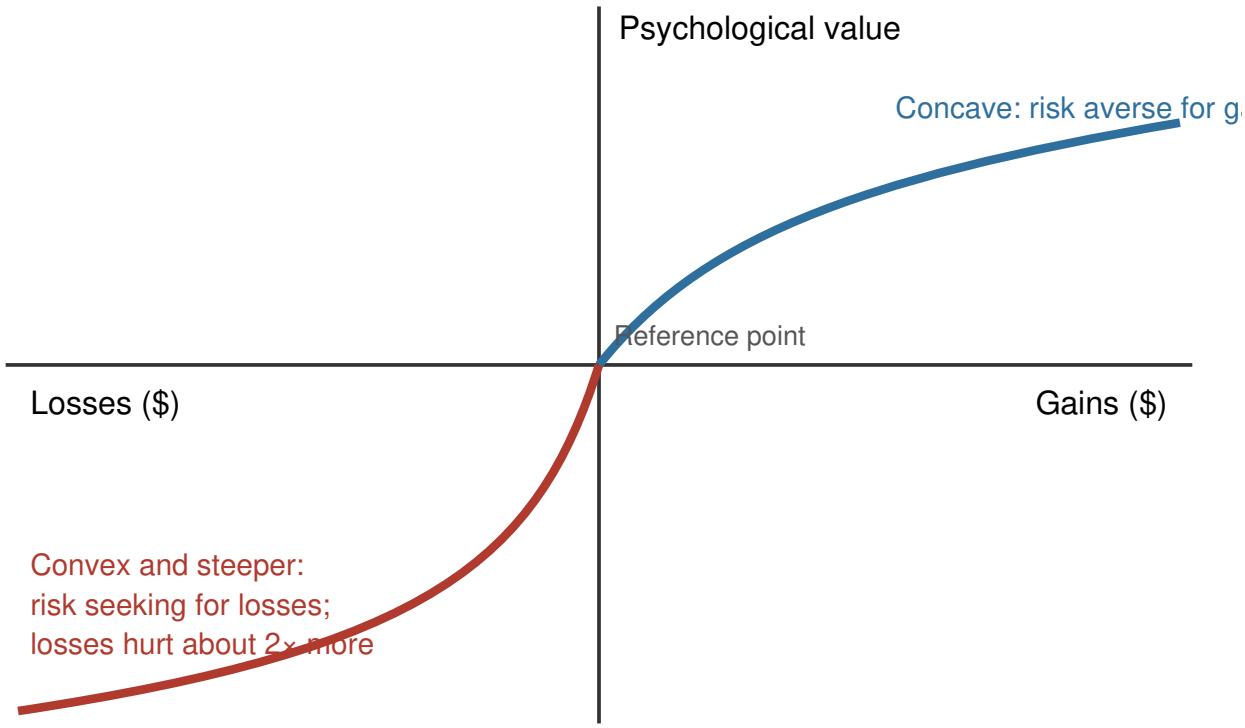
Kahneman and Tversky noticed that experiments on the utility of money used tiny gains and losses of a few cents. Surely people don't think about such gambles in terms of final wealth. When Amos asked "How about losses?", they found that risk aversion flips to **risk seeking**. *Problem 1:* get $900 for sure, or a 90% chance to get $1,000? Most people take the sure thing. *Problem 2:* lose $900 for sure, or a 90% chance to lose $1,000? Most people gamble.

The decisive demonstration of Bernoulli's error: *Problem 3:* in addition to whatever you own, you've been given $1,000; choose a 50% chance to win another $1,000 or a sure $500. *Problem 4:* you've been given $2,000; choose a 50% chance to lose $1,000 or a sure loss of $500. In final-wealth terms the problems are identical (a sure $1,500 vs a 50/50 chance of $1,000 or $2,000). Yet most people choose the sure thing in Problem 3 and the gamble in Problem 4. The reference point decides.

Prospect theory has three cognitive features, all operating characteristics of System 1:

1. **Reference dependence.** Evaluation is relative to a neutral reference point (the status quo or an expectation). Put one hand in cold water and one in warm, then both in lukewarm water: the same water feels warm to one hand and cold to the other.
2. **Diminishing sensitivity.** The difference between $100 and $200 feels bigger than between $900 and $1,000, for both gains and losses. So the value function is concave for gains and convex for losses.
3. **Loss aversion.** Losses loom larger than gains. The function is steeper for losses. For a 50/50 coin toss risking $100, most people want to win about $200. The loss-aversion ratio is usually 1.5–2.5.

Kahneman acknowledges prospect theory's **blind spots**. It can't account for regret and disappointment. A 90% chance to win $1 million versus nothing *feels* different from a 90% chance of $1 million versus $12, because missing out with the first is disappointing. Matthew Rabin's theorem showed that loss aversion cannot be explained by the utility of wealth: anyone who rejects a small favourable gamble because of curvature of wealth utility must also reject absurdly favourable large gambles.



The prospect theory value function (schematic redrawing of the book's Figure 10). It is S-shaped, steeper for losses than for gains, and flattens as amounts grow in both directions.

Key quotes

> *“Evaluation is relative to a neutral reference point, which is sometimes referred to as an "adaptation level."”* p. 273

> *“A principle of diminishing sensitivity applies to both sensory dimensions and the evaluation of changes of wealth.”* p. 274

> *“The slope of the function changes abruptly at the reference point: the response to losses is stronger than the response to corresponding gains. This is loss aversion.”* p. 274

> *“The "loss aversion ratio" has been estimated in several experiments and is usually in the range of 1.5 to 2.5.”* p. 276

> *“He weighs losses about twice as much as gains, which is normal.”* p. 280

> *“He suffers from extreme loss aversion, which makes him turn down very favorable opportunities.”* p. 280

Concepts and experiments

- **Problems 1–2.** Risk aversion for gains, risk seeking for losses.
- **Problems 3–4.** The same final states give opposite choices, which disproves final-wealth utility.
- **Three bowls of water.** A physical demonstration of reference dependence.
- **Value function.** S-shaped: concave for gains, convex for losses, kinked at the reference point.
- **Mixed gambles and loss aversion.** Win $150 / lose $100 is rejected by most people. A ratio of about 2:1 is needed.
- **Professional risk takers.** Traders in financial markets are more tolerant of losses, probably because they don't react emotionally to every fluctuation.
- **Blind spots.** Disappointment and regret are not modelled. Reference points can be expectations, not just the status quo.
- **Rabin's theorem.** Loss aversion can't be explained by the curvature of wealth utility.

Connections

- **Earlier:** fixes Bernoulli's error (Chapter 25). Its features are System 1 properties (Chapter 9's list).
- **Later:** loss aversion leads to the endowment effect (Chapter 27) and status-quo bias (Chapter 28). Decision weights make up the second half of prospect theory (Chapter 29). Narrow framing combined with loss aversion is costly (Chapter 31). Framing (Chapter 34) and Appendix B.

Markets, money and people

- **"Getting back to even."** Losing traders often increase risk to recover losses, the risk seeking in the domain of losses that Problem 2 predicts. Nick Leeson's hidden losses at Barings Bank grew as he doubled down. Losses of about £827 million brought down the 233-year-old bank in February 1995.
- **Holding losers, selling winners.** Diminishing sensitivity and loss aversion predict the **disposition effect** (Chapter 32). Odean's data confirmed it.
- **Equity premium puzzle.** Benartzi & Thaler (1995) argued that *myopic loss aversion*, meaning loss aversion combined with frequent evaluation, helps explain why investors demand such a large premium to hold stocks (Chapter 31).
- **Insurance and warranties.** People overpay to avoid small losses because losses loom large.
- **Hypothetical:** you're down 30% on a stock. A sure exit "locks in" the loss, so System 1 prefers to gamble on recovery even when the odds are poor.

Takeaways

- People are risk averse for gains and risk seeking for losses.

- Losses hurt about twice as much as equivalent gains please.
- Sensitivity diminishes: the first $100 matters more than the next.
- The reference point (status quo or expectation) determines what counts as a gain or loss.

How to apply it

- 💰 **Investing:** when you're losing, be aware that you'll be biased toward risk seeking, and use pre-set stops and position limits. Evaluate gambles by their expected value across your *whole* portfolio, not by isolated feelings of loss.
- 🤝 **People:** losses persuade more than gains: "Don't lose $500 a year" is more powerful than "Save $500 a year." Use it ethically.
- 🎬 **Content:** loss framing in titles and hooks ("The mistake costing you…") draws more attention. The prospect theory curve is one of the best visuals for explaining human money behaviour.
- 🧠 **Thinking:** before deciding, identify your reference point and check whether a different reasonable one would change your choice.

Chapter 27 · The Endowment Effect

PDF pp. 281–291

The argument

Standard economics represents preferences by indifference curves, which assume that your value for a good doesn't depend on whether you already own it. Richard Thaler observed otherwise. His professor Richard Rosett, a wine lover, would not sell bottles from his collection for less than $100, but would not pay more than $35 for a bottle of similar quality. Owning something makes it more valuable to you. Thaler called this the **endowment effect**, and prospect theory explains it: giving up the good is a **loss**, which looms larger than the **gain** of acquiring it.

The classic experiments, by Kahneman, Jack Knetsch and Thaler, used coffee mugs. Sellers (given a mug) asked $7.12 on average. Choosers (who could choose a mug or money) valued it at $3.12. Buyers would pay only $2.87. The gap between Sellers and Choosers is remarkable because they face the same choice: go home with a mug or with money. Owning the mug creates a reference point. Reluctance to give up goods held "for use" (mugs, wine, vacation days) differs from goods held "for exchange." A shoe merchant doesn't feel a loss when selling shoes, because money given up in exchange is anticipated. **Thinking like a trader** eliminates the effect. John List found that experienced baseball-card traders showed little endowment effect. Poor people are an exception in another way: Kahneman suggests that for the poor, every expenditure is a loss, because they live below their reference point.

Key quotes

> *"The fundamental ideas of prospect theory are that reference points exist, and that losses loom larger than corresponding gains."* p. 288

> *"The gap between Sellers and Choosers is remarkable, because they actually face the same choice!"* p. 287

> *"Being poor, in prospect theory, is living below one's reference point."* p. 289

Concepts and experiments

- **Indifference curves vs reference dependence.** Standard theory ignores the current position.
- **Professor R's wine (Thaler).** Selling price far above buying price.
- **Mug experiments (Kahneman, Knetsch & Thaler, 1990).** Sellers $7.12, Choosers $3.12, Buyers $2.87.

- **Goods for use vs goods for exchange.** No endowment effect for routine trading goods or for money.
- **Thinking like a trader (List).** Experienced traders show little endowment effect.
- **Poverty.** Decisions are always among losses.
- **Brain imaging.** Selling goods one would normally use activates regions associated with disgust and pain.

Connections

- **Earlier:** loss aversion and reference points (Chapter 26) and Bernoulli's error (Chapter 25).
- **Later:** status-quo bias and loss aversion in negotiations (Chapter 28), the disposition effect (Chapter 32) and broad framing ("think like a trader," Chapter 31).

Markets, money and people

- **Holding inherited or long-held stocks.** Investors often refuse to sell positions they've owned for years or inherited, valuing them above the market price. The endowment effect plus loss aversion.
- **Founders and shares.** Founders and early employees often overvalue their equity and hold concentrated positions too long. Many dot-com employees saw paper fortunes disappear in 2000–2002 because they held.
- **Real estate.** Homeowners often set asking prices above market value and hold out, especially after a price decline.
- **Hypothetical:** you wouldn't buy this stock today at this price, but you won't sell it. That's the endowment effect.

Takeaways

- Owning something makes you value it more, because giving it up feels like a loss.
- The effect disappears for goods held for exchange and among experienced traders.
- Ask "Would I buy it now at this price?" to neutralize ownership.

How to apply it

- 💰 **Investing:** review every holding as if you held cash: "If I had the money instead, would I buy this?" If not, consider selling, minding taxes. Rotating positions periodically can help you think like a trader.
- 🤝 **People:** free trials and "try it at home" offers create endowment. Once people own something, they're reluctant to give it up. In negotiations, the other side will overvalue what they're giving up.
- 🎬 **Content:** viewers who "own" something (a free template, a community membership, a named series) become more loyal.
- 🧠 **Thinking:** separate attachment from value.

Chapter 28 · Bad Events

PDF pp. 292–301

The argument

The brain gives priority to bad news. Pictures of fearful eyes, flashed so briefly that people don't consciously see them, still activate the amygdala (Paul Whalen's study). Angry faces "pop out" in a crowd of happy faces. Words like *war* and *crime* attract attention faster than *peace* and *love*. Paul Rozin notes that a single cockroach completely ruins a bowl of cherries, but a cherry does nothing for a bowl of cockroaches. In the paper "Bad Is Stronger Than Good" (Baumeister and colleagues), bad emotions, bad parents and bad feedback have more impact than good ones. John Gottman found that stable marriages need good interactions to outnumber bad ones by at least 5 to 1. This **negativity dominance** is the broad basis of loss aversion.

Goals are reference points. Falling short of a goal is a loss and exceeding it is a gain. Professional golfers putt more accurately for par (avoiding a bogey, a loss) than for birdie (a gain). Pope and Schweitzer's analysis of more than 2.5 million putts estimated that this costs top players strokes and money. New York cabdrivers who set a daily income target quit early on busy rainy days and work long hours on slow days, the opposite of what maximizes income.

Loss aversion defends the status quo. In negotiations, both sides feel their own concessions as losses, so agreements are hard to reach, and reforms are fought harder by those who lose than they are supported by those who gain. Kahneman, Knetsch and Thaler's **fairness** studies show that people judge firms by a reference transaction. Raising the price of snow shovels from $15 to $20 after a blizzard was rated unfair or very unfair by 82% of respondents. Cutting wages is unfair; failing to raise them is acceptable. In law, losses get more protection than forgone gains ("possession is nine-tenths of the law").

Key quotes

> *"The brains of humans and other animals contain a mechanism that is designed to give priority to bad news."* p. 293

> *"Bad emotions, bad parents, and bad feedback have more impact than good ones, and bad information is processed more thoroughly than good."* p. 293

> *"Gottman estimated that a stable relationship requires ... that good interactions outnumber bad interactions by at least 5 to 1."* p. 294

> *"Animals, including people, fight harder to prevent losses than to achieve gains."* p. 296

> *"Loss aversion is a powerful conservative force that favors minimal changes from the status quo in the lives of both institutions and individuals."* p. 297

> *"This reform will not pass. Those who stand to lose will fight harder than those who stand to gain."* p. 300

Concepts and experiments

- **Negativity dominance.** Threats get priority (fearful eyes, angry faces, the cockroach in the cherries).
- **Bad is stronger than good (Baumeister et al.).** Across emotions, feedback and relationships.
- **Gottman's 5:1 ratio.**
- **Goals as reference points.** Golfers' par vs birdie putts (Pope & Schweitzer) and NYC cabdrivers' daily targets (Camerer et al.).
- **Status quo and negotiations.** Concessions feel like losses, which makes reform hard.
- **Fairness (Kahneman, Knetsch & Thaler).** Snow shovels and the "reference transaction." Firms can't impose losses on customers or workers without justification.
- **Loss aversion in law.** Actual losses are compensated more than forgone gains.

Connections

- **Earlier:** loss aversion (Chapter 26) and the endowment effect (Chapter 27).
- **Later:** the fourfold pattern (Chapter 29), narrow framing (Chapter 31), regret and the omission/commission asymmetry (Chapter 32), and framing (Chapter 34). Bad experiences dominate memory in Part 5 (peak-end).

Markets, money and people

- **Bad news moves markets faster.** Markets often fall faster than they rise. The saying "stairs up, elevator down" reflects this asymmetry, and implied volatility typically rises when markets fall.
- **Price targets and income goals.** Traders who set daily P&L targets behave like the cabdrivers: they stop after hitting a target on a good day and keep trading to hit it on a bad day, often making things worse.
- **Fairness and pricing backlash.** Surge pricing (Uber's surge pricing during emergencies drew public criticism) and price hikes during crises trigger backlash, as the snow-shovel study predicts.
- **Policy reform.** Pension and tax reforms meet fierce resistance from those who would lose.
- **Hypothetical:** you set a goal of +20% for the year and you're at +18% in December. The goal is now your reference, and you may take excessive risk to "hit" it.

Takeaways

- Bad is stronger than good, in perception, emotion, relationships and memory.
- Goals become reference points: falling short feels like a loss and shapes behaviour.
- Loss aversion protects the status quo and makes negotiation and reform hard.
- Perceived fairness depends on reference transactions.

How to apply it

- 💰 **Investing:** avoid short-horizon P&L targets that become reference points. Expect markets to react more strongly to bad news, and treat panics as possible opportunities if your thesis is intact.
- 🤝 **People:** for healthy relationships and teams, aim for at least 5 positive interactions for every negative one. In negotiation, frame your requests as their gains, not their losses. Explain price rises with cost justifications to preserve fairness.
- 🎬 **Content:** negative hooks grab attention, but a channel built only on fear burns trust. Balance warnings with practical wins. Handle criticism knowing one hateful comment weighs more than five kind ones.
- 🧠 **Thinking:** when you feel strong resistance to a change, ask whether it's the merits or just loss aversion toward the status quo.

Chapter 29 · The Fourfold Pattern

PDF pp. 302–313

The argument

When you evaluate a gamble, you assign **decision weights** to outcomes, and they don't equal probabilities. Going from 0% to 5% (the **possibility effect**) and from 95% to 100% (the **certainty effect**) feel like much bigger changes than going from 5% to 10% or from 60% to 65%. Imagine inheriting $1 million that your greedy stepsister contests in court. Your lawyer gives you a 95% chance of winning. You'd likely accept a settlement well below the expected value, because the 5% chance of losing everything looms large. Overweighting of small probabilities raises the appeal of both lotteries and insurance.

Allais's paradox. At a 1952 meeting in Paris, Maurice Allais presented choices to leading economists, including the statistician Jimmie Savage. Most preferred a 100% chance of $500,000 to a 98% chance of $520,000 in one problem, but in a scaled-down version, where both probabilities were reduced, they chose the riskier option. Their choices violated expected utility theory. The certainty effect explains it.

Kahneman and Tversky estimated decision weights: a 1% chance gets a weight of about 5.5, 10% about 18.6, 50% about 42.1, 90% about 71.2 and 99% about 91.2. Small probabilities are overweighted and large ones underweighted. Combining decision weights with the value function produces the **fourfold pattern**:

	GAINS	LOSSES
HIGH PROBABILITY (certainty effect)	**95% chance to win $10,000** Fear of disappointment **RISK AVERSE** Accept an unfavourable settlement	**95% chance to lose $10,000** Hope to avoid loss **RISK SEEKING** Reject a favourable settlement
LOW PROBABILITY (possibility effect)	**5% chance to win $10,000** Hope of large gain **RISK SEEKING** Buy lottery tickets	**5% chance to lose $10,000** Fear of large loss **RISK AVERSE** Buy insurance

The bottom-right and top-left cells were known before (insurance, risk aversion for sure gains). The top-right cell, risk seeking for high-probability losses, is the source of many disasters: people facing a nearly sure loss gamble desperately to avoid it, turning bad situations into catastrophes. Kahneman applies the pattern to **litigation**. Plaintiffs with strong cases are risk averse (they settle). Defendants facing near-certain losses are risk seeking (they fight on). In frivolous suits, plaintiffs with a small chance of a big win are risk seeking, while defendants are risk averse and settle to avoid the small risk of a big loss. Over many cases, these deviations from expected value are costly.

Key quotes

> *"The conclusion is straightforward: the decision weights that people assign to outcomes are not identical to the probabilities of these outcomes, contrary to the expectation principle."* p. 304

> *"Outcomes that are almost certain are given less weight than their probability justifies."* p. 303

> *"people attach values to gains and losses rather than to wealth, and the decision weights that they assign to outcomes are different from probabilities."* p. 308

Concepts and experiments

- **Decision weights vs probabilities.** Non-linear weighting of probabilities.
- **Possibility effect.** Going from 0 to 5% has a big impact (lotteries, fears).
- **Certainty effect.** Going from 95% to 100% has a big impact (settlements, insurance against residual risk).
- **Allais paradox (1952).** Leading economists violated expected utility.
- **Viscusi, Magat & Huber's insecticide study.** Parents would pay an extra $2.38 per bottle to cut a poisoning risk from 15 to 5 per 10,000 bottles, but $8.09, more than three times as much, to eliminate it completely. That is a certainty premium.
- **Fourfold pattern.** Risk attitudes depend on gains vs losses and high vs low probability.
- **Litigation.** Frivolous suits succeed because defendants overweight small risks of large losses.

Connections

- **Earlier:** the value function and loss aversion (Chapter 26).
- **Later:** rare events, vividness and denominator neglect (Chapter 30), and risk policies to counter the costs of these patterns (Chapter 31). Appendix B's "Psychophysics of Chances."

Markets, money and people

- **Lottery-like stocks.** Research (for example Boyer, Mitton & Vorkink, 2010; Bali, Cakici & Whitelaw, 2011) finds that stocks with lottery-like payoffs (high skewness or very large recent daily gains) tend to have lower average future returns. Investors overpay for the possibility effect. The same logic applies to short-dated out-of-the-money options popular with retail traders since 2020.
- **Tail-risk insurance.** Investors pay up for crash protection after crashes. Some funds have profited from it in specific crises (Universa Investments reported very large gains in March 2020), though steady premium costs can drag on returns in calm years.

- **Doubling down on losers.** The top-right cell (risk seeking for near-certain losses) is the psychology behind rogue-trader disasters (Barings 1995; Société Générale's Jérôme Kerviel, with about €4.9 billion in losses in 2008) and companies betting the firm to avoid acknowledging failure.
- **Hypothetical:** a startup's bank account will run out in three months. Founders facing a near-certain loss take wild bets they would never take from a position of strength.

Takeaways

- Small probabilities are overweighted (hope and fear), and near-certainties are underweighted.
- Risk attitudes depend on gains vs losses and high vs low probabilities, which gives four distinct patterns.
- Risk seeking when facing almost certain losses is a major source of disasters.

How to apply it

- 💰 **Investing:** be suspicious of your attraction to lottery-like payoffs (penny stocks, meme stocks, far-out options). Know when you're in the "top right" cell, losing and tempted to double down, and have a rule against it. Price insurance and hedges rationally, not emotionally.
- 🤝 **People:** a counterpart facing an almost certain loss is likely to be risk seeking, so offer face-saving ways out. A counterpart close to a sure gain will accept less to lock it in.
- 🎬 **Content:** use the fourfold table as a visual. It explains lotteries, insurance, lawsuits and trading behaviour in one image.
- 🧠 **Thinking:** when a probability is near 0 or 100%, deliberately ask what the expected value is.

Chapter 30 · Rare Events

PDF pp. 314–325

The argument

Kahneman visited Israel during a period of suicide bus bombings: 23 bombings between December 2001 and September 2004, with 236 deaths, against about 1.3 million daily bus riders. The risk to any rider was tiny. Kahneman knew this, yet he found himself uncomfortable stopping next to a bus at a red light. Terrorism speaks directly to System 1. Vivid images of a rare event trigger an emotional response that System 2 cannot fully override, even when it knows the numbers.

Kahneman revises the original prospect theory. **People overestimate the probabilities of unlikely events and overweight unlikely events in their decisions.** Overestimation is driven by focused attention, confirmation and cognitive ease. When Craig Fox asked basketball fans to estimate the probability of each of the eight teams winning the NBA playoffs, the probabilities summed to 240%, because each team, once focused on, seemed plausible. Fans also paid more for bets on all teams than the total payoff. Overweighting is strongest when outcomes are vivid and emotional. Rottenstreich and Hsee found that the value of a chance to kiss a favourite movie star (or to avoid a painful shock) was much less sensitive to probability than cash prizes.

Vivid probabilities. Seymour Epstein's urn experiment: urn A has 10 marbles with 1 red (10% chance); urn B has 100 marbles with 8 red (8%). Many people choose urn B because 8 winning marbles is more vivid than 1. This is **denominator neglect**. Frequency formats make outcomes vivid. Forensic psychologists were more likely to refuse to discharge a patient when told that "of every 100 patients similar to Mr. Jones, 10 are estimated to commit an act of violence" than when told that such patients "have a 10% probability." A disease that "kills 1,286 people out of every 10,000" was judged more dangerous than one that "kills 24.14% of the population," although the risk is only half as large. **Choice from experience** reverses the pattern: people who learn probabilities by sampling outcomes, rather than from a description, *underweight* rare events, often because they have never experienced them. Most Californians have never experienced a major earthquake, and in 2007 no banker had personally experienced a devastating financial crisis.

Key quotes

> *"People overestimate the probabilities of unlikely events. People overweight unlikely events in their decisions."* p. 315

> *"The emotional arousal is associative, automatic, and uncontrolled, and it produces an impulse for protective action."* p. 314

> *"The probability of a rare event is most likely to be overestimated when the alternative is not fully specified."* p. 316

> *"Most Californians have never experienced a major earthquake, and in 2007 no banker had personally experienced a devastating financial crisis."* p. 323

> *"It's the familiar disaster cycle. Begin by exaggeration and overweighting, then neglect sets in."* p. 325

> *"They want people to be worried by the risk. That's why they describe it as 1 death per 1,000. They're counting on denominator neglect."* p. 325

> *"We shouldn't focus on a single scenario, or we will overestimate its probability. Let's set up specific alternatives and make the probabilities add up to 100%."* p. 325

(This links back to Chapter 13's availability cascades: "Terrorism speaks directly to System 1" (p. 142).)

Concepts and experiments

- **Overestimation vs overweighting.** A judgment of probability vs the decision weight given to it.
- **Bus bombings in Israel.** An emotional response despite known tiny risks.
- **Focused attention (Fox's NBA study).** Probabilities summed to 240%. Explicit alternatives reduce overestimation.
- **Vivid outcomes (Rottenstreich & Hsee).** Emotional outcomes make decisions insensitive to probability.
- **Denominator neglect (Epstein's urns; Slovic's Mr. Jones).** Counts of cases are more vivid than percentages.
- **Description vs experience (Hertwig, Erev and others).** Rare events are overweighted when described and underweighted when learned from experience.
- **Global impressions.** If a rare event doesn't get attention, it may be ignored completely.

Connections

- **Earlier:** availability and affect (Chapters 12–13) and the fourfold pattern and decision weights (Chapter 29).
- **Later:** risk policies (Chapter 31). Framing effects of formats (Chapter 34). Disaster cycles (also Chapter 13).

Markets, money and people

- **Pre-2008 neglect.** The book's own point: in 2007 bankers had not experienced a crisis, so the risk of a nationwide housing decline was underweighted, a choice from experience. Then, after 2008, investors overweighted crash risk for years. Both errors come from the same mind.
- **Black swans and tail hedging.** Taleb's *The Black Swan* (2007) became popular after the crisis. Tail-hedging demand typically rises *after* crashes, when protection is expensive.
- **Meme and lottery behaviour.** Vivid stories of 100× gains (Dogecoin rose more than 10,000% from early 2020 to its May 2021 peak) make tiny probabilities feel large.
- **Pandemic (2020).** A global pandemic had been discussed as a risk for years but was underweighted until it happened, after which fear spiked.
- **Hypothetical:** a financial ad says "1 in 5 retirees will run out of money." The frequency format feels more alarming than "20%," which is denominator neglect at work.

Takeaways

- Vivid, emotional or focused-on rare events are overweighted.
- Rare events we've never experienced are underweighted.
- Frequencies ("10 out of 100") feel more vivid than percentages (10%).
- Listing all alternatives and making them add up to 100% curbs overestimation.

How to apply it

- 💰 **Investing:** for tail risks, set a standing policy (hedging budget, maximum leverage) instead of reacting to how vivid the risk feels. Model scenarios that sum to 100%. Stress-test portfolios against events no one in the firm has personally experienced.
- 🤝 **People:** to make a risk feel real, use frequencies and stories. To calm fear, use percentages and comparisons. Be honest about which you're doing.
- 🎬 **Content:** thumbnails with vivid tail events (crash, collapse) attract clicks because of overweighting. Use them with responsible context and real probabilities.
- 🧠 **Thinking:** when you notice a vivid image driving your estimate, ask for the actual numbers and the alternatives.

Chapter 31 · Risk Policies

PDF pp. 326–333

The argument

Consider two concurrent decisions. (i) Choose between A, a sure gain of $240, and B, a 25% chance to gain $1,000. (ii) Choose between C, a sure loss of $750, and D, a 75% chance to lose $1,000. Most people choose A and D, following the fourfold pattern. Combined, A and D give "a 25% chance to win $240 and 75% chance to lose $760." The rejected combination, B and C, gives "a 25% chance to win $250 and 75% chance to lose $750," which **dominates** AD. People make dominated choices because they evaluate each decision separately. This is **narrow framing**. **Broad framing** looks at decisions together, as a portfolio.

Paul Samuelson's friend refused a single coin toss (lose $100, win $200) but said he'd accept 100 such tosses. Samuelson showed that this was inconsistent, but Kahneman argues the friend was right in one sense: as Matthew Rabin and Richard Thaler pointed out, 100 such tosses have an expected return of $5,000 with only about a 1-in-2,300 chance of losing any money. The mistake is treating each small gamble in isolation. Since life offers many small favourable gambles, a rational person should accept them all and think of each as one of many. A trader's mantra, "you win a few, you lose a few," is broad framing. The combination of loss aversion and narrow framing is costly. Investors who check their portfolios often see many small losses and feel more pain than pleasure, which leads them to take less risk than they should. Benartzi and Thaler call this **myopic loss aversion**.

A **risk policy** is a standing rule for a class of decisions. "Never buy extended warranties" and "always take the highest deductible on insurance" are risk policies. They aggregate decisions so that the occasional loss is offset by many gains. Combined with the outside view (a policy for forecasting), a risk policy counters both optimism and loss aversion. Richard Thaler asked the managers of a company's 25 divisions whether each would accept a risky option with equal chances of losing a large amount of their division's capital or earning double that amount. None would. The CEO, who would face all 25 bets together, wanted them all accepted.

Key quotes

> *"The combination of loss aversion and narrow framing is a costly curse."* p. 331

> *"Closely following daily fluctuations is a losing proposition, because the pain of the frequent small losses exceeds the pleasure of the equally frequent small gains."* p. 331

> *"Tell her to think like a trader! You win a few, you lose a few."* p. 332

> *"I decided to evaluate my portfolio only once a quarter. I am too loss averse to make sensible decisions in the face of daily price fluctuations."* p. 333

> *"They never buy extended warranties. That's their risk policy."* p. 333

> *"Each of our executives is loss averse in his or her domain. That's perfectly natural, but the result is that the organization is not taking enough risk."* p. 333

Concepts and experiments

- **Concurrent decisions (A/B and C/D).** Narrow framing leads to dominated choices.
- **Narrow vs broad framing.** Isolated vs aggregated evaluation.
- **Samuelson's problem.** One bet refused, 100 bets accepted. Aggregation reduces the felt risk of loss.
- **Myopic loss aversion (Benartzi & Thaler).** Frequent evaluation leads to excessive risk aversion.
- **Risk policies.** Standing rules that aggregate decisions.
- **Thaler's 25 division managers.** Individual managers are loss averse, but the CEO sees the portfolio.

Connections

- **Earlier:** loss aversion (Chapter 26), the fourfold pattern (Chapter 29) and the outside view (Chapter 23).
- **Later:** mental accounting (Chapter 32), joint vs single evaluation (Chapter 33) and frames (Chapter 34). WYSIATI (Chapter 7) is behind narrow framing.

Markets, money and people

- **Checking less often.** Benartzi & Thaler's work and later experiments (Thaler, Tversky, Kahneman & Schwartz, 1997) found that people who received investment feedback less frequently invested more in stocks and earned more. Many advisers recommend that long-term investors limit how often they check their accounts.
- **Portfolio vs position thinking.** A macro investor running a book of uncorrelated bets thinks like Thaler's CEO. Dalio's "Holy Grail" principle, combining 15 or more good, uncorrelated return streams to reduce risk dramatically, is broad framing turned into strategy.
- **Insurance policies.** Taking high deductibles and skipping extended warranties are positive-expected-value risk policies for anyone who can absorb small losses.
- **Corporate underinvestment.** Divisional managers reject positive-expected-value projects because they fear individual blame, the "organization is not taking enough risk" problem.

- **Hypothetical:** a creator rejects every experimental video format because each one might flop. Over 50 experiments, a few hits would more than pay for the misses.

Takeaways

- Evaluating decisions one at a time (narrow framing) combined with loss aversion leads to too little risk taking and dominated choices.
- Broad framing, seeing each decision as one of many, is closer to rational.
- Adopt risk policies: standing rules for repeated types of decisions.
- Check investments less often.

How to apply it

- 💰 **Investing:** write a risk policy covering position sizing, maximum loss per trade, diversification, rebalancing schedule and how often you check your accounts (for a long-term portfolio, quarterly). Evaluate performance at the portfolio level, not trade by trade.
- 🤝 **People:** in organizations, reward good risk-taking decisions across a portfolio of bets rather than punishing each individual failure. Leaders should "aggregate" their teams' bets.
- 🎬 **Content:** treat content like a portfolio. Most uploads perform modestly, a few explode. Commit to a volume of experiments rather than agonizing over each.
- 🧠 **Thinking:** before any decision, ask "Is this one of many similar decisions? What policy would I want for all of them?"

Chapter 32 · Keeping Score

PDF pp. 334–344

The argument

Except for the very poor, money is mainly a proxy for points on a scale of self-regard and achievement. We keep score in our heads, and those scores shape our preferences. Richard Thaler's **mental accounting** describes how we organize money into separate accounts: spending money, savings, the vacation fund, each stock position. Mental accounts are a form of narrow framing. They help self-control but lead to foolish decisions.

The **disposition effect**: investors are much more likely to sell winning stocks than losing ones. Selling a winner "closes the account" with a gain, a pleasure. Selling a loser closes it with a loss, which hurts. The behaviour is costly. Terry Odean found that the stocks investors sold (winners) went on to outperform the losers they held, and in the U.S. it's tax-inefficient, since selling losers produces tax deductions. More sophisticated investors are less prone to the effect. The **sunk-cost fallacy**: two fans plan to drive 40 miles to a basketball game, and a blizzard is announced. The one who paid for his ticket is more likely to brave the storm than the one who got it free. Economists would say the ticket price is irrelevant, since it's spent either way. The sunk-cost fallacy keeps people in poor jobs, unhappy marriages and unpromising projects. Executives who began a failing project are less likely to cut it, which is one reason boards replace CEOs: the new CEO doesn't carry the old mental account.

Regret is a counterfactual emotion triggered when an alternative to reality is easy to imagine. Mr. Brown, who almost never picks up hitchhikers, and Mr. Smith, who frequently does, are both robbed by a hitchhiker. 88% of respondents say Mr. Brown will feel more regret, because his action was abnormal. Paul considered switching from stock A to stock B and didn't; George switched from B to A. Both would have been $1,200 better off owning B. 92% say George feels more regret, because regret is stronger for *actions* (commission) than for *inactions* (omission), and stronger for departures from the default. People are also reluctant to accept even tiny increases in risk for money (the "**taboo tradeoff**"), especially regarding children's safety. Kahneman's advice: anticipate regret explicitly, and either be thorough or accept that hindsight will hurt. Daniel Gilbert and colleagues argue that people generally anticipate more regret than they actually experience, because they underestimate their own psychological defences (their "psychological immune system").

Key quotes

> *“For the billionaire looking for the extra billion, and indeed for the participant in an experimental economics project looking for the extra dollar, money is a proxy for points on a scale of self-regard and achievement.”* p. 334

> *"The sunk-cost fallacy keeps people for too long in poor jobs, unhappy marriages, and unpromising research projects."* p. 337

> *"Regret is one of the counterfactual emotions"* p. 338

> *"people expect to have stronger emotional reactions (including regret) to an outcome that is produced by action than to the same outcome when it is produced by inaction."* p. 339

> *"He has separate mental accounts for cash and credit purchases. I constantly remind him that money is money."* p. 343

> *"We are hanging on to that stock just to avoid closing our mental account at a loss. It's the disposition effect."* p. 344

Concepts and experiments

- **Mental accounts (Thaler).** Money is separated into categories, so it is not treated as fungible.
- **Disposition effect (Shefrin & Statman; Odean).** Investors sell winners and hold losers.
- **Sunk-cost fallacy.** The blizzard ticket and escalation of commitment to failing projects.
- **Regret and normality.** The hitchhiker scenarios (Mr. Brown vs Mr. Smith). Regret is felt more for abnormal actions.
- **Omission vs commission.** Paul vs George: 92% say George feels more regret.
- **Defaults.** Deviating from the default produces more regret, and so more blame.
- **Taboo tradeoffs.** Refusing to trade small increases in children's risk for money.
- **Insurance against regret.** Anticipated regret drives conservative, conventional choices.

Connections

- **Earlier:** narrow framing (Chapter 31), loss aversion (Chapter 26), norms and counterfactuals (Chapter 6) and the endowment effect (Chapter 27).
- **Later:** framing and default effects (Chapter 34: organ donation) and regret in policy (Conclusions). The "keeping score" idea returns in Part 5's remembering self.

Markets, money and people

- **Disposition effect in practice.** Odean (1998) found that investors realized gains much more readily than losses, and that the winners they sold outperformed the losers they kept over the following year. A habit Druckenmiller often describes is the opposite: cut losers fast and let winners run.

- **Sunk costs in corporate investment.** Companies keep pouring money into failing projects to avoid writing off past spending. Shareholders often reward new CEOs who announce write-downs, which break the old mental account.
- **"Career risk" and regret.** Professional investors fear the regret and blame of being wrong *unconventionally* more than being wrong with the crowd. This is part of why herding persists. Keynes's famous line, that it is better for reputation to fail conventionally than to succeed unconventionally, expresses the same idea.
- **Hypothetical:** you hold two stocks, one up 40% and one down 40%, and you need cash. Mental accounting says sell the winner. Tax logic and forward-looking analysis may say sell the loser.

Takeaways

- We keep score in mental accounts, and closing an account at a loss is painful, so we avoid it.
- Investors tend to sell winners too early and hold losers too long.
- Past costs are irrelevant to future decisions. Ignore sunk costs.
- Regret is stronger for actions than inactions and for departures from the norm, which pushes us to conventional, passive choices.

How to apply it

- 💰 **Investing:** set pre-committed sell rules for losers (thesis broken, or a price/time stop). Review positions without looking at purchase price. Harvest tax losses. Ask "If I were starting fresh, would I own this?"
- 🤝 **People:** when someone won't let go of a failing project, offer a "fresh start" frame (a new budget year, a new owner) to close the mental account. Anticipate that people fear the regret of acting; making action the default reduces that barrier.
- 🎬 **Content:** don't keep producing a failing series because of what you've already invested in it. Each new video should earn its slot on future expected value.
- 🧠 **Thinking:** before deciding, imagine both regrets, from acting and from not acting, so that the asymmetry doesn't decide for you.

Chapter 33 · Reversals

PDF pp. 345–353

The argument

You set compensation for a man shot during a robbery at a convenience store. Scenario (i): the robbery happened at his regular store. Scenario (ii): his regular store was closed for a funeral, so he went to another store, where he was shot. Considered separately (**single evaluation**), people award more in the second scenario because it evokes stronger poignancy ("if only..."). Considered together (**joint evaluation**), almost everyone agrees that the location shouldn't matter. The judgments reverse.

Preference reversals were first documented by Sarah Lichtenstein and Paul Slovic. People chose bet A (high probability of a modest win) over bet B (low probability of a big win), but put a higher selling price on B. Economists, including David Grether and Charles Plott, set out to show the effect was an artefact and ended up confirming it. This was a challenge to economic theory, which assumes stable preferences.

Reversals happen because single evaluation is governed by the norms of the category the object belongs to, while joint evaluation brings comparison across categories. Asked to contribute to saving dolphins or to providing skin-cancer check-ups for farmworkers, people give more to dolphins when each is evaluated alone (dolphins rank high among endangered species, farmworkers' skin cancer lower among public-health issues), but more to farmworkers in joint evaluation (humans matter more than dolphins). Christopher Hsee's **evaluability hypothesis**: some attributes can only be evaluated in comparison. A music dictionary with 10,000 entries in "like new" condition is valued more than one with 20,000 entries and a torn cover in single evaluation. Together, the larger dictionary wins easily. The legal system often decides cases in single evaluation, so punitive damages and penalties are inconsistent across categories. Joint evaluation is usually, though not always, more reasonable, and it can be exploited by salespeople who control what you compare.

Key quotes

> *"The result illustrates Hsee's evaluability hypothesis: The number of entries is given no weight in single evaluation, because the numbers are not "evaluable" on their own."* p. 351

> *"The BTU units meant nothing to me until I saw how much air-conditioning units vary. Joint evaluation was essential."* p. 353

> *"You say this was an outstanding speech because you compared it to her other speeches. Compared to others, she was still inferior."* p. 353

> *"It is often the case that when you broaden the frame, you reach more reasonable decisions."* p. 353

> *"When you see cases in isolation, you are likely to be guided by an emotional reaction of System 1."* p. 353

Concepts and experiments

- **Single vs joint evaluation.** The same options are evaluated differently alone than side by side.
- **The gunshot victim.** Poignancy affects single but not joint judgments.
- **Preference reversals (Lichtenstein & Slovic; Grether & Plott).** Choice vs pricing of bets.
- **Categories and norms.** Single evaluation is within a category, joint evaluation across categories (dolphins vs farmworkers).
- **Evaluability (Hsee).** Attributes like number of entries become meaningful only in comparison (dictionaries).
- **Unjust reversals in law.** Mock jurors in Texas, judging one case at a time, awarded higher punitive damages to a bank defrauded of $10 million than to a child burned by flammable pajamas. Seen side by side, the award to the child rose above the bank's. Across pairs of cases, awards for personal injury were more than twice as large in joint evaluation. Yet jurors are explicitly prohibited from considering other cases. Sunstein found that penalties across U.S. agencies (such as OSHA and the EPA) are sensible within each agency but incoherent across them. The fine for a "serious violation" of worker-safety rules was capped at $7,000, while a violation of the Wild Bird Conservation Act could bring up to $25,000.

Connections

- **Earlier:** Hsee's dinnerware sets (Chapter 15) and broad vs narrow framing (Chapter 31).
- **Later:** framing (Chapter 34). In the Conclusions, structured comparison and broader frames are offered as remedies.

Markets, money and people

- **Investment comparisons.** A fund's 12% return looks great alone, but less so next to a benchmark that returned 15%. Fund marketing often presents performance in single evaluation, without the benchmark.
- **Valuation in isolation.** A P/E of 30 means nothing alone. Next to peers and history it becomes evaluable.
- **Product choice.** Retailers place a deliberately expensive "decoy" next to the item they want to sell, which manipulates joint evaluation.
- **Hypothetical:** you evaluate a job offer of $150,000 alone and feel great, then learn that peers at the same level earn $180,000.

Takeaways

- Judgments made one at a time are driven by emotion and category norms. Comparisons bring in other considerations.
- Some attributes only become meaningful in comparison.
- Broader frames (joint evaluation) usually lead to more reasonable decisions, but watch who chose what you're comparing.

How to apply it

- 💰 **Investing:** never evaluate a fund, stock or strategy alone. Compare it to a benchmark, peers, its history and alternatives. Build comparison tables before deciding.
- 🤝 **People:** in negotiation, control the comparison set, and when you're the buyer, insist on your own comparisons. To evaluate performance, compare to relevant peers.
- 🎬 **Content:** side-by-side comparisons (Fund A vs Index, Then vs Now) make numbers evaluable and are highly engaging.
- 🧠 **Thinking:** "Compared to what?" is a simple question that prevents many mistakes.

Chapter 34 · Frames and Reality

PDF pp. 354–365

The argument

"Italy won" and "France lost" describe the 2006 World Cup final equally well. For an Econ they mean the same thing, but for a Human they evoke different associations. **Framing effects** are unjustified influences of formulation on beliefs and preferences. Would you accept a gamble with a 10% chance to win $95 and a 90% chance to lose $5? Would you pay $5 for a lottery ticket with a 10% chance to win $100 and a 90% chance to win nothing? The two are identical, but the second is far more popular, because a "cost" is less painful than a "loss."

Benedetto De Martino and colleagues ran a brain-imaging study in London. Participants received £50 and then chose between a sure option framed as "KEEP £20" or "LOSE £30" and a gamble. People chose the sure option more when it was framed as KEEP. The amygdala (associated with emotion) was active when choices conformed to the frame, and the anterior cingulate (associated with conflict and self-control) was active when they resisted it. People most resistant to framing showed more activity in a frontal area involved in combining emotion and reasoning. Framing isn't a superficial slip. It is System 1 at work. McNeil and colleagues gave physicians statistics on surgery vs radiation for lung cancer. "The one-month survival rate is 90%" led 84% to choose surgery. "There is 10% mortality in the first month" led only 50% to choose it. Physicians were just as susceptible as patients and business-school students.

The **Asian disease problem** (Appendix B): with 600 lives at stake, Program A saves 200 for sure, while Program B has a 1/3 chance of saving all 600 and a 2/3 chance of saving none. Most choose A. Reframed as deaths (C: 400 will die; D: 1/3 chance nobody dies, 2/3 chance 600 die), most choose D. When people are confronted with their inconsistency, they often have no answer, because System 2 has no moral intuitions of its own about the frame. Thomas Schelling's example of a child tax exemption showed that moral intuitions depend on whether the frame is a "bonus for children" or a "penalty for the childless."

Some frames are better than others. **Good frames** help: a theatregoer who loses $160 in cash is more willing to buy tickets than one who loses the tickets themselves, because the lost tickets are charged to the same mental account. Larrick and Soll's **MPG illusion** showed that miles per gallon misleads. Gallons per mile reveals that replacing a 12-mpg car with a 14-mpg one saves more fuel than going from 30 to 40 mpg. U.S. fuel-economy labels later added gallons-per-100-miles information. **Organ donation:** rates are close to 100% in Austria but 12% in Germany, and 86% in Sweden but 4% in Denmark, because some countries use opt-out and others opt-in. Defaults are frames, and they are powerful.

Key quotes

> *"Our preferences are about framed problems, and our moral intuitions are about descriptions, not about substance."* p. 361

> *"System 1, as we have gotten to know it, is rarely indifferent to emotional words: mortality is bad, survival is good, and 90% survival sounds encouraging whereas 10% mortality is frightening."* p. 358

> *"An article published in 2003 noted that the rate of organ donation was close to 100% in Austria but only 12% in Germany, 86% in Sweden but only 4% in Denmark."* p. 363

> *"Let's reframe the problem by changing the reference point. Imagine we did not own it; how much would we think it is worth?"* p. 365

> *"They ask you to check the box to opt out of their mailing list. Their list would shrink if they asked you to check a box to opt in!"* p. 365

Concepts and experiments

- **Framing effects.** Logically equivalent descriptions lead to different choices.
- **Costs vs losses.** A lottery ticket price ($5) is more acceptable than a $5 loss in a gamble.
- **De Martino et al. (2006).** KEEP vs LOSE, and the neural correlates of framing and of resisting it.
- **McNeil et al. (1982).** Survival vs mortality framing for physicians: 84% vs 50% chose surgery.
- **Asian disease problem (Tversky & Kahneman, 1981).** The lives-saved vs lives-lost frame reverses risk preference.
- **Schelling's child exemption.** Moral intuitions are frame-bound.
- **Good frames.** Lost cash vs lost tickets (mental accounts), the MPG illusion (Larrick & Soll, 2008).
- **Defaults and organ donation (Johnson & Goldstein, 2003).** Opt-out vs opt-in.

Connections

- **Earlier:** reference points and loss aversion (Chapters 25–26), mental accounts (Chapter 32) and WYSIATI (Chapter 7).
- **Later:** nudges and libertarian paternalism in the Conclusions. Appendix B is the original framing paper.

Markets, money and people

- **Retirement defaults.** Madrian & Shea (2001) found that automatic enrollment raised 401(k) participation among new employees from about 37% to 86%. The U.S. Pension Protection Act of 2006 encouraged auto-enrollment, and Thaler and Benartzi's "Save More Tomorrow" program uses defaults and future raises to increase saving.
- **Framing in investing.** "This fund lost 10% this year" vs "This fund has returned 8% annually over 10 years" can describe the same fund. Fund marketing chooses the frame.
- **Central bank communication.** Words like "transitory," "patient" or "whatever it takes" (Mario Draghi, July 2012) are frames that moved markets enormously. After Draghi's pledge, euro-area bond spreads narrowed sharply.
- **Hypothetical:** a broker describes a structured product as "100% capital protected with upside participation" instead of "you give up most of the upside and dividends in exchange for limited downside protection." Same product, different frame.

Takeaways

- How a choice is described changes the choice. Humans are frame-bound, not reality-bound.
- Losses vs costs, survival vs mortality and keep vs lose frames push choices in predictable directions.
- Defaults are frames, and they powerfully shape behaviour.
- Some frames are better than others. Seek the frame that reveals reality (like gallons per mile).

How to apply it

- 💰 **Investing:** reframe every decision at least twice (as gain and as loss; as a percentage and in dollars; per year and cumulatively). If your choice flips, think again. Set good defaults for yourself, such as automatic contributions, automatic rebalancing and pre-set stops.
- 🤝 **People:** you can't avoid framing, so choose frames ethically. Make the right thing the default (opt-out), and frame offers as gains or as avoided losses depending on the goal.
- 🎬 **Content:** your title is a frame. "Why you'll keep 80% of your gains" and "Why you'll lose 20% to taxes" describe the same facts and have very different effects. Choose frames that inform rather than mislead.
- 🧠 **Thinking:** ask "How else could this be described, and would I decide the same way?"

Part 5 · Two Selves

Part 5 turns the machinery of the book onto happiness and memory. Bentham's "experienced utility" differs from the "decision utility" of economics. We have an experiencing self that lives each moment and a remembering self that keeps score and makes choices. The remembering self follows the peak-end rule and neglects duration, so we choose for memories, not for experiences. Kahneman then explores measuring well-being (the Day Reconstruction Method, the U-index) and the focusing illusion that distorts our forecasts of what will make us happy.

Chapter 35 · Two Selves

PDF pp. 368–376

The argument

Jeremy Bentham meant "utility" as the experience of pleasure and pain. Economists later used it to mean "wantability," what people choose. Kahneman calls the first **experienced utility** and the second **decision utility**. If people always chose what they would enjoy most, the two would coincide. Often they don't.

With Don Redelmeier, Kahneman studied 154 patients undergoing colonoscopy in the early 1990s, before routine anaesthesia. Patients reported their pain every 60 seconds. Patient A's procedure lasted 8 minutes and ended at a peak of pain. Patient B's lasted 24 minutes, with more total pain, but ended gently. Afterwards, B rated the experience as *less* bad than A. Two rules explain this. The **peak-end rule**: retrospective ratings were predicted by the average of the worst moment and the final moment. **Duration neglect**: the length of the procedure had no effect on the ratings of total pain. Kahneman draws the dilemma that follows. If the goal is to reduce the *memory* of pain, lowering the peak and providing gradual relief at the end may matter more than shortening the procedure. If the goal is to reduce pain actually *experienced*, speed is better. (A later randomized trial not described in the book, Redelmeier, Katz & Kahneman, 2003, found that extending colonoscopies with a short, milder ending made patients remember them as less unpleasant.)

The **cold-hand experiment** made the conflict sharp. Participants held a hand in 14°C water for 60 seconds (the short episode). In the long episode they held it for 60 seconds at 14°C and then for 30 more seconds while the water warmed by about 1°C, slightly less painful. Asked which to repeat, 80% of those who reported that their pain diminished at the end chose the *long* one, accepting 30 seconds of extra pain. The remembering self chose against the experiencing self's interests. Kahneman calls this **the tyranny of the remembering self**. The **experiencing self** answers "Does it hurt now?" The **remembering self** answers "How was it, on the whole?" Memories are all we keep, so the remembering self makes the decisions, and it may optimize the wrong thing. Kahneman suggests evolutionary reasons: memory represents prototypes and averages (Chapter 8), not sums.

Key quotes

> *"Peak-end rule: The global retrospective rating was well predicted by the average of the level of pain reported at the worst moment of the experience and at its end."* p. 370

> *"Duration neglect: The duration of the procedure had no effect whatsoever on the ratings of total pain."* p. 370

> *"The experiencing self is the one that answers the question: "Does it hurt now?" The remembering self is the one that answers the question: "How was it, on the whole?""* p. 371

> *"Memories are all we get to keep from our experience of living"* p. 371

> *"Confusing experience with the memory of it is a compelling cognitive illusion"* p. 372

> *"What we learn from the past is to maximize the qualities of our future memories, not necessarily of our future experience. This is the tyranny of the remembering self."* p. 372

> *"You are thinking of your failed marriage entirely from the perspective of the remembering self. A divorce is like a symphony with a screeching sound at the end—the fact that it ended badly does not mean it was all bad."* p. 375

Concepts and experiments

- **Experienced vs decision utility.** What we feel vs what we choose.
- **Colonoscopy study (Redelmeier & Kahneman).** 154 patients; peak-end rule and duration neglect.
- **Cold-hand experiment (Kahneman, Fredrickson, Schreiber & Redelmeier, 1993).** Most participants preferred to repeat the longer episode with the better ending.
- **Two selves.** The experiencing self lives; the remembering self keeps score and chooses.
- **Tyranny of the remembering self.** Decisions maximize remembered utility.
- **"A symphony with a screeching end."** A bad ending distorts the memory of a good experience.

Connections

- **Earlier:** averages vs sums (Chapter 8); reference points and the neglect of time in prospect theory (Part 4). Bad events dominate memory (Chapter 28).
- **Later:** life as a story (Chapter 36), measuring experienced well-being (Chapter 37), and the focusing illusion (Chapter 38). The Conclusions discuss the policy implications of the two selves.

Markets, money and people

- **Peak-end in investing.** Investors remember a year by its worst drawdown and its ending. A year that ended with a sharp sell-off (Q4 2018, for example, when the S&P 500 fell nearly 20% from its September peak to December 24) feels worse in memory than its full path warrants, and that memory shapes risk appetite.

- **Exit decisions.** Investors who sold at a panic low remember the ending, and the memory may keep them out of markets for years. People who lived through 2008 took years to return.
- **Client experience.** Advisers and businesses know that how a meeting, product or quarter *ends* shapes how clients remember it.
- **Hypothetical:** a trader has a great year but loses money in the final two weeks. They remember it as a bad year and cut risk the next year, which is duration neglect in action.

Takeaways

- We remember experiences by their peak and their end, and largely ignore duration.
- The remembering self makes our choices, sometimes against the experiencing self's interests.
- A bad ending can ruin the memory of a good experience, and a better ending can improve the memory of a bad one.

How to apply it

- 💰 **Investing:** evaluate past periods with full data (time-weighted returns, the whole path), not by the memorable peak or ending. Don't let a bad end to a period dictate next year's risk.
- 🤝 **People:** end meetings, negotiations and customer experiences on a high note. Fix the end of a difficult process.
- 🎬 **Content:** peak-end is a content design rule. Build one clear emotional peak and a strong ending (a payoff, an insight, a call to action). Viewers remember the end and judge the whole video by it.
- 🧠 **Thinking:** when reflecting on an experience, ask "Am I judging the whole, or just the peak and the end?"

Chapter 36 · Life as a Story

PDF pp. 377–381

The argument

At the end of Verdi's *La Traviata*, Violetta lies dying while her lover rushes to see her. Will he arrive in time? We care intensely about those last ten minutes, even though she has lived a whole life. We care whether the story ends well, not how long it lasted. A story is about significant events and memorable moments, not time passing. Duration neglect is normal in narratives. We even care about the stories of the dead and of people we will never meet, as when we feel sorry for a man who died believing his wife loved him when she had been unfaithful.

Ed Diener and his students tested this with the life of Jen, a fictitious woman who died in an accident. Doubling the length of her very happy life (from 30 to 60 years) had no effect on judgments of its desirability or total happiness. Adding five years that were pleasant but less happy than before made her life seem *less* desirable. That is the peak-end effect for whole lives ("less is more" again).

Amnesic vacations. Many vacations are designed for memories: photos, stories, "bucket list" sightseeing. Kahneman asks you to imagine that at the end of a vacation all pictures will be destroyed and you'll take a drug that erases your memories. Would you choose the same trip? Many people would change their plans, and some say they wouldn't bother going. That shows how much we choose for the remembering self. Kahneman confesses that he identifies with his remembering self, and that the experiencing self feels like a stranger to him.

Key quotes

> *"A story is about significant events and memorable moments, not about time passing."* p. 377

> *"Caring for people often takes the form of concern for the quality of their stories, not for their feelings."* p. 377

> *"Odd as it may seem, I am my remembering self, and the experiencing self, who does my living, is like a stranger to me."* p. 380

> *"You seem to be devoting your entire vacation to the construction of memories. Perhaps you should put away the camera and enjoy the moment, even if it is not very memorable?"* p. 381

Concepts and experiments

- **Narrative and endings.** Stories are judged by their key moments and endings (*La Traviata*).
- **Jen's life (Diener, Wirtz & Oishi).** Duration neglect and the peak-end effect apply to whole lives.
- **Amnesic vacation.** A thought experiment that separates choosing for experience from choosing for memory.
- **Identifying with the remembering self.** We see ourselves as the author of our story.

Connections

- **Earlier:** two selves, peak-end and duration neglect (Chapter 35); the narrative fallacy (Chapter 19) is the same storytelling machinery applied to the past.
- **Later:** experienced well-being (Chapter 37), life satisfaction and the focusing illusion (Chapter 38), and the Conclusions.

Markets, money and people

- **Legacy and the remembering self.** Some investors and executives make decisions to protect their life story ("I never sold at a loss," "I called the crash"), not to maximize outcomes. The "Speaking of" line about protecting "the narrative of a life of integrity" applies to reputational decisions in finance.
- **Retirement planning.** People plan retirement for the story (travel, milestones) and underweight the daily experience (health, social connection, routine).
- **Hypothetical:** a fund manager holds a losing position into year-end to avoid "ending the year" on a realized loss. The story beats the economics.

Takeaways

- We judge lives and episodes like stories: by key moments and endings, not duration.
- Much of what we choose (vacations, purchases, careers) is for the remembering self.
- Ask whether you're living for the experience or for the story.

How to apply it

- 💰 **Investing:** watch for "story protection," holding or trading to preserve a narrative about yourself. Spend on experiences you'll enjoy *while they happen*, not just those you can show.
- 🤝 **People:** people care about the stories of their lives, their reputations and their endings. Respect that in negotiations (give face-saving endings).
- 🎬 **Content:** your brand is a story. Audiences remember arcs, turning points and endings. Series with a clear narrative arc keep viewers.
- 🧠 **Thinking:** try the amnesic-vacation test on big decisions: "Would I still do this if I couldn't remember it or tell anyone?"

Chapter 37 · Experienced Well-Being

PDF pp. 382–388

The argument

Most happiness research asks a remembering-self question: "All things considered, how satisfied are you with your life as a whole these days?" Kahneman, suspicious after the colonoscopy and cold-hand studies, wanted to measure the experiencing self directly. **Experience sampling** (Mihaly Csikszentmihalyi's method) beeps people at random times and asks what they are doing and feeling. It is accurate but expensive and burdensome. So Kahneman's team (including Alan Krueger, David Schkade, Norbert Schwarz and Arthur Stone) developed the **Day Reconstruction Method (DRM)**. Participants relive the previous day in episodes, like scenes in a film, and report what they did, who they were with and how they felt.

The **U-index** is the percentage of time a person spends in an unpleasant state, where a negative feeling is rated higher than any positive one. It measures time, not a rating scale. American women spent about 19% of their time in an unpleasant state, compared with 16% for French women and 14% for Danish women. Activities differed. For 1,000 American women in a Midwestern city, the U-index was 29% for the morning commute, 27% for work, 24% for child care, 18% for housework, 12% for socializing, 12% for TV watching and 5% for sex. Unhappiness is concentrated: a minority of people do most of the suffering. Situational factors matter greatly. Being with friends lifts mood, and being with a boss lowers it. Paying attention also matters, since an enjoyable activity is less pleasant when your mind is on other things.

Gallup's surveys (more than 450,000 responses in the U.S.) let Kahneman and Angus Deaton compare experienced well-being with life evaluation. Life evaluation keeps rising with income. Experienced well-being rose with income up to a **satiation level of about $75,000** of household income in high-cost areas and, on average, not at all beyond it. Poverty amplifies the effects of misfortunes like illness and divorce. Kahneman's practical conclusion: the easiest way to increase happiness is to control your use of time, spending more of it on things you enjoy, with people you like.

Update note. The $75,000 plateau has since been revised. Matthew Killingsworth (2021) found experienced well-being continuing to rise with income. In a 2023 adversarial collaboration, Killingsworth, Kahneman and Barbara Mellers (published in *PNAS*) concluded that for most people happiness keeps rising with log income well past $75,000. A plateau appears only among the least happy group (roughly the unhappiest 20%), for whom extra income stops helping beyond a point. Kahneman also later noted that $75,000 in 2010 dollars is considerably more today.

Key quotes

> *“We called the percentage … of time that an individual spends in an unpleasant state the U-index.”* p. 384

> *"It is only a slight exaggeration to say that happiness is the experience of spending time with people you love and who love you."* p. 386

> *"The satiation level beyond which experienced well-being no longer increases was a household income of about $75,000 in high-cost areas (it could be less in areas where the cost of living is lower)."* p. 387

> *"The easiest way to increase happiness is to control your use of time. Can you find more time to do the things you enjoy doing?"* p. 388

> *"Beyond the satiation level of income, you can buy more pleasurable experiences, but you will lose some of your ability to enjoy the less expensive ones."* p. 388

Concepts and experiments

- **Life satisfaction vs experienced well-being.** Remembering self vs experiencing self.
- **Experience sampling (Csikszentmihalyi).** Random real-time reports.
- **Day Reconstruction Method (Kahneman, Krueger, Schkade, Schwarz & Stone, 2004).** Reconstructing yesterday episode by episode.
- **U-index.** The percentage of time spent in an unpleasant state. It varies by activity, person and country.
- **Attention and situation.** Who you're with and whether you're absorbed shape moment-to-moment experience.
- **Income and well-being (Kahneman & Deaton, 2010).** Life evaluation rises with income, while experienced well-being plateaued at about $75,000 in their data (see the update note).

Connections

- **Earlier:** experiencing vs remembering selves (Chapters 35–36).
- **Later:** life evaluation, goals and the focusing illusion (Chapter 38). The Conclusions discuss using experienced well-being in policy.

Markets, money and people

- **Money and time.** Wealthy professionals often trade experienced well-being (commutes, long hours, stress) for life-evaluation gains (status, income). The DRM's activity rankings suggest that commuting and work with bosses are among the least pleasant parts of the day.
- **Retirement and the U-index.** Retirement decisions should consider daily experience: social connection, purpose and health, not just portfolio size.
- **Wealth effects on mood.** Research on stock market swings and mood has found associations between large market declines and worse self-reported mood and health outcomes. The experiencing self feels volatility.

- **Hypothetical:** a creator earning well but spending 8 hours a day editing alone may raise experienced well-being more by hiring an editor than by earning an extra $50,000.

Takeaways

- Life satisfaction (remembering) and moment-to-moment happiness (experiencing) are different things.
- Time use, who we spend it with, and attention drive experienced well-being.
- Kahneman and Deaton's original finding was that money stopped improving daily mood above about $75,000. Later research, including Kahneman's own 2023 collaboration, revised this: for most people the relationship continues. Lack of money clearly harms both measures.

How to apply it

- 💰 **Investing:** set your portfolio's risk level so that you can sleep, a real experiencing-self cost. Spend money to buy time (outsourcing chores, shorter commutes) and to share experiences with people you like.
- 🤝 **People:** time with people you love is the most reliable source of happiness. Protect it.
- 🎬 **Content:** design your work week for experienced well-being. Batch the activities that drain you and spend more time on those that put you in flow.
- 🧠 **Thinking:** try a personal DRM. Reconstruct yesterday hour by hour and rate each episode. Change the worst one.

Chapter 38 · Thinking About Life

PDF pp. 389–398

The argument

A graph from Andrew Clark, Ed Diener and Yannis Georgellis's analysis of a German panel shows life satisfaction rising in the years before marriage, peaking around the wedding and declining in the following years. Kahneman's reading is not that marriage makes people unhappy. When asked about their life, newlyweds think of their marriage, a new and salient event, and that raises their ratings. As the novelty fades, it comes to mind less often. Life-satisfaction answers are heavily influenced by whatever comes to mind (current mood, recent events, a good or bad day). Many decisions to marry reflect **affective forecasting** errors (Daniel Gilbert and Timothy Wilson's term).

Goals matter. In a survey of about 12,000 people who had entered elite colleges in 1976, those who at 18 rated "being very well-off financially" as essential were much more likely to achieve high incomes nineteen years later. Those who wanted money and didn't get it were particularly dissatisfied. Life satisfaction is partly about achieving the goals you set. Kahneman and David Schkade asked students in California and the Midwest about life satisfaction and also asked how satisfied they thought the *other* group was. Both groups expected Californians to be happier, because of the climate. In fact life satisfaction was about the same. This is the **focusing illusion**: **"Nothing in life is as important as you think it is when you are thinking about it."** When asked about climate, you focus on it, but daily life in California is mostly not about climate.

The focusing illusion explains why people overestimate the effect of buying a car or moving, and of disabilities. Paraplegics, after a period of adjustment, are in a good mood for most of the day. Over time attention shifts away from their condition, and the mood tracks what they are attending to. It also explains **miswanting**: errors in choosing goods and experiences because we expect them to matter more than they will. Adaptation makes new cars and bigger houses fade from attention, while some activities (socializing, hobbies like playing the cello) keep holding attention and pleasure. Kahneman concludes that well-being has two dimensions, how people experience their lives and how they evaluate them, and that a sensible view of happiness must consider both.

Key quotes

> *"Nothing in life is as important as you think it is when you are thinking about it."* p. 393

> *"The focusing illusion creates a bias in favor of goods and experiences that are initially exciting, even if they will eventually lose their appeal."* p. 397

> *"She thought that buying a fancy car would make her happier, but it turned out to be an error of affective forecasting."* p. 398

> *"Buying a larger house may not make us happier in the long term. We could be suffering from a focusing illusion."* p. 398

> *"His car broke down on the way to work this morning and he's in a foul mood. This is not a good day to ask him about his job satisfaction!"* p. 398

Concepts and experiments

- **Marriage and life satisfaction (Clark, Diener & Georgellis).** Salience, not lasting happiness, drives the spike.
- **Life satisfaction as a heuristic judgment.** Mood and recent events substitute for an evaluation of the whole life (Chapter 9's substitution).
- **Goals and satisfaction.** The 1976 elite-college cohort and financial aspirations.
- **California vs Midwest (Schkade & Kahneman, 1998).** Climate dominates forecasts but not experience.
- **Focusing illusion.** Whatever you're thinking about seems more important than it is.
- **Adaptation.** Paraplegics' moods, and the novelty of new possessions fading.
- **Miswanting (Gilbert & Wilson).** Choosing on the basis of mistaken forecasts.
- **Time and attention.** Activities that keep demanding attention retain their value. Possessions that stop drawing attention lose theirs.

Connections

- **Earlier:** substitution and mood (Chapter 9), WYSIATI (Chapter 7), the two selves (Chapters 35–37) and the endowment effect (Chapter 27).
- **Later:** the Conclusions argue that policy should consider both experienced well-being and life evaluation.

Markets, money and people

- **Wealth and the focusing illusion.** Many people believe that reaching a portfolio number ($1M, $10M) will change how they feel day to day. The goal does matter for life evaluation, but daily experience adapts. Many lottery winners and newly rich founders report that happiness gains fade.
- **Macro focusing illusion.** Investors focus on one dominant macro variable (the Fed, inflation, an election) and overestimate its importance for long-term returns. "Nothing is as important as you think when you're thinking about it" is a useful check on headline-driven positioning.

- **Location and lifestyle choices.** Moves for climate or tax reasons (many U.S. moves to Florida and Texas during 2020–22) may deliver less happiness than expected if daily life (friends, work, commute) changes less, or for the worse.
- **Hypothetical:** a creator believes reaching 1 million subscribers will make them happy. It will improve life evaluation, but daily mood will depend on how they spend their time.

Takeaways

- What you're focused on seems more important than it really is.
- Life satisfaction answers are swayed by what comes to mind. Experienced happiness depends on attention and time use.
- We mispredict what will make us happy (affective forecasting, miswanting) because we focus on changes that fade from attention.

How to apply it

- 💰 **Investing:** when one factor dominates your thinking, list five others that also affect your position. Before big purchases, ask "How much will I think about this in six months?" Spend on experiences and relationships that keep drawing attention.
- 🤝 **People:** don't ask people to evaluate something on a bad day. Moods spill into judgments. Help others see beyond their current focus.
- 🎬 **Content:** "Nothing in life is as important as you think it is when you are thinking about it" is one of the book's most shareable lines. Pair it with market examples (the Fed meeting that "changes everything").
- 🧠 **Thinking:** a focusing check. "If I weren't thinking about this right now, how important would it seem?"

Conclusions

PDF pp. 399–409

The argument

Kahneman ties the book together through its three distinctions, in reverse order.

Two selves. The remembering self and the experiencing self have different interests, and the remembering self makes decisions. Should we care about how people's lives are experienced or how they're remembered? He argues that neither alone is enough. A purely experiencing-self view would ignore people's goals and stories; a purely remembering-self view would accept people choosing more pain (as in the cold-hand experiment). Policy should consider both. Measuring experienced well-being, as some national statistics now do, could direct attention to reducing suffering, especially from depression, extreme poverty and loneliness.

Econs and Humans. Rationality in economics means **logical coherence**, consistency of beliefs and preferences, not reasonableness. By that definition Humans can't be rational. Kahneman emphasizes that he and Tversky did not show that humans are irrational, only that they are not well described by the rational-agent model. The Chicago school (Milton Friedman's *Free to Choose*) holds that people should be free to make their own choices, and the state shouldn't protect them from their own decisions. Behavioural economists counter that freedom has costs for individuals who choose badly and for the society that feels obliged to help them. Thaler and Sunstein's **libertarian paternalism** (*Nudge*, 2008) offers a middle way: design choice architecture (defaults, disclosure, simple formats) that helps people make better decisions while preserving their freedom. Examples are automatic enrollment in pension plans, the Save More Tomorrow plan, clear mortgage disclosures and fuel-economy labels.

Two systems. System 1 is the source of most of what we do right and of many of our errors. It is not readily educable. Kahneman says his own intuitions are no less prone to overconfidence, extreme predictions and the planning fallacy than before he did this research, though he has improved at recognizing situations where errors are likely. The way to block errors that originate in System 1 is to **recognize the signs that you are in a cognitive minefield, slow down, and ask for reinforcement from System 2.** It is easier to see others in a minefield than yourself. **Organizations** are better than individuals at avoiding errors because they think more slowly and can impose procedures: checklists, reference-class forecasting, premortems, and decision processes viewed as a factory where each stage (framing the problem, collecting information, reviewing) can be improved. A richer vocabulary, the purpose of the "Speaking of" sections, helps organizations criticize decisions constructively, at the watercooler.

Key quotes

> *"The two selves are the experiencing self, which does the living, and the remembering self, which keeps score and makes the choices."* p. 399

> *"Rationality is logical coherence—reasonable or not."* p. 401

> *"Although Humans are not irrational, they often need help to make more accurate judgments and better decisions, and in some cases policies and institutions can provide that help."* p. 402

> *"System 1 is not readily educable."* p. 407

> *"The way to block errors that originate in System 1 is simple in principle: recognize the signs that you are in a cognitive minefield, slow down, and ask for reinforcement from System 2."* p. 407

> *"Organizations are better than individuals when it comes to avoiding errors, because they naturally think more slowly and have the power to impose orderly procedures."* p. 408

> *"it is much easier to identify a minefield when you observe others wandering into it than when you are about to do so."* p. 408

> *"The voice of reason may be much fainter than the loud and clear voice of an erroneous intuition"* p. 408

Concepts

- **Policy implications of the two selves.** Measure and reduce suffering (U-index). Respect people's goals.
- **Rationality as coherence.** Humans fall short of it. That isn't the same as being irrational.
- **Chicago school vs behavioural economics.** Freedom vs protection from errors.
- **Libertarian paternalism / nudges.** Defaults, simplification and disclosure.
- **Debiasing individuals.** Recognize minefields and slow down. It is hard, because System 1 can't be trained away.
- **Debiasing organizations.** Procedures, checklists, premortems, outside views, quality control in the decision "factory," and a shared vocabulary.

Connections

- Pulls together all five parts: System 1/2 (Part 1), heuristics (Part 2), overconfidence remedies (Part 3), Econs vs Humans (Part 4) and the two selves (Part 5).

Markets, money and people

- **Nudges in personal finance.** Automatic enrollment and auto-escalation in retirement plans, now widespread in the U.S. and in the UK (where workplace pension auto-enrolment began in 2012), have greatly increased participation.
- **Investment firm processes.** Bridgewater's documented principles, Buffett and Munger's checklists (Munger's "Psychology of Human Misjudgment" talk lists many of the same biases), and investment committees that require written pre-mortems are organizational debiasing in practice.
- **Druckenmiller and Dalio as "organizations of one."** Both describe building external checks: partners who challenge them, written theses, explicit risk rules. Individuals can borrow the organization's advantage by imposing procedures on themselves.

Takeaways

- Humans aren't irrational, but they aren't Econs either. They need better choice architecture.
- You can't switch off System 1. You can learn to recognize minefields and slow down.
- Organizations, and individuals who adopt organizational procedures, can catch errors that individuals miss.
- A shared vocabulary makes criticism constructive.

How to apply it

- 💰 **Investing:** build your personal "decision factory": a written thesis, base rate, premortem, scorecard, risk policy, scheduled reviews and a decision journal.
- 🤝 **People:** use the vocabulary with your team and audience. "Are we anchoring?" and "What's the outside view?" Design defaults that help people.
- 🎬 **Content:** your channel can be the "watercooler" that improves how people talk about money decisions. Teach the vocabulary.
- 🧠 **Thinking:** learn the signs of a cognitive minefield: high stakes, unfamiliar situations, strong emotion, time pressure, a coherent story and high confidence. When you see them, slow down.

Appendix A · Judgment Under Uncertainty: Heuristics and Biases

PDF pp. 410–430 · Amos Tversky & Daniel Kahneman, *Science* 185 (1974)

The argument

This is the original paper that launched the field. Many decisions rest on beliefs about uncertain events, such as the outcome of an election, a defendant's guilt or the future value of the dollar. People assess these probabilities using a **limited number of heuristic principles** that reduce complex tasks to simpler judgments. The heuristics are generally useful but sometimes lead to **severe and systematic errors**, much as judging distance by clarity works well but makes things look farther away on hazy days. The paper describes three heuristics and their biases.

1. Representativeness (judging the probability that A belongs to B by how much A resembles B). It produces:

- *Insensitivity to prior probability of outcomes.* Told that a group contains 70 engineers and 30 lawyers (or the reverse), people judging a description ignore the base rates. With *no* description they use the base rates correctly. With a *worthless* description they ignore them.
- *Insensitivity to sample size.* In a town where a large hospital has about 45 births a day and a small one about 15, which hospital records more days on which over 60% of babies are boys? Most people say both are about the same. The correct answer is the small hospital.
- *Misconceptions of chance.* People expect short random sequences to look random (the gambler's fallacy). Even research psychologists believe in a "law of small numbers."
- *Insensitivity to predictability.* Predictions are as extreme as descriptions are favourable, regardless of how reliable the description is.
- *The illusion of validity.* Confidence comes from the fit between input and outcome, and redundant, correlated inputs *raise* confidence while *lowering* accuracy.
- *Misconceptions of regression.* People don't expect regression and invent causal explanations for it (the flight instructors).

2. Availability (judging frequency by ease of recall or construction). It produces:

- *Biases due to the retrievability of instances* (famous names make a sex seem more numerous in a list).
- *Biases due to the effectiveness of a search set* (words starting with *r* vs words with *r* third).
- *Biases of imaginability* (committees of 2 vs 8 from 10 people; the two numbers are actually equal, but small committees are easier to imagine).
- *Illusory correlation* (Chapman & Chapman: judges "saw" correlations between diagnoses and drawing features that matched their associations).

3. Adjustment and anchoring (estimates start from an initial value and adjust insufficiently). It produces:

- *Insufficient adjustment.* The wheel of fortune and the UN percentage. Estimating 8×7×6×5×4×3×2×1 within five seconds gave a median of 2,250, while 1×2×3×4×5×6×7×8 gave 512. The correct answer is 40,320.
- *Biases in evaluating conjunctive and disjunctive events.* People overestimate the probability of conjunctive events (all steps of a plan succeeding) and underestimate disjunctive ones (at least one component failing). This is why plans are optimistic and the risk of system failure is underestimated.
- *Anchoring in subjective probability distributions.* Confidence intervals are too narrow.

The discussion stresses that experts are vulnerable too, that people don't learn these rules from experience because the relevant evidence is rarely coded, and that internal consistency is not enough for a subjective probability to be rational.

Key quotes

> *"This article shows that people rely on a limited number of heuristic principles which reduce the complex tasks of assessing probabilities and predicting values to simpler judgmental operations."* p. 410

> *"In general, these heuristics are quite useful, but sometimes they lead to severe and systematic errors."* p. 410

> *"When no specific evidence is given, prior probabilities are properly utilized; when worthless evidence is given, prior probabilities are ignored."* p. 412

> *"Thus, redundancy among inputs decreases accuracy even as it increases confidence, and people are often confident in predictions that are quite likely to be off the mark."* p. 416

> *"In many situations, people make estimates by starting from an initial value that is adjusted to yield the final answer."* p. 421

> *"The reliance on heuristics and the prevalence of biases are not restricted to laymen."* p. 425

> *"These heuristics are highly economical and usually effective, but they lead to systematic and predictable errors."* p. 427

Connections

- **The book's Part 2 restates this paper** with the later framework of System 1, substitution and WYSIATI. Representativeness → Chapters 14–15. Small samples → Chapter 10. Regression → Chapter 17. Availability → Chapters 12–13. Anchoring → Chapter 11. Validity → Chapter 20. Conjunctive events → the planning fallacy (Chapter 23).

Markets, money and people

- **Conjunctive events in deals and projects.** An M&A deal that needs regulatory approval, financing, integration and customer retention to all go right is a conjunctive event, systematically overestimated.
- **Disjunctive risk in portfolios.** The chance that *at least one* leveraged position or counterparty blows up is higher than intuition suggests, which is the disjunctive-event bias. In 2008, risks that looked unlikely one by one (a mortgage originator failure, a monoline insurer downgrade, a money-market fund "breaking the buck," as the Reserve Primary Fund did in September 2008) were not independent.
- **Narrow confidence intervals.** Forecasters and CFOs (Chapter 24) give intervals that are too narrow, the anchoring in probability distributions the paper describes.

Takeaways

- Three heuristics (representativeness, availability, anchoring) underlie many judgments under uncertainty.
- They're efficient but produce predictable biases: base-rate neglect, sample-size neglect, regression blindness, ease-of-recall distortions and insufficient adjustment.
- Experts are not immune.

How to apply it

- 💰 **Investing:** for every multi-step thesis, multiply the probabilities of its steps (conjunctive). For risk, add up the probabilities of failure modes (disjunctive). Widen your confidence intervals.
- 🤝 **People:** when someone confidently predicts on the basis of a redundant set of signals ("all the indicators agree"), ask whether the signals are independent.
- 🎬 **Content:** the hospital problem and the 8×7×6… anchoring demo are perfect audience-participation segments.
- 🧠 **Thinking:** in any estimate, ask: What resembles what? What comes to mind easily? What number did I start from?

Appendix B · Choices, Values, and Frames

PDF pp. 431–452 · Daniel Kahneman & Amos Tversky, *American Psychologist* 39 (1984)

The argument

This paper, originally Kahneman and Tversky's Distinguished Scientific Contributions Award address, summarizes their work on decisions. It covers the cognitive and psychophysical determinants of choice in risky and riskless contexts.

Risky choice. The psychophysics of value produce risk aversion for gains and risk seeking for losses. The value function is defined on gains and losses, not total wealth. It is concave for gains, convex for losses and steeper for losses (loss aversion). The **weighting function** overweights low probabilities and underweights moderate and high ones. Certainty is overweighted relative to high probabilities. The paper notes that all analyses of rational choice include two principles: **dominance** (if A is at least as good as B in every respect and better in one, choose A) and **invariance** (the preference order shouldn't depend on how the options are described).

Framing of outcomes. The Asian disease problem: 72% choose Program A ("200 people will be saved") over the risky Program B. When the same options are framed as deaths ("400 people will die" vs a one-third probability that nobody will die), 78% choose the risky option. This violates invariance. People, including sophisticated respondents, often wish to stay consistent when the inconsistency is pointed out but can't resolve it, because the frames evoke different responses.

The psychophysics of chances introduces the **pseudo-certainty effect**: in sequential (two-stage) gambles, an outcome that is certain only *within* a stage is weighted as if it were certain. Protective actions that eliminate one risk entirely are overvalued relative to those that reduce overall risk more (probabilistic insurance is unattractive).

Formulation effects. Changing the description changes choices, as in the survival/mortality frames for physicians and in the difference between a "discount" and a "surcharge" for cash vs credit card payments.

Transactions and trades (mental accounting). People are willing to drive 20 minutes to save $5 on a $15 calculator (68%) but not on a $125 calculator (29%). Savings are evaluated relative to the price in a *topical* mental account. Losing a $10 ticket makes people less willing to buy another (46% would) than losing a $10 bill (88% would still buy a ticket), because the lost ticket is charged to the play's account.

Losses and costs. Many choices are between the status quo and an alternative. Disadvantages are evaluated as losses, which biases decisions toward the status quo. This is Thaler's **endowment effect**, and the reason buying and selling prices differ. Paying a premium for insurance is accepted as a cost, while an equivalent sure loss is rejected. Treating outlays as costs rather than losses changes decisions.

Concluding remarks. The paper distinguishes **experience value** (the pleasure or pain actually felt) from **decision value** (the contribution of an anticipated outcome to a choice). Decision theory assumes the two coincide. They often don't, which anticipates Part 5's experienced vs decision utility.

Key quotes

> *"We discuss the cognitive and the psychophysical determinants of choice in risky and riskless contexts."* p. 431

> *"The psychophysics of value induce risk aversion in the domain of gains and risk seeking in the domain of losses."* p. 431

> *"However, all analyses of rational choice incorporate two principles: dominance and invariance."* p. 434

> *"If Program A is adopted, 200 people will be saved."* p. 435

> *"If Program C is adopted, 400 people will die."* p. 436

> *"We call this phenomenon the pseudo-certainty effect because an event that is actually uncertain is weighted as if it were certain."* p. 441

> *"Because losses loom larger than gains, the decision maker will be biased in favor of retaining the status quo."* p. 446

Connections

- **The book's Part 4 explains this paper** in fuller form. Value function → Chapter 26. Weighting and the fourfold pattern → Chapter 29. Framing → Chapter 34. Mental accounts → Chapter 32. Endowment → Chapter 27. Experience vs decision value → Chapter 35.

Markets, money and people

- **The calculator effect in finance.** Investors haggle over small brokerage fees on small trades but ignore much larger percentage costs on big decisions (fund expense ratios, advisory fees), because savings are judged relative to the "topical" amount.
- **Pseudo-certainty in products.** Products that "guarantee" one stage (principal-protected notes, "risk-free" yield in crypto lending platforms like Celsius, which froze withdrawals in June 2022 and filed for bankruptcy in July 2022) exploit the overweighting of apparent certainty.
- **Costs vs losses.** Framing hedges and insurance as a "cost of doing business" makes them acceptable. The same money framed as a "loss" would be resisted.

Takeaways

- Frames change choices even when outcomes are identical, violating invariance.
- Mental accounts are topical: we judge savings and losses relative to the specific transaction.
- Experience value and decision value differ, which anticipates the two selves.

How to apply it

- 💰 **Investing:** judge savings and costs in absolute dollars across your whole portfolio, not relative to a transaction. Reframe hedges as insurance costs, budgeted in advance.
- 🤝 **People:** a $5 discount on a cheap item motivates more than the same discount on an expensive one, so frame savings relative to small bases when you want them to feel big.
- 🎬 **Content:** the Asian disease problem is a superb poll. Split your audience across two frames (community post A vs B) and reveal the results.
- 🧠 **Thinking:** check invariance. "Would I choose the same if this were described the other way?"

Master List of Biases & Effects, with Investing Applications

Every major bias, heuristic and effect in the book, in roughly the order they appear. "Ch" refers to the chapter where it is mainly discussed. The investing column gives a concrete defence or application.

Bias / effect	What it is	Ch	Investing application / defence
Inattentional blindness	Missing obvious things outside your focus (the gorilla)	1	Single-metric focus (e.g. VaR) hides growing risks. Review a broad risk dashboard, not one number.
Law of least effort	Defaulting to the least demanding route	2	Ratings and headlines stand in for analysis. Force yourself to read the footnotes on complex products.
Ego depletion / decision fatigue*	Self-control and judgment weaken with load and fatigue (*weak replication)	3	Don't make big portfolio decisions late, tired or multitasking.
Lazy System 2 / belief bias	Accepting intuitive answers or plausible conclusions without checking	3	Recompute percentages, break-evens and compounding on paper ("bat-and-ball" checks).
Priming*	Prior exposure shapes later thoughts and actions (*behavioural priming weakly replicated)	4	Strip hype labels from ideas. Describe the business neutrally before reading the narrative.
Cognitive ease / illusion of truth	Repetition, fluency and clarity feel like truth	5	Be extra sceptical of consensus trades and polished pitches. Reformat key numbers plainly.
Mere exposure effect	Familiarity breeds liking	5	Home bias and "brands I know." Check whether familiarity is substituting for analysis.
Mood and gullibility	Good mood loosens System 2	5	After big wins, slow down. Euphoria makes you careless.
Causal over-interpretation	Seeing causes and intentions in random events	6	Most daily moves are noise. Know your positions' normal volatility.
Confirmation bias	Searching for evidence that supports what you're testing	7	Write the strongest opposing case. Ask "What would change my mind?"
Halo effect	Liking one trait colours judgments of everything	7, 19	Charismatic CEOs and famous backers (Theranos, Madoff). Score attributes separately.
WYSIATI	Building a confident story from only the information at hand	7	List what you don't know. Hunt for the missing data before sizing up.

Bias / effect	What it is	Ch	Investing application / defence
Correlated errors / groupthink	Discussion before independent judgment makes errors correlated	7	Collect written, independent views before investment committee debate.
Scope insensitivity	System 1 thinks in averages and prototypes, not sums	8	Convert risks to dollars and portfolio percentages so that size registers.
Mental shotgun	Computing more than asked, so irrelevant assessments leak in	8	"Love the product" ≠/"good investment at this price" (e.g. Peloton 2021).
Substitution	Answering an easier question instead of the hard one	9	Write the exact target question. Check that your analysis answers it.
Affect heuristic	Feelings determine judgments of risk and benefit	9, 13	Assets you love seem low-risk. Ask a neutral colleague for the bear case.
Law of small numbers	Trusting small samples, seeing patterns in noise	10	Short track records and backtests mislead. Demand long out-of-sample evidence.
Hot-hand belief	Expecting streaks to continue	10	Fund and trader streaks are mostly luck (SPIVA persistence). Don't chase.
Anchoring	Estimates pulled toward any number considered first	11	Ignore purchase price, 52-week high and price targets. Estimate value before seeing the price.
Availability heuristic	Judging frequency by ease of recall	12	Recent crashes and booms feel likelier. Consult a base-rate file.
Self-serving contribution bias	Overestimating one's own share of joint work	12	Partners and team members each overcredit themselves. Use objective attribution.
Availability cascade	Media and public feedback loops inflate risks	13	Bank runs (SVB 2023), meme squeezes. Verify the data, keep liquidity for panics.
Probability neglect	Vivid harms make probability irrelevant	13	Fear trades. Ask "how likely?" not just "could it happen?"
Representativeness / base-rate neglect	Judging probability by similarity to a stereotype	14	"Next Amazon" stories vs the low base rate of success for similar firms.
Conjunction fallacy	Detailed scenarios judged more probable than simpler ones	15	Multiply step probabilities in multi-step macro theses.
Less is more (averages vs sums)	Adding lower-quality items lowers perceived value	15	Bundles and padded offerings. Judge totals, not average quality.
Statistical vs causal base rates	Statistics ignored unless they tell a causal story	16	Turn base rates into stories: keep a "graveyard file" of similar failures.
Regression neglect	Inventing causes for regression to the mean	17	Last year's top funds and sectors tend to regress. Expect extremes to fade.

Bias / effect	What it is	Ch	Investing application / defence
Nonregressive prediction	Predictions as extreme as the evidence feels	18	Regress growth forecasts toward base rates in proportion to the evidence's validity.
Narrative fallacy	Simple causal stories of the past that underweight luck	19	Success stories (and post-mortems) overstate skill. Ask where luck mattered.
Hindsight bias	"I knew it all along"	19	Decision journals written before the outcome.
Outcome bias	Judging decisions by results, not process	19	Evaluate process separately from P&L. Good bets can lose.
Illusion of validity	Confidence from coherence, not accuracy	20	Track your hit rates. Confidence is not evidence.
Illusion of skill (stock picking)	Believing in skill where outcomes are mostly luck	20	Ask "Why isn't this in the price?" Index the core. Trade less (Odean).
Hedgehog thinking / expert overconfidence	One big theory, confident forecasts, rare admissions of error	20	Prefer fox-like probabilistic forecasters with scored records.
Algorithm aversion	Preferring human judgment to better-performing formulas	21	Use rules and scorecards for entries, exits and risk.
Inconsistency (noise)	The same expert gives different answers to the same case	21	Standardize decision criteria. Checklists reduce noise.
Overconfidence in intuition	Trusting gut feel in irregular or low-feedback environments	22	Trust intuition only where feedback is fast and the environment regular (execution, not 5-year forecasts).
Planning fallacy / inside view	Best-case plans, ignoring the reference class	23	Reference-class forecasting for timelines, M&A synergies, turnarounds.
Sunk-cost fallacy	Continuing because of past investment	23, 32	Ask "Would I enter this position today?" Past costs are irrelevant.
Optimistic bias	Overrating one's odds and abilities	24	Premortems. Widen confidence intervals. Size for being wrong.
Competition neglect / excess entry	Ignoring rivals when estimating success	24	Crowded trades and hot sectors (SPAC 2021). Ask "Who else is doing this?"
Overconfidence (narrow intervals)	Confidence intervals far too narrow	24	CFOs' 80% intervals were missed 67% of the time. Plan for wider ranges.
Theory-induced blindness	Accepted theories hide their own flaws	25	Question model assumptions (normal distributions, stable correlations).
Reference dependence	Valuing changes relative to a reference point, not final states	25–26	Notice your reference (cost basis, peak, last statement). Decide on total wealth.

Bias / effect	What it is	Ch	Investing application / defence
Loss aversion	Losses weigh about 1.5–2.5× equivalent gains	26	Turning down good bets. Use expected value and broad framing.
Risk seeking in losses ("get back to even")	Gambling to avoid sure losses	26, 29	Pre-set stops. Never double down to recover (Barings, SocGen).
Endowment effect	Valuing what you own more than you'd pay for it	27	Review holdings as if in cash. "Would I buy it now?"
Negativity dominance	Bad is stronger than good	28	Markets often fall faster than they rise. Panics can create opportunities if the thesis is intact.
Goals as reference points	Falling short of a target feels like a loss	28	Avoid daily or annual P&L targets that distort risk (the cabdriver trap).
Status quo bias	Loss aversion favours no change	28	Schedule regular rebalancing so inertia doesn't decide.
Fairness / reference transactions	Price changes judged against past terms	28	Pricing-power analysis: customers resist "unfair" hikes.
Possibility effect	Small probabilities overweighted	29	Lottery-like stocks and far-OTM options tend to be overpriced.
Certainty effect	Near-certainties underweighted relative to certainty	29	Overpaying for "guaranteed" or "protected" products.
Fourfold pattern	Risk attitudes flip by gain/loss and high/low probability	29	Diagnose which cell you are in before acting.
Overweighting vivid rare events	Vivid or focused-on rare events weighted too much	30	Crash fear after crashes. Set tail-hedge budgets by policy.
Denominator neglect	"1 in 10" vs "8 in 100"; counts feel bigger than rates	30	Convert all claims to the same format and base.
Underweighting unexperienced rare events	Rare events never experienced are neglected	30	Stress-test against events nobody on the team has lived through (2007 bankers).
Narrow framing	Evaluating each decision in isolation	31	Think portfolio, not position. Use risk policies.
Myopic loss aversion	Frequent evaluation + loss aversion = too little risk	31	Check long-term accounts quarterly at most.
Mental accounting	Treating money differently by label or account	32	Money is fungible. Optimize the whole balance sheet.
Disposition effect	Selling winners, holding losers	32	Cut losers by rule. Harvest tax losses. Let winners run.
Regret aversion / omission bias	More regret for actions than inactions	32	Inaction is also a decision. Anticipate both regrets.

Bias / effect	What it is	Ch	Investing application / defence
Taboo tradeoffs	Refusing to trade safety for money at any price	32	Recognize when "safety" products are overpriced.
Preference reversals (single vs joint evaluation)	Judgments flip when options are compared	33	Always compare to a benchmark and alternatives.
Evaluability	Some attributes are meaningful only in comparison	33	P/E, fees and yields mean little without context.
Framing effects	Equivalent descriptions lead to different choices	34	Reframe every decision (gain/loss, %, $, annual/cumulative).
Default effects	Pre-set options stick	34	Automate good defaults (contributions, rebalancing, stops).
Peak-end rule	Experiences remembered by peak and end	35	Don't let a bad December set next year's risk. Use full-period data.
Duration neglect	Length of experience ignored in memory	35	A long, good bull run can be overshadowed by a short crash at the end.
Focusing illusion	Whatever you're thinking of seems most important	38	One macro variable isn't everything. List other drivers.
Affective forecasting errors / miswanting	Mispredicting future feelings	38	Hitting a wealth number changes life evaluation more than daily mood.
Pseudo-certainty effect	Stage-wise certainty treated as real certainty	App B	"Principal-protected" and "risk-free yield" offerings (e.g. Celsius 2022).
Conjunctive/disjunctive event bias	Overestimating chains of success; underestimating any-failure risk	App A	Multi-step deals fail more often, and any one counterparty can blow up.
Illusory correlation	Seeing correlations that match expectations	App A	Indicators that "always work" because you remember the hits.

One-Page Cheat Sheet

The model

System 1: fast, automatic, associative, emotional, coherence-seeking. **System 2**: slow, effortful, lazy, limited. **Econs** vs **Humans**. **Experiencing** vs **remembering** self (peak-end, duration neglect).

The 5 engines of error

1. **WYSIATI**: what you see is all there is.
2. **Substitution**: answering an easier question.
3. **Cognitive ease**: fluent feels true.
4. **Causal stories** over statistics.
5. **Reference points + loss aversion** (about 2×).

Before every investment

☐ Exact target question written?
☐ Base rate / reference class?
☐ Fair value estimated *before* seeing price?
☐ Bear case + missing info listed?
☐ Steps multiplied (conjunction)?
☐ Regressed forecast vs story forecast?
☐ Premortem done?
☐ Size set for being wrong?

While holding

☐ "Would I buy it today?" (endowment, sunk cost)
☐ Ignore cost basis and 52-week high (anchors)
☐ Cut losers by rule (disposition effect)
☐ Check quarterly, not daily (myopic loss aversion)
☐ Is this move noise? (causal over-reading)

Minefield warning signs → SLOW DOWN

High stakes · unfamiliar · strong emotion · time pressure · a beautiful story · high confidence · everyone agrees · just won or lost big · tired or rushed.

When to trust intuition

Only when the environment is **regular** AND you have had **long practice with fast, clear feedback**. Confidence ≠ accuracy.

Prediction formula

Forecast = Baseline + r × (Intuitive − Baseline).
Weak evidence ⇒ stay near the base rate.
Extreme results ⇒ expect regression.

Risk policy

Think portfolio, not position (broad framing). Accept many small +EV bets. Skip extended warranties. Pre-commit stops and limits. Never "get back to even" by doubling down.

Persuasion & people

Clear + simple + repeated + rhymes = believed. Lead with a person, then the statistic. Anchor first. Frame as gains or avoided losses. Make good choices the default. End on a high (peak-end). 5 positives : 1 negative.

Content creation

One idea per video · hook = a puzzle System 1 gets wrong · fluent visuals · a story before data · a clear peak and strong ending · honest frames · show the base rate · score your past calls (be a fox).

Organizational defences

Independent views before discussion · checklists and scorecards · reference-class forecasts · premortems · decision journals · shared vocabulary ("Are we anchoring?").

Five lines to remember

"What you see is all there is." · "Nothing in life is as important as you think it is when you are thinking about it." · "Intuition is nothing more and nothing less than recognition." · "Losses loom larger than gains." · "Whenever we can replace human judgment by a formula, we should at least consider it."